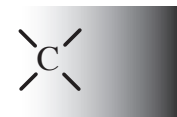


5 ^1H NMR Spectroscopy



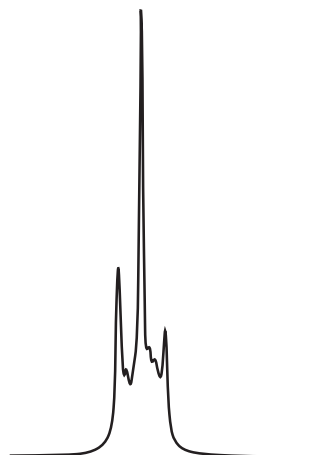
5.1 Alkanes

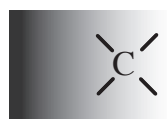
5.1.1 Chemical Shifts

^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

CH_4 0.23 $^2J_{\text{gem}}$ -12.4	CH_3 0.86 CH_3	CH_3 0.91 CH_2 1.33 CH_3	$^3J_{\text{vic}}$ 7.4	CH_3 a 0.91 CH_2 b 1.31 CH_2 CH_3	$^3J_{\text{ab}}$ 7.3 $^2J_{\text{bb'}}$ -12.4 $^3J_{\text{bc}}$ 5.7 $^3J_{\text{bc'}}$ 8.5
CH_3 0.89 CH 1.74 H_3C CH_3	$^3J_{\text{vic}}$ 6.8	CH_3 0.90 $\text{H}_3\text{C}-\text{C}-\text{CH}_3$ CH_3		CH_3 0.88 CH_2 1.27 CH_2 1.27 CH_2 CH_3	

In long-chain alkanes, the methyl groups at δ ca. 0.8 ppm typically show distorted triplets due to second-order effects:

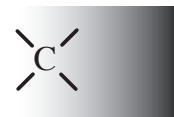


^1H Chemical Shifts of Monosubstituted Alkanes (δ in ppm)


	Substituent	Methyl		Ethyl		1-Propyl	
		$-\text{CH}_3$	$-\text{CH}_2$	$-\text{CH}_3$	$-\text{CH}_2$	$-\text{CH}_2$	$-\text{CH}_3$
C	$-\text{H}$	0.23	0.86	0.86	0.91	1.33	0.91
	$-\text{CH}=\text{CH}_2$	1.71	2.00	1.00	2.02	1.43	0.91
	$-\text{C}\equiv\text{CH}$	1.80	2.16	1.15	2.10	1.50	0.97
	$-\text{phenyl}$	2.35	2.63	1.21	2.59	1.65	0.95
X	$-\text{F}$	4.27	4.55	1.35	4.30	1.68	0.97
	$-\text{Cl}$	3.06	3.47	1.33	3.47	1.81	1.06
	$-\text{Br}$	2.69	3.37	1.66	3.35	1.89	1.06
	$-\text{I}$	2.16	3.16	1.88	3.16	1.88	1.03
O	$-\text{OH}$	3.48	3.71	1.24	3.59	1.59	0.94
	$-\text{O-alkyl}$	3.24	3.37	1.15	3.27	1.55	0.93
	$-\text{OCH}=\text{CH}_2$	3.16	3.66	1.21			
	$-\text{O-phenyl}$	3.73	3.98	1.38	3.86	1.70	1.05
	$-\text{OCOCH}_3$	3.67	4.12	1.26	4.02	1.65	0.95
	$-\text{OCO-phenyl}$	3.88	4.37	1.38	4.25	1.76	1.07
	$-\text{OS(O)}_2\text{-4-tolyl}$	3.70	4.07	1.30	3.94	1.60	0.95
N	$-\text{NH}_2$	2.47	2.66	1.11	2.65	1.46	0.91
	$-\text{NHCH}_3$	2.30					
	$-\text{N(CH}_3)_2$	2.22	2.32	1.06			
	$-\text{NHCOCH}_3$	2.79	3.26	1.14	3.18	1.55	0.96
	$-\text{NO}_2$	4.29	4.37	1.58	4.28	2.01	1.03
	$-\text{C}\equiv\text{N}$	1.98	2.35	1.31	2.34	1.70	1.08
	$-\text{NC}$	2.85	3.39	1.28			
S	$-\text{SH}$	2.00	2.44	1.31	2.50	1.63	0.99
	$-\text{S-alkyl}$	2.09	2.49	1.25	2.43	1.59	0.98
	$-\text{SS-alkyl}$	2.30	2.67	1.35	2.63	1.71	1.03
	$-\text{S(O)CH}_3$	2.50					
	$-\text{S(O)}_2\text{CH}_3$	2.84	2.94	2.80			
O C	$-\text{CHO}$	2.20	2.46	1.13	2.37	1.64	0.97
	$-\text{COCH}_3$	2.17	2.44	1.06	2.40	1.60	0.93
	$-\text{CO-phenyl}$	2.55	2.92	1.18	2.86	1.72	1.02
	$-\text{COOH}$	2.10	2.36	1.16	2.31	1.68	1.00
	$-\text{COOCH}_3$	2.01	2.32	1.15	2.22	1.65	0.98
	$-\text{CONH}_2$	2.02	2.23	1.13	2.19	1.68	0.99
	$-\text{COCl}$	2.66	2.93	1.24	2.87	1.74	1.00

¹H Chemical Shifts of Monosubstituted Alkanes (δ in ppm, contd.)

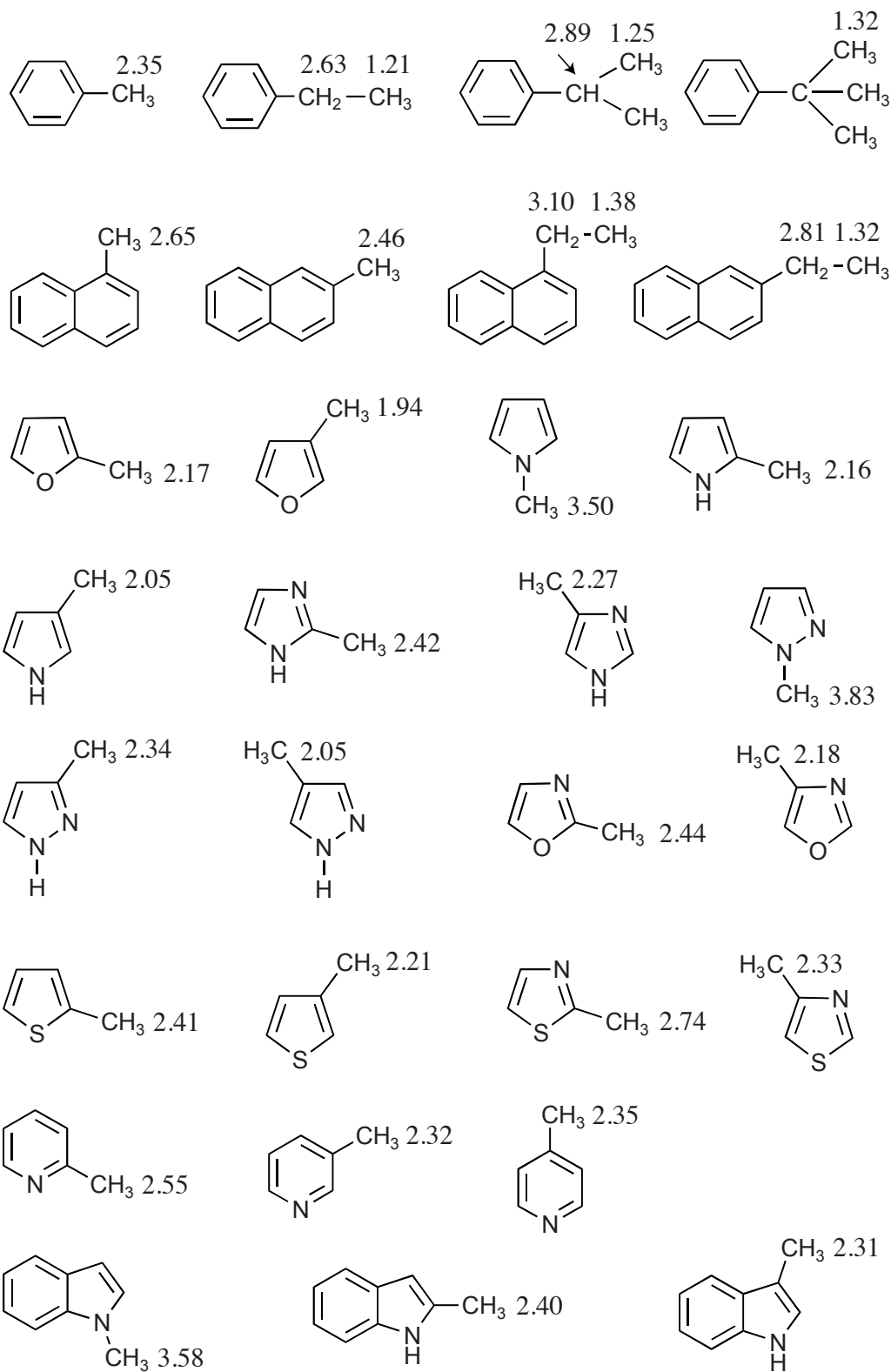
	Substituent	2-Propyl		<i>n</i> -Butyl			<i>tert</i> -Butyl	
		-CH	-CH ₃	-CH ₂	-CH ₂	-CH ₂	-CH ₃	-CH ₃
C	-H	1.33	0.91	0.91	1.31	1.31	0.91	0.89
	-CH=CH ₂			2.06	≈1.5	≈1.2	0.90	1.02
	-C≡C-	2.59	1.15	2.18	1.52	1.41	0.92	1.24
	-phenyl	2.89	1.25	2.61	1.60	1.34	0.93	1.32
X	-F	4.84	1.34	4.34	1.65		0.95	1.34
	-Cl	4.14	1.55	3.42	1.68	1.41	0.92	1.60
	-Br	4.21	1.73	3.42	1.84	1.46	0.93	1.76
	-I	4.24	1.89	3.20	1.80	1.42	0.93	1.95
O	-OH	4.02	1.21	3.64	1.56	1.39	0.94	1.26
	-O-alkyl	3.55	1.08	3.40	1.54	1.38	0.92	1.24
	-OCH=CH ₂	4.06	1.23	3.68	1.61	1.39	0.94	
	-O-phenyl	4.51	1.31	3.94	1.76	1.47	0.97	
	-OCOCH ₃	4.99	1.23	4.06	1.60	1.39	0.94	1.45
	-OCO-phenyl	5.22	1.37					1.58
	-OS(O) ₂ -4-tolyl	4.70	1.25	4.03	1.62	1.36	0.88	
N	-NH ₂	3.07	1.03	2.69	1.43	1.35	0.92	1.15
	-NHCOCH ₃	4.01	1.13	3.21	1.49	1.35	0.92	1.28
	-NO ₂	4.44	1.53	4.47	2.07	1.50	1.07	1.59
	-C≡N	2.67	1.35	2.34	1.63	1.50	0.96	1.37
	-NC	3.87	1.45					1.44
S	-SH	3.16	1.34	2.52	1.59	1.43	0.92	1.43
	-S-alkyl	2.93	1.25	2.49	1.56	1.42	0.92	1.39
	-SS-alkyl			2.69	1.64	1.42	0.93	1.32
	-S(O) ₂ CH ₃	3.13	1.41					1.44
O C	-CHO	2.39	1.13	2.42	1.59	1.35	0.93	1.08
	-COCH ₃	2.58	1.11					1.13
	-CO-phenyl	3.58	1.22	2.95	1.72	1.41	0.96	
	-COOH	2.56	1.21	2.35	1.62	1.39	0.93	1.23
	-COOCH ₃	2.56	1.17	2.31	1.61	1.33	0.92	1.20
	-CONH ₂	2.44	1.18	2.22	1.60	1.37	0.93	1.22
	-COCl	2.97	1.31	2.88	1.67	1.40	0.93	



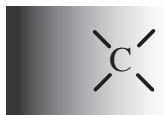
Estimation of ¹H Chemical Shifts of Substituted Alkanes (δ in ppm)

$$\begin{aligned} \delta_{\text{CH}_3\text{R}^1} &= 0.86 + Z_\alpha \\ \delta_{\text{CH}_2} &= 1.37 + \sum_i Z_{\alpha_i} + \sum_j Z_{\beta_j} \\ \delta_{\text{CH}} &= 1.50 + \sum_i Z_{\alpha_i} + \sum_j Z_{\beta_j} \\ \delta_{\text{CH}_3\text{CR}^1\text{R}^2\text{R}^3} &= 0.86 + \sum_i Z_{\beta_i} \end{aligned}$$

	Substituent (R ¹ , R ² , R ³)	CH ₃		CH ₂		CH	
		Z _α	Z _β	Z _α	Z _β	Z _α	Z _β
C	–C≡	0.00	0.05	0.00	-0.06	0.17	-0.01
	–C=C<	0.85	0.20	0.63	0.00	0.68	0.03
	–C≡C–	0.94	0.32	0.70	0.13	1.04	
	–phenyl	1.51	0.38	1.22	0.29	1.28	0.38
X	–F	3.41	0.41	2.76	0.16	1.83	0.27
	–Cl	2.20	0.63	2.05	0.24	1.98	0.31
	–Br	1.83	0.83	1.97	0.46	2.44	0.41
	–I	1.30	1.02	1.80	0.53	2.46	0.15
O	–OH	2.53	0.25	2.20	0.15	1.73	0.08
	–O–C≡	2.38	0.25	2.04	0.13	1.85	0.32
	–OC=C<	2.64	0.36	2.63	0.33	2.00	0.30
	–O–phenyl	2.87	0.47	2.61	0.38	2.20	0.50
	–O–CO–	2.81	0.44	2.83	0.24	2.47	0.59
N	–N<	1.61	0.14	1.32	0.22	1.13	0.23
	–N ⁺ ≡	2.44	0.39	1.91	0.40	1.78	0.56
	–N–CO–	1.88	0.34	1.63	0.22	2.10	0.62
	–NO ₂	3.43	0.65	3.08	0.58	2.31	
	–C≡N	1.12	0.45	1.08	0.33	1.00	
	–NCS	2.51	0.54	2.20	0.36	1.94	0.60
S	–S–	1.14	0.45	1.23	0.26	1.06	0.31
	–S–CO–	1.41	0.37	1.54	0.63	1.31	0.19
	–S(O)–	1.64	0.36	1.24	0.30	1.25	
	–S(O) ₂ –	1.98	0.42	2.08	0.52	1.50	0.40
	–SCN	1.75	0.66	1.62		1.64	
O C	–CHO	1.34	0.21	1.07	0.29	0.86	0.22
	–CO–	1.23	0.20	1.12	0.24		
	–COOH	1.22	0.23	0.90	0.23	0.87	0.32
	–COO–	1.15	0.28	0.92	0.35	0.83	0.63
	–CO–N<	1.16	0.28	0.85	0.24	0.94	0.30
	–COCl	1.94	0.22	1.51	0.25		

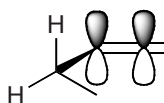
¹H Chemical Shifts of Aromatically Substituted Alkanes (δ in ppm)

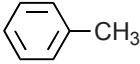
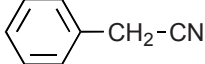
5.1.2 Coupling Constants

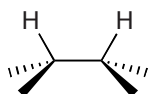
**Geminal Coupling Constants (2J in Hz)**

$^2J_{\text{HCH}} -8 \text{ to } -18$

Electronegative substituents cause a decrease in $|J|_{\text{gem}}$ while a double or triple bond next to the CH_2 group causes an increase. The latter effect is strongest if one of the C-H bonds is parallel to the π orbitals:

**Influence of Substituents on the Geminal Coupling Constant**

Compound	J_{gem}	Compound	J_{gem}
CH_4	-12.4	CH_3COCH_3	-14.9
CH_3Cl	-10.8	CH_3COOH	-14.5
CH_2Cl_2	-7.5	CH_3CN	-16.9
CH_3OH	-10.8	$\text{CH}_2(\text{CN})_2$	-20.3
	-14.3		-18.5

Vicinal Coupling Constants (3J in Hz)

$$\begin{array}{l} \text{conformation not fixed: } ^3J_{\text{HCCH}} \approx 7 \\ \text{fixed: } ^3J_{\text{HCCH}} \approx 0-18 \end{array}$$

Influence of Substituents on the Vicinal Coupling Constant

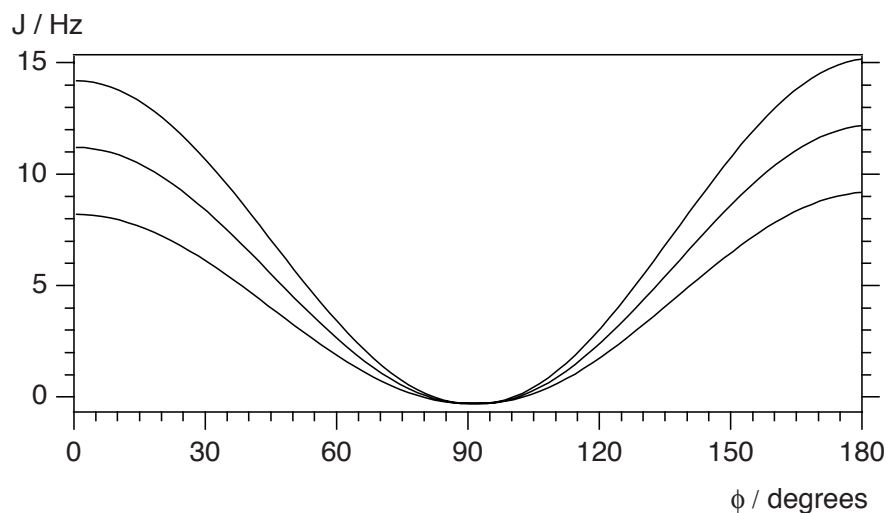
Compound	J_{vic}	Compound	J_{vic}	Compound	J_{vic}
CH_3CHF_2	4.5	$\text{CH}_3\text{CH}_2\text{OH}$	6.9	$\text{CH}_3\text{CH}_2\text{CN}$	7.6
CH_3CHCl_2	6.1	$(\text{CH}_3\text{CH}_2)_3\text{O}^+ \text{BF}_4^-$	7.2	$(\text{CH}_3\text{CH}_2)_2\text{S}$	7.4
$\text{CH}_3\text{CH}_2\text{F}$	6.9	$(\text{CH}_3\text{CH}_2)_3\text{N}$	7.1	$(\text{CH}_3\text{CH}_2)_4\text{Si}$	8.0
$\text{CH}_3\text{CH}_2\text{Cl}$	7.2	$(\text{CH}_3\text{CH}_2)_4\text{N}^+ \text{I}^-$	7.3	$\text{CH}_3\text{CH}_2\text{Li}$	8.4

Vicinal coupling constants strongly depend on the dihedral angle, ϕ (Karplus equation):

$$\begin{aligned} {}^3J &= J^0 \cos^2 \phi - 0.3 & 0^\circ \leq \phi \leq 90^\circ \\ {}^3J &= J^{180} \cos^2 \phi - 0.3 & 90^\circ \leq \phi \leq 180^\circ \end{aligned}$$

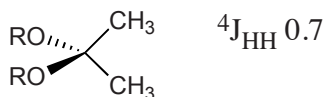


The same relationship between torsional angle and vicinal coupling constant holds for substituted alkanes if appropriate values are used for J^0 and J^{180} . These limiting values depend on the electronegativity and orientation of substituents, the hybridization of carbon atoms, bond lengths, and bond angles.



Long-Range Coupling Constants ($|J|$ in Hz)

Coupling constants through more than three bonds (long-range coupling) in alkanes are generally much smaller than 1 Hz and, thus, not visible in routine 1D NMR spectra. They are, however, much larger than 1 Hz for fixed conformations (e.g., in condensed alicyclic systems, see Chapter 5.4) and in unsaturated compounds (see Chapter 5.2). They are also significant when electronegative substituents are present between the coupling partners, as e.g.:

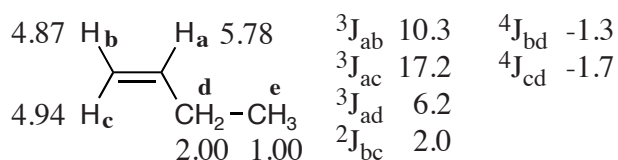
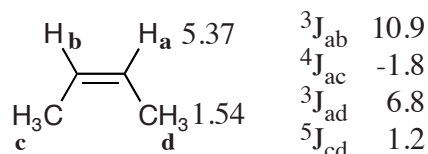
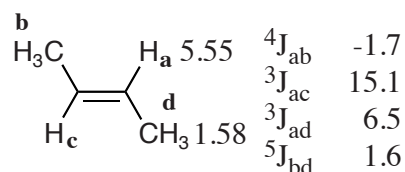
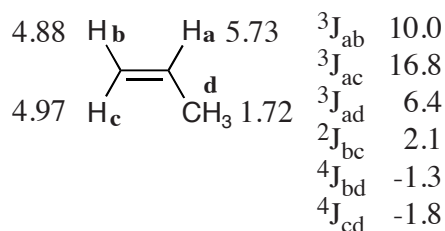
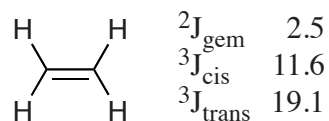


5.2 Alkenes

5.2.1 Substituted Ethylenes

 ^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

C=C

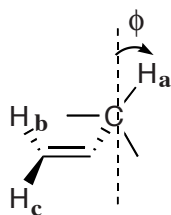
**Geminal and Vicinal Couplings of Alkenes (J in Hz)**

The values of the coupling constants strongly depend on the electronegativity of the substituents (see Table on pp 166, 167). They decrease with increasing electronegativity and number of electronegative substituents. The same trend holds for the signed values of geminal coupling constants but not for the absolute values because J_{gem} can be positive or negative. Although the total ranges of cis and trans vicinal coupling constants overlap, $J_{\text{trans}} > J_{\text{cis}}$ always holds for given substituents.

Typical ranges:	J_{gem}	-4 to 4
	J_{cis}	4 to 12
	J_{trans}	14 to 19

Coupling Over More than Three Bonds in Alkenes (Long-Range Coupling, J in Hz)

Allylic Coupling



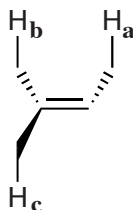
$$\begin{aligned} \text{cisoid: } & {}^4J_{ab} -3.0 \text{ to } +2.0 \\ \text{transoid: } & {}^4J_{ac} -3.5 \text{ to } +2.5 \end{aligned}$$



In acyclic systems, the coupling constants range from ca. -0.8 to -1.8 Hz and, usually, $|J|_{\text{cisoid}}$ is larger than $|J|_{\text{transoid}}$. The magnitudes of the coupling constants depend on the conformation. Largest absolute values are observed if the C-H bond of the substituents overlaps with the π electrons of the double bond ($\phi = 0$ or 180°):

ϕ	${}^4J_{ab}$	${}^4J_{ac}$
0°	-3.0	-3.5
90°	+1.8	+2.2
180°	-3.0	-3.5
270°	0.0	0.8

Homoallylic Coupling

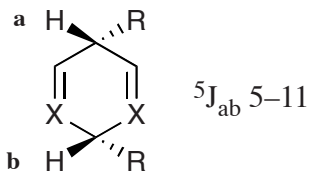


$$\begin{aligned} \text{cisoid: } & |{}^5J_{ab}| 0-3 \\ \text{transoid: } & |{}^5J_{ac}| 0-3 \end{aligned}$$

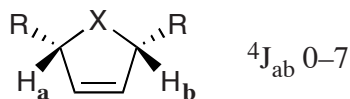
The values of homoallylic coupling constants between methyl groups and of allylic ones are comparable:

$${}^5J_{\text{H}_3\text{C}-\text{C}=\text{C}-\text{CH}_3} \approx {}^4J_{\text{H}-\text{C}=\text{C}-\text{CH}_3}$$

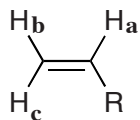
In acyclic systems, $|J|_{\text{cisoid}} < |J|_{\text{transoid}}$ usually holds. Large homoallylic coupling constants are occasionally observed in cyclic systems with fixed conformation between the protons:



X: CH, N
R: any substituent



X: O, NH
R: any substituent

¹H Chemical Shifts and Coupling Constants of Monosubstituted Ethylenes
(δ in ppm, J in Hz)**C = C**

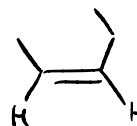
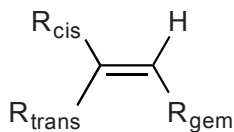
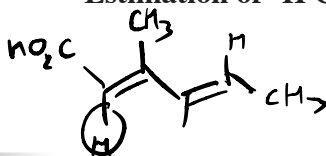
	Substituent R	H _a	H _b	H _c	J _{ab}	J _{ac}	J _{bc}	Other
	-H	5.28	5.28	5.28	11.6	19.1	2.5	
C	-CH ₃	5.73	4.88	4.97	10.0	16.8	2.1	CH ₃ 1.72
	-CH ₂ CH=CH ₂	5.71	4.92	4.95	10.3	16.9	2.2	CH ₂ 2.72
	-CH ₂ -phenyl	5.89	5.00	5.01	10.0	17.0	1.9	CH ₂ 3.19
	-cyclopropyl	5.32	4.84	5.04	10.4	17.1	1.8	
	-cyclohexyl	5.79	4.88	4.95	10.5	17.6	1.9	
	-CF ₃	5.90	5.56	5.85	11.1	17.5	0.2	
	-CH=C=CH ₂	6.31	4.99	5.19	10.1	17.2	1.6	
	-C≡C-CH ₃	5.62	5.24	5.39	11.1	17.0	2.3	
	-phenyl	6.72	5.20	5.72	11.1	17.9	1.0	
	-2-naphthyl	6.87	5.32	5.86				
	-2-nitrophenyl	7.19	5.45	5.68	10.7	17.4	1.1	
	-3-nitrophenyl	6.74	5.42	5.86	10.9	17.5	0.4	
	-4-nitrophenyl	6.77	5.48	5.90	10.9	17.4	0.8	
	-2-pyridyl	6.84	5.45	6.22	11.3	18.5	1.4	
	-4-pyridyl	6.61	5.42	5.91	10.8	17.6	0.7	
X	-F	6.17	4.03	4.37	4.7	12.8	-3.2	
	-Cl	6.26	5.39	5.48	7.5	14.5	-1.4	
	-Br	6.44	5.97	5.84	7.1	14.9	-1.9	
	-I	6.53	6.23	6.57	7.8	15.9	-1.5	
O	-OH	6.45	3.82	4.18	6.4	14.2	-1.0	
	-OCH ₃	6.44	3.88	4.03	7.0	14.1	-2.0	CH ₃ 3.16
	-OCH=CH ₂	6.49	4.21	4.52	6.4	14.0	-1.8	
	-O-phenyl	6.64	4.40	4.74	6.1	13.7	-1.6	
	-OCHO	7.33	4.66	4.96	6.4	13.9	-1.7	CHO 8.07
	-OCOCH ₃	7.28	4.56	4.88	6.3	14.1	-1.6	CH ₃ 2.13
	-OCOCH=CH ₂	7.39	4.62	4.96	6.4	14.2	-1.6	
	-OCO-phenyl	7.52	4.67	5.04	6.3	13.8	-1.7	
	-OP(O)(O-ethyl) ₂	6.58	4.59	4.91	6.0	13.8	-2.1	
N	-NH ₂	≈6.05	≈3.99	≈4.04				
	-N ⁺ (CH ₃) ₃ Br ⁻	6.50	5.54	5.76	8.2	15.1	-4.3	
	-NHCOCH ₃	≈7.33	≈4.68	≈4.53				
	-NO ₂	7.12	5.87	6.55	7.0	14.6	1.4	
	-C≡N	5.69	6.11	6.24	11.8	17.9	0.9	
	-NC	5.90	5.35	5.58	8.6	15.6	-0.5	
	-NCO	6.12	4.77	5.01	7.6	15.2	-0.1	

	Substituent R	H _a	H _b	H _c	J _{ab}	J _{ac}	J _{bc}	Other
S	–SCH ₃	6.43	5.18	4.95	10.3	16.4	-0.3	CH ₃ 2.25
	–S–phenyl	6.53	5.32	5.32	9.6	16.7	-0.2	
	–S(O)CH ₃	6.77	5.92	6.08	9.8	16.7	-0.6	CH ₃ 2.61
	–S(O) ₂ CH ₃	6.76	6.14	6.43	10.0	16.5	-0.5	CH ₃ 2.96
	–S(O) ₂ CH=CH ₂	6.67	6.17	6.41	10.0	16.4	-0.6	
	–S(O) ₂ OH	6.73	6.13	6.41	10.2	16.8	-1.2	
	–S(O) ₂ OCH ₃	6.57	6.22	6.43	10.1	16.9	-0.6	CH ₃ 3.85
	–S(O) ₂ NH ₂	6.93	5.98	6.17	10.0	16.3	0	NH ₂ 6.7
	–S(O) ₂ NH–phenyl	6.56	5.86	6.18	10.1	16.7	-0.3	NH 9.07
	–SF ₅	6.63	5.64	5.96	9.8	16.6	0.4	
	–SCN	6.19	5.70	5.66				
O C	–CHO	6.37	6.52	6.35	10.0	17.4	1.0	CHO 9.59
	–COCH ₃	6.30	5.91	6.21	10.7	18.7	1.3	CH ₃ 2.29
	–COCH=CH ₂	6.67	5.82	6.28	11.0	17.9	1.4	
	–CO–phenyl	7.20	5.81	6.52	9.9	17.7	2.3	
	–COOH	6.15	5.95	6.53	10.5	17.2	1.8	COOH 12.8
	–COOCH ₃	6.12	5.83	6.41	10.6	17.4	1.5	CH ₃ 3.77
	–CONH ₂	6.48	5.71	6.17	7.9	17.3	5.0	NH ₂ 7.55
	–CON(CH ₃) ₂	6.64	5.55	6.12	9.8	17.0	3.4	
	–COF	6.14	6.25	6.60	10.7	17.3	0.8	
	–COCl	6.35	6.16	6.63	10.6	17.4	0.2	
P	–P(CH ₃) ₂	6.23	5.51	5.39	11.8	18.3	2.0	CH ₃ 0.95
	–P(CH=CH ₂) ₂	6.16	5.64	5.59	11.8	18.4	2.0	
	–P(phenyl) ₂	7.38	6.31	7.07	12.5	18.2	0	
	–PCl ₂	7.48	6.68	6.64	11.7	18.6	0.4	
	–P(O)(phenyl) ₂	6.72	6.21	6.25	12.9	18.9	1.8	
	–PSCl ₂	6.42	5.90	6.13	11.0	17.5	0.3	
	–P(S)(CH ₃) ₂	6.60	6.14	6.26	11.8	17.9	1.8	
	–P(S)(phenyl) ₂	6.82	6.17	6.34	11.7	17.9	1.6	
M	–Li	7.29	6.65	5.91	19.3	23.9	7.1	
	–MgBr	6.66	6.15	5.51	17.7	23.3	7.6	
	–Si(CH ₃) ₃	6.11	5.88	5.63	14.6	20.2	3.8	CH ₃ 0.06
	–Sn(CH=CH ₂) ₃	6.39	6.21	5.75	13.4	20.7	3.1	
	–Pb(CH=CH ₂) ₃	6.70	6.19	5.46	12.2	19.8	2.1	
	–HgBr	6.45	5.92	5.52	11.9	18.7	3.1	

C = C

$$5.25 + 0.45 - 0.28 + 1.12$$

Estimation of ^1H Chemical Shifts of Substituted Ethylenes (δ in ppm)



C=C

$$\delta_{\text{C=CH}} = 5.25 + Z_{\text{gem}} + Z_{\text{cis}} + Z_{\text{trans}}$$

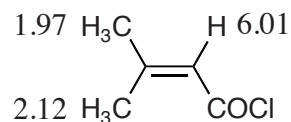
$$5.25 + -0.09 + 0.97 + -0.28$$

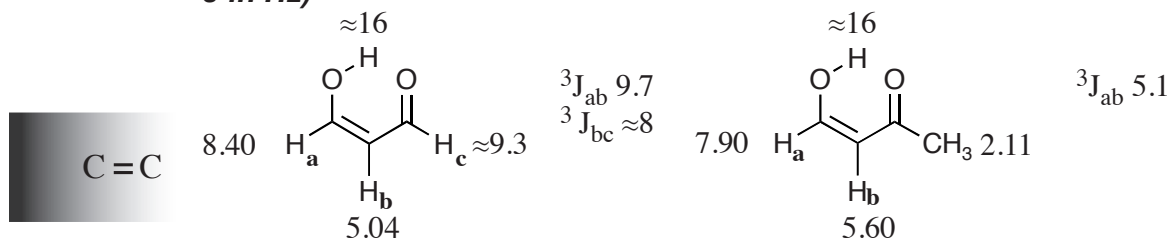
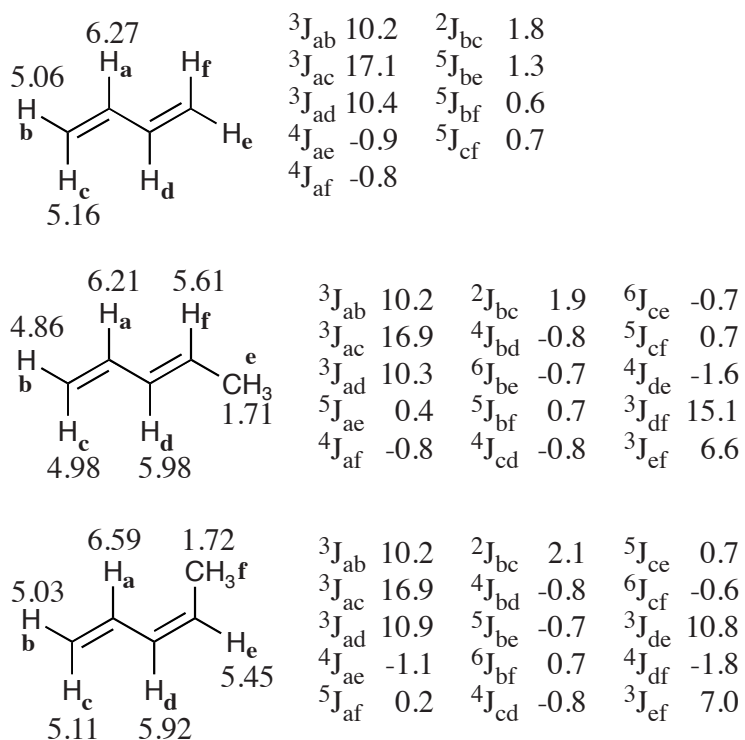
$$5.25 + -0.23 + 0.97 + -0.22$$

Substituent R	Z_{gem}	Z_{cis}	Z_{trans}
-H	0.00	0.00	0.00
C -alkyl	0.45	-0.22	-0.28
-alkyl ring ¹	0.69	-0.25	-0.28
-CH ₂ -aromatic	1.05	-0.29	-0.32
-CH ₂ X, X: F, Cl, Br	0.70	0.11	-0.04
-CHF ₂	0.66	0.32	0.21
-CF ₃	0.66	0.61	0.32
-CH ₂ O-	0.64	-0.01	-0.02
-CH ₂ N<	0.58	-0.10	-0.08
-CH ₂ CN	0.69	-0.08	-0.06
-CH ₂ S-	0.71	-0.13	-0.22
-CH ₂ CO-	0.69	-0.08	-0.06
-C=C<	1.00	-0.09	-0.23
-C=C< conjugated ²	1.24	0.02	-0.05
-C≡C-	0.47	0.38	0.12
-aromatic	1.38	0.36	-0.07
-aromatic, fixed ³	1.60	-	-0.05
-aromatic, <i>o</i> -substituted	1.65	0.19	0.09
X -F	1.54	-0.40	-1.02
-Cl	1.08	0.18	0.13
-Br	1.07	0.45	0.55
-I	1.14	0.81	0.88
O -OC≡ (<i>sp</i> ³)	1.22	-1.07	-1.21
-OC= (<i>sp</i> ²)	1.21	-0.60	-1.00
-OCO-	2.11	-0.35	-0.64
-OP(O)(OCH ₂ CH ₃) ₂	1.33	-0.34	-0.66
N -NR ₂ ; R: H, C≡ (<i>sp</i> ³)	0.80	-1.26	-1.21
-NR-; R: C= (<i>sp</i> ²)	1.17	-0.53	-0.99
-NCO-R	2.08	-0.57	-0.72
-N=N-phenyl	2.39	1.11	0.67
-NO ₂	1.87	1.30	0.62
-C≡N	0.27	0.75	0.55

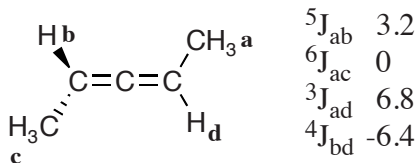
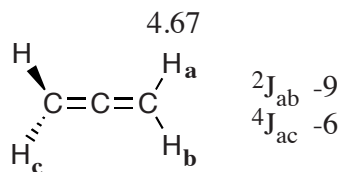
$$C=C$$

- Influence of cis- and trans-Substituents on the ^1H Chemical Shift of Methyl Groups at the Double Bond in Isobutenes (δ in ppm)***

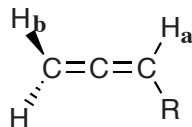


^1H Chemical Shifts and Coupling Constants of Enols (δ in ppm, J in Hz)**5.2.2 Conjugated Dienes** **^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)**

5.2.3 Allenes

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

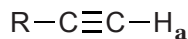
C=C

¹H Chemical Shifts and Coupling Constants of Monosubstituted Allenes (δ in ppm, J in Hz)

Substituent R	H_a	H_b	J_{ab}	Other
-H	4.67	4.67	-9.0	
-CH ₃	4.94	4.50	-6.7	CH ₃ (c) 1.59, $^3J_{\text{ac}}$ 7.2, $^5J_{\text{bc}}$ 3.4
-CH ₂ CH ₃	5.03	4.55	-6.8	CH ₂ (c) 1.95, $^3J_{\text{ac}}$ 6.2, $^5J_{\text{bc}}$ 3.5
-CH ₂ Cl	5.4.3	4.92	-6.6	CH ₂ (c) 4.11, $^3J_{\text{ac}}$ 7.7, $^5J_{\text{bc}}$ 2.2
-CH=CH ₂	5.96	4.92	-6.6	CH (c) 6.31, $^3J_{\text{ac}}$ 10.4, $^5J_{\text{bc}}$ 1.1 ^a
-Cl	5.76	5.17	-6.1	
-Br	5.85	4.83	-6.1	
-I	5.63	4.48	-6.3	
-phenyl	5.91	4.92		
-OCH ₃	6.77	5.48	-5.9	
-COCH ₃	5.77	5.25	-6.4	
-C $\equiv$ N	4.97	5.04		
-Si(CH ₃) ₃	4.92	4.31		
-SiCl ₃	5.35	4.92	-5.9	
-SnCl ₃	4.98	4.11	-7.2	

^a =CH₂, H_{cis} (d) 5.19, $^4J_{\text{ad}}$ -0.8, $^6J_{\text{bd}}$ -1.5, $^3J_{\text{cd}}$ 17.2; CH₂, H_{trans} (e) 4.99, $^4J_{\text{ae}}$ -0.9, $^6J_{\text{be}}$ -1.8, $^3J_{\text{ce}}$ 10.1, $^2J_{\text{de}}$ 1.6

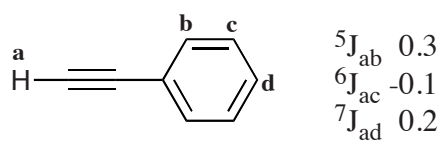
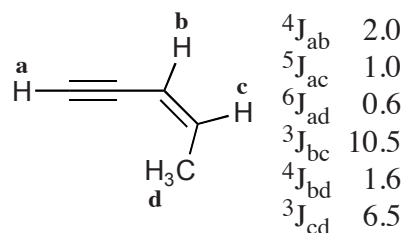
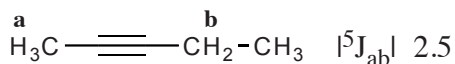
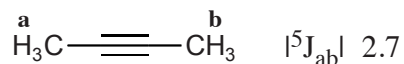
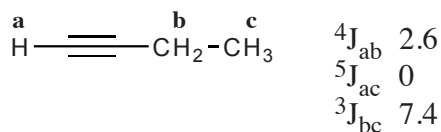
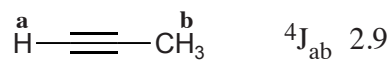
5.3 Alkynes

¹H Chemical Shifts of Substituted Alkynes (δ in ppm)

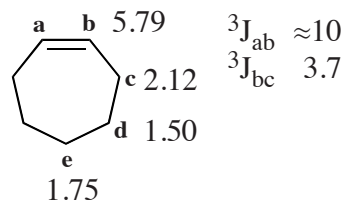
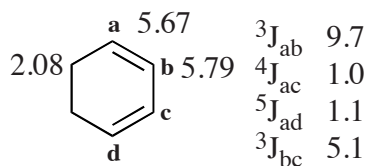
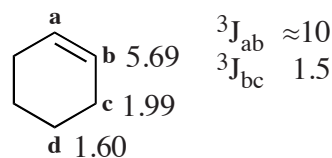
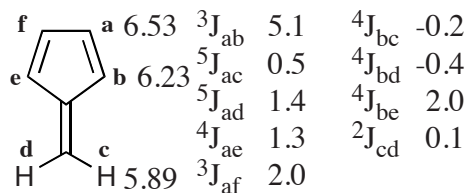
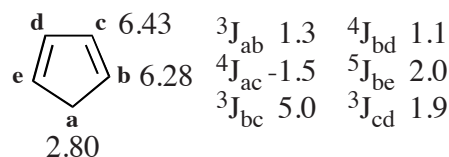
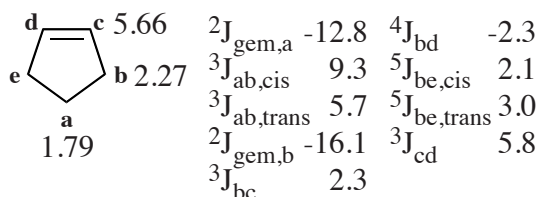
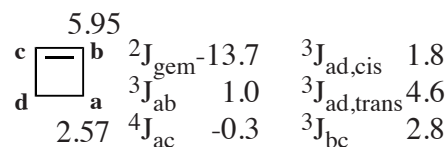
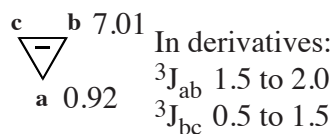
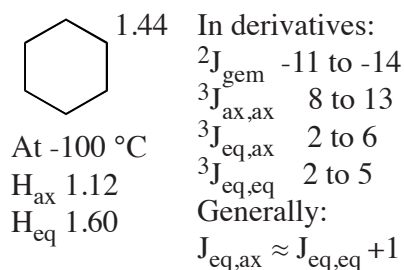
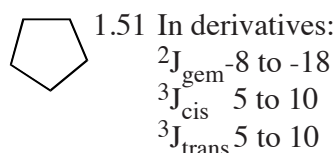
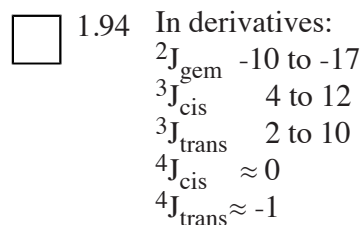
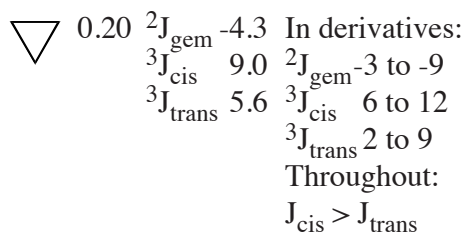
C≡C

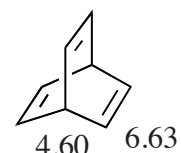
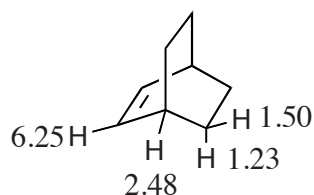
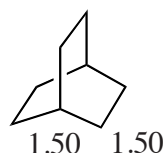
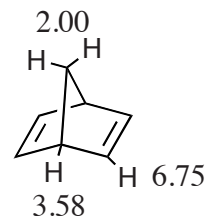
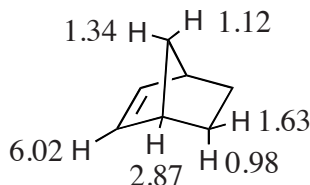
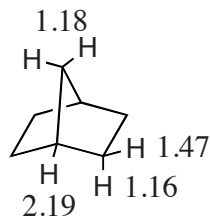
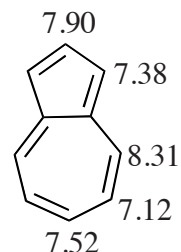
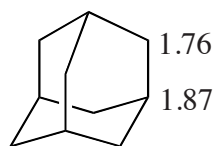
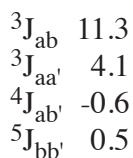
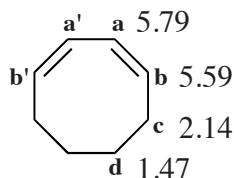
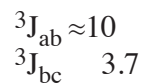
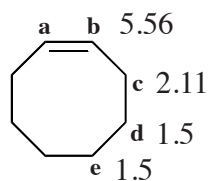
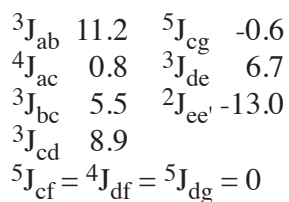
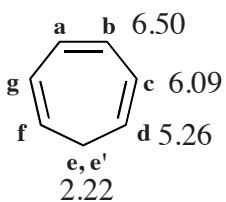
	Substituent R	H _a
C	-H	1.91
	-CH ₃	1.91
	-CH ₂ CH ₃	1.97
	-C(CH ₃) ₃	2.07
	-CF ₃	2.95
	-CH=CH ₂	3.07
	-C≡CH	2.16
	-phenyl	3.07
	-1-naphthyl	3.43
X	-F	1.74
	-Cl	2.05
	-Br	2.32
	-I	2.34
O	-OCH ₂ CH ₃	1.48
	-OCH=CH ₂	2.04
	-O-phenyl	2.07
N	-N(CH ₂ CH ₃) ₂	2.30
	-N(phenyl) ₂	2.86
	-C≡N	2.63

	Substituent R	H _a
S	-SCH ₂ CH ₃	2.79
	-SCH=CH ₂	3.26
	-S-phenyl	3.28
	-S(O) ₂ - <i>n</i> -butyl	3.95
O=C	-COCH ₃	3.65
	-CO-phenyl	3.48
	-COOH	3.17
	-COOCH ₂ CH ₃	2.90
	-CONH ₂	3.05
Si	-Si(CH ₃) ₃	2.34
	-Si(phenyl) ₃	2.47
P	-P(CH ₂ CH ₃) ₂	2.85
	-P(phenyl) ₂	3.22
	-P(O)(CH ₂ CH ₃) ₂	3.33
	-P(O)(phenyl) ₂	3.48

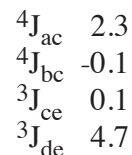
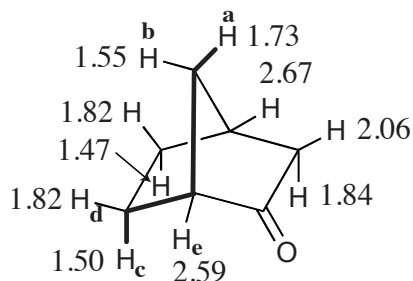
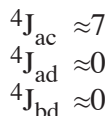
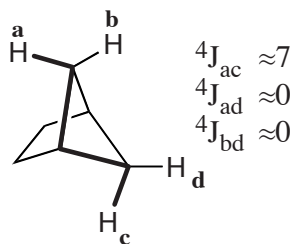
¹H, ¹H Coupling Constants of Substituted Alkynes (J in Hz)

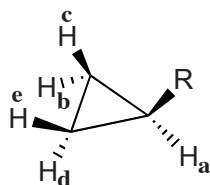
5.4 Alicyclics

¹H Chemical Shifts and Coupling Constants of Saturated Alicyclic Hydrocarbons (δ in ppm, J in Hz)



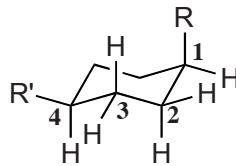
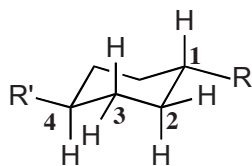
In condensed alicyclics, couplings over four bonds are often observed. Such long-range couplings are particularly large if the arrangement of the bonds between the two protons is W-shaped (cf. J_{ac} vs. J_{ad} and J_{bd} below left and J_{ac} vs. J_{bc} below right). Owing to the rigid arrangement, vicinal coupling constants (3J) may assume unusually small values when the torsional angles are close to 90° (J_{ce} below right).



^1H Chemical Shifts and Coupling Constants of Monosubstituted Cyclopropanes (δ in ppm, J in Hz)

	Substituent R	H_a	$H_{b;d}$	$H_{c;e}$	$^3J_{ab}$	$^3J_{ac}$	$^2J_{bc}$	$^3J_{bd}$	$^3J_{be}$	$^3J_{ce}$
	-H	0.20	0.20	0.20	9.0	5.6	-4.3	9.0	5.6	9.0
C	-CH ₃	1.00	0.35	0.15						
	-CH ₂ OH	1.14	0.40	0.30						
	-CH=CH ₂	1.35	0.64	0.34	8.2	4.9	-4.5	9.3	6.2	9.0
	-phenyl	1.83	0.89	0.65	9.5	6.3	-4.5	9.5	5.2	8.9
X	-F	4.32	0.69	0.27	5.9	2.4	-6.7	10.8	7.7	12.0
	-Cl	2.55	0.87	0.74	7.0	3.6	-6.0	10.3	7.1	10.6
	-Br	2.83	0.96	0.81	7.1	3.8	-6.1	10.2	7.0	10.5
	-I	2.31	1.04	0.76	7.5	4.4	-5.9	9.9	6.6	10.0
O	-OH	3.35	0.40	0.48	6.2	2.9	-5.4	10.3	6.8	10.9
	-NH ₂	2.23	0.32	0.20	6.6	3.6	-4.3	9.7	6.2	9.9
	-NH ₃ ⁺	1.06	0.52	0.34						
N	-NO ₂	4.21	1.13	1.60	7.0	3.4	-5.5	10.1	8.3	11.3
	-C≡N	1.29	0.96	1.04	8.4	5.1	-4.7	9.2	7.1	9.5
O C	-CHO	1.79	0.99	1.03	8.0	4.6	-4.5	8.8	7.0	9.6
	-COCH ₃	1.83	0.77	0.93	7.9	4.6	-3.5	9.2	7.0	9.5
	-CO-cyclopropyl	1.70	0.56	1.02	7.9	4.6	-3.5	9.1	7.0	9.5
	-CO-phenyl	2.65	1.01	1.23						
	-COOH	1.59	0.91	1.05	8.0	4.6	-4.0	9.3	7.1	9.7
	-COOCH ₃	1.61	0.86	0.98	8.0	4.6	-3.4	8.8	6.9	9.6
	-CONH ₂	1.39	0.70	0.95						
	-COF	1.66	1.11	1.20	8.0	4.6	-4.5	10.1	7.5	9.3
	-COCl	2.11	1.18	1.28	7.9	4.4	-4.5	9.2	7.6	10.0
M	-Li	-2.53	0.43	-0.12	10.3	9.1	-1.6	7.7	3.2	6.5
	-MgBr	-2.04	0.25	-0.13	11.0	8.5	-1.7	7.8	3.5	6.6
	-B(cyclopropyl) ₂	-0.25	0.61	0.66	8.9	5.8	-3.3	8.2	5.9	8.4
	-Si(cyclopropyl) ₃	-0.67	0.49	0.36	9.7	6.9	-3.4	8.4	5.1	8.1
	-P ⁺ (phenyl) ₃	3.28	1.82	0.63						
	-Hg-cyclopropyl	0.00	0.75	0.47	9.6	6.9	-3.7	8.5	4.8	7.9



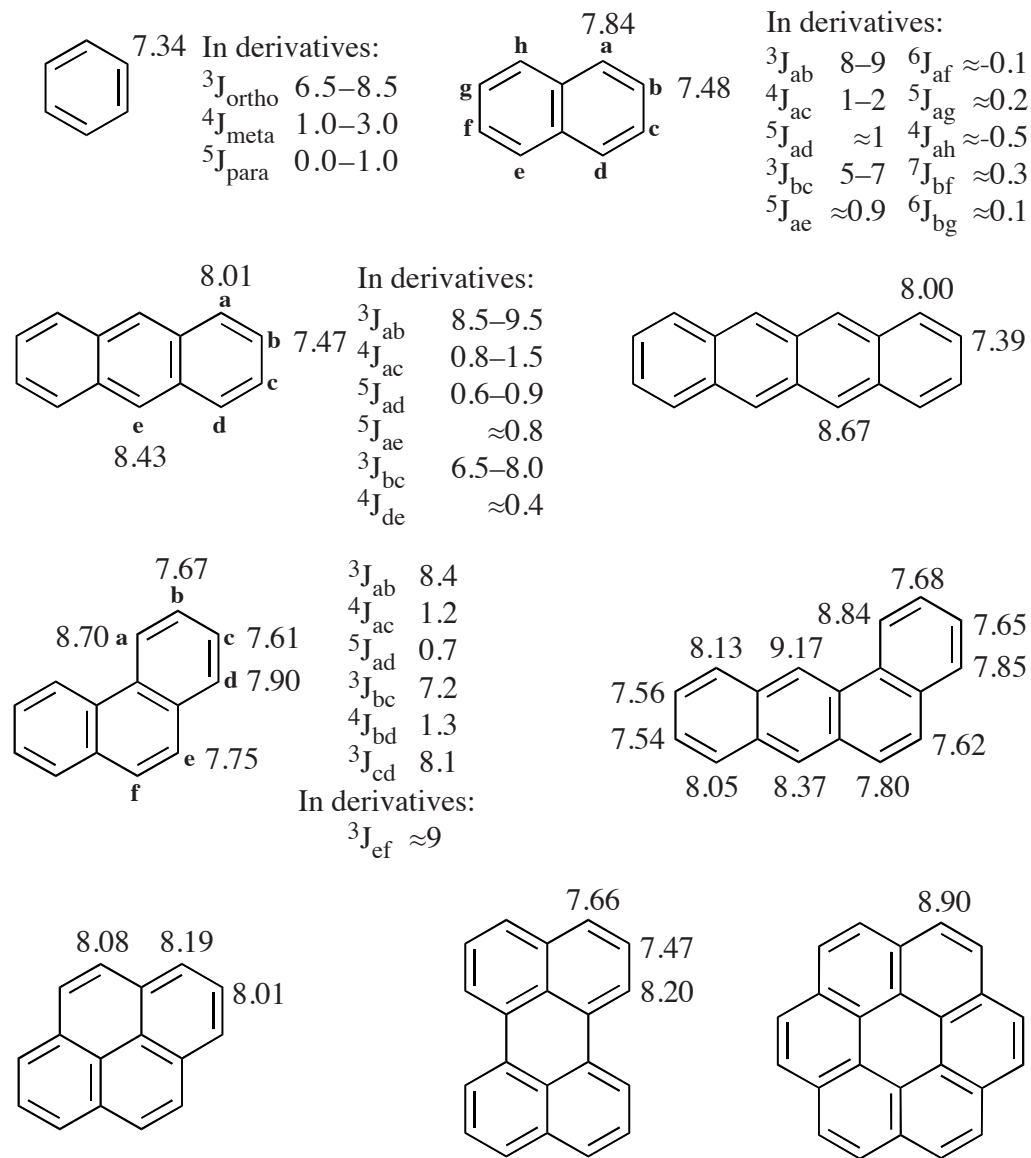
^1H Chemical Shifts and Coupling Constants of Equatorially and Axially Substituted Cyclohexanes (δ in ppm, J in Hz)

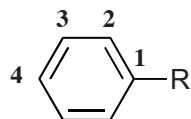
	Eq. substituent R	$\text{H}_{1,\text{ax}}$	$\text{H}_{2,\text{ax}}$	$\text{H}_{2,\text{eq}}$	$\text{H}_{3,\text{ax}}$	$\text{H}_{3,\text{eq}}$	$\text{H}_{4,\text{ax}}$	$\text{H}_{4,\text{eq}}$
C	–D*	1.12	1.12	1.60	1.12	1.60	1.12	1.60
	–C*	1.27	0.81	1.57	1.15	1.60	1.06	1.58
	–C $\equiv$ C*	2.25	1.36	1.98	1.20	1.73	1.17	1.67
	–phenyl*	2.46						
X	–F*	4.49	1.42	2.15	1.28	1.86	1.12	1.65
	–Cl*	3.88	1.58	2.22	1.33	1.84	1.18	1.68
	–Br*	4.09	1.75	2.33	1.35	1.80	1.22	1.72
	–I*	4.18	1.97	2.45	1.36	1.67	1.30	1.80
O	–OH**	3.52	1.22	2.01	1.05	1.78	0.97	
	–OCOCH ₃ *	4.74	1.72	1.85	1.35	1.41	1.25	1.55
N	–NH ₂ **	2.55	1.03	1.89	1.03	1.76	0.96	
	–NHCOCH ₃ **	3.67	1.07	2.01	1.11	1.78	1.01	
	–NO ₂ *	4.38	2.23	1.85	1.38	1.85	1.28	1.67
	–C $\equiv$ N**	2.31	1.53	2.16	0.98	1.86	1.03	
S	–SH*	2.79	1.34	2.01	1.31	1.75	1.22	1.61
	–COOCH ₃ *	2.30	1.44	1.90	1.27	1.75	1.24	1.64

	Ax. substituent R	$\text{H}_{1,\text{eq}}$	$\text{H}_{2,\text{ax}}$	$\text{H}_{2,\text{eq}}$	$\text{H}_{3,\text{ax}}$	$\text{H}_{3,\text{eq}}$	$\text{H}_{4,\text{ax}}$	$\text{H}_{4,\text{eq}}$
C	–D*	1.60	1.12	1.60	1.12	1.60	1.12	1.60
	–C*	1.93	1.37	1.40	1.39	1.34	1.06	1.58
	–C $\equiv$ C*	2.87	1.48	1.78				
	–phenyl*	3.16		2.42				
X	–F*	4.94	1.43	2.03	1.63	1.75	1.28	1.58
	–Cl*	4.59	1.76	2.00	1.77	1.75	1.26	1.75
	–Br*	4.80	1.81	2.08	1.79	1.60	1.24	1.78
	–I*	4.96	1.53	2.06	1.72	1.62	1.26	1.73
O	–OH**	4.03	1.49	1.83	1.35	1.54	0.99	
	–OCOCH ₃ *	5.31	1.49	2.51				
N	–NH ₂ **	3.15	1.54	1.65	1.27	1.53	0.96	
	–NHCOCH ₃ **	4.11	1.51	1.85	1.03	1.66	1.04	
	–NO ₂ **	4.43	1.6	2.6				
	–C $\equiv$ N**	2.96	1.54	2.00	1.50	1.70	1.20	
S	–SH**	3.43	1.5	1.9				

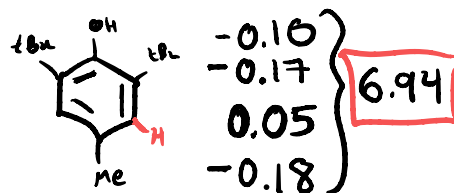
* R': –H; ** R': –*tert*-butyl

5.5 Aromatic Hydrocarbons

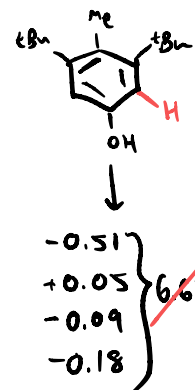
¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

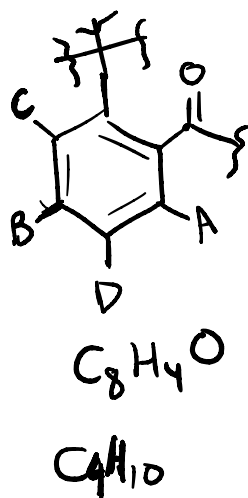
Effect of Substituents on ¹H Chemical Shifts of Monosubstituted Benzenes (in ppm)

$$\delta_{H_i} = 7.34 + Z_i$$



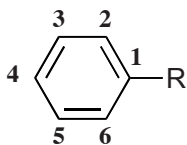
	Substituent R	Z ₂	Z ₃	Z ₄
C	-CH ₃	-0.17	-0.09	-0.17
	-CH ₂ CH ₃	-0.14	-0.05	-0.18
	-CH(CH ₃) ₂	-0.13	-0.08	-0.18
	-C(CH ₃) ₃	0.05	-0.04	-0.18
	-CF ₃	0.19	-0.07	0.00
	-CCl ₃	0.55	-0.07	-0.09
	-CH ₂ OH	-0.07	-0.07	-0.07
	-CH=CH ₂	0.08	-0.02	-0.09
	-CH=CH-phenyl (<i>trans</i>)	0.16	0.00	-0.15
	-C≡CH	0.16	-0.01	-0.01
	-C≡C-phenyl	0.20	-0.04	-0.07
	-phenyl	0.22	0.06	-0.04
	-2-pyridyl	0.73	0.09	0.02
X	-F	-0.31	-0.03	-0.21
	-Cl	-0.01	-0.06	-0.12
	-Br	0.15	-0.12	-0.06
	-I	0.36	-0.24	-0.02
O	-OH	-0.51	-0.10	-0.41
	-OCH ₃	-0.44	-0.05	-0.40
	-OCH ₂ CH=CH ₂	-0.45	-0.13	-0.43
	-O-phenyl	-0.33	-0.02	-0.25
	-OCOCH ₃	-0.26	0.03	-0.12
	-OCO-phenyl	-0.12	0.10	-0.06
	-OS(O) ₂ CH ₃	-0.05	0.07	-0.01
N	-NH ₂	-0.67	-0.20	-0.59
	-NHCH ₃	-0.73	-0.16	-0.64
	-N(CH ₃) ₂	-0.60	-0.10	-0.62
	-N(phenyl) ₂	-0.26	-0.10	-0.34
	-N ⁺ (CH ₃) ₃ I ⁻	0.72	0.40	0.34
	-NHCHO (<i>trans</i> to O)	-0.25	0.03	-0.13
	-NHCHO (<i>cis</i> to O)	-0.20	0.21	-0.01
	-N(CH ₃)CHO	-0.16	0.07	-0.05
	-NHCOCH ₃	0.15	-0.02	-0.23
	-NHCSNH ₂	0.14	0.07	-0.14



**S****O**
||
C**M**

Substituent R	Z_2	Z_3	Z_4
-NHNH ₂	-0.60	-0.08	-0.55
-N=N-phenyl	0.67	0.20	0.20
-NO	0.55	0.29	0.35
-NO ₂	0.93	0.26	0.39
-C≡N	0.32	0.14	0.28
-NCS	-0.11	0.04	-0.02
-SH	-0.08	-0.16	-0.22
-SCH ₃	-0.08	-0.10	-0.24
-S-phenyl	-0.06	-0.20	-0.26
-S-S-phenyl	0.13	-0.05	-0.10
-S(O)-CH=CH ₂	0.28	0.15	0.15
-S(O)-phenyl	0.29	0.09	0.13
-S(O) ₂ CH ₃	0.70	0.37	0.41
-S(O) ₂ OCH ₃	0.60	0.26	0.28
-S(O) ₂ Cl	0.68	0.27	0.37
-S(O) ₂ NH ₂	0.51	0.28	0.24
-CHO	0.54	0.19	0.29
-COCH ₃	0.62	0.12	0.22
-COCH ₂ CH ₃	0.61	0.11	0.21
-CO-phenyl	0.56	0.12	0.23
-CO-(2-pyridyl)	0.86	0.11	0.20
-COOH	0.79	0.14	0.28
-COOCH ₃	0.70	0.09	0.21
-COOCH(CH ₃) ₂	0.73	0.11	0.20
-COO-phenyl	0.87	0.18	0.30
-CONH ₂	0.48	0.11	0.19
-COF	0.71	0.21	0.38
-COCl	0.77	0.15	0.35
-COBr	0.70	0.15	0.32
-CH=N-phenyl	0.64	0.24	0.24
-Li	0.77	0.26	-0.29
-MgBr	0.40	-0.19	-0.26
-Mg-phenyl	-0.49	0.18	0.25
-Si(CH ₃) ₃	0.19	0.00	0.00
-Si(phenyl) ₂ Cl	0.32	0.07	0.12
-SiCl ₃	0.52	0.20	0.20
-P(phenyl) ₂	0.0	0.0	0.0
-P(O)(OCH ₃) ₂	0.46	0.14	0.22
-Pb ⁺ (phenyl) Cl ⁻	0.30	0.49	0.61
-Zn-phenyl	-0.36	0.02	0.05
-Hg-phenyl	0.06	0.10	-0.10

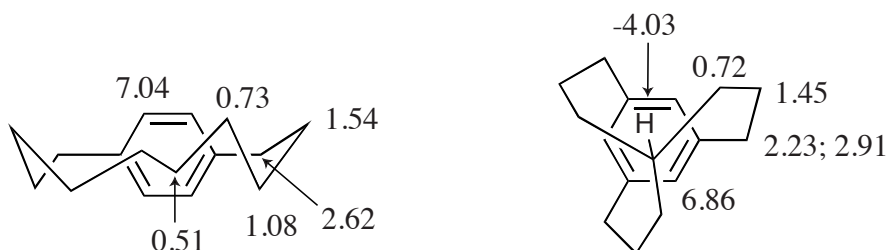
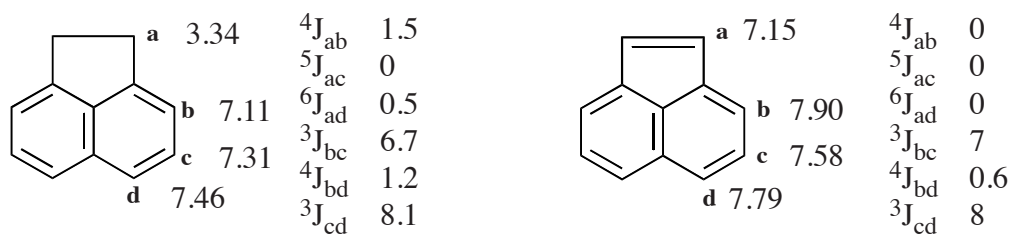
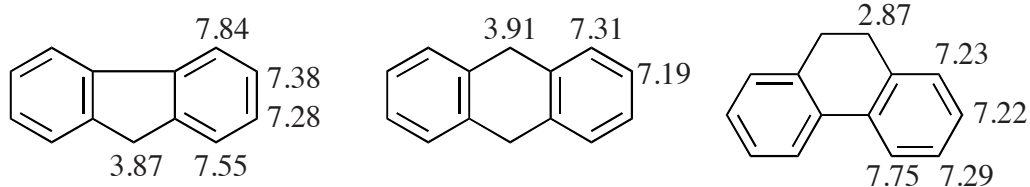
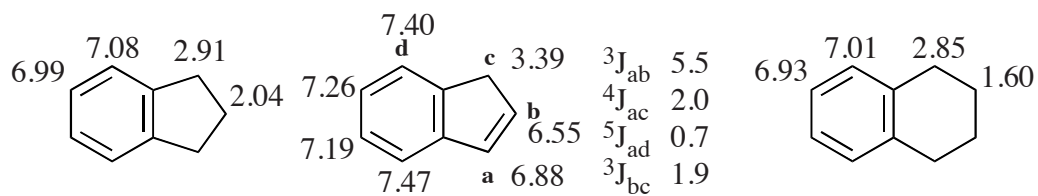


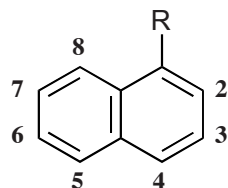
^1H - ^1H Coupling Constants in Selected Monosubstituted Benzenes ($|J|$ in Hz)

Substituent R		J_{23}	J_{24}	J_{25}	J_{26}	J_{34}	J_{35}
C	$-\text{CH}_3$	7.7	1.3	0.6	2.0	7.5	1.5
	$-\text{CH}=\text{CH}_2$	7.8	1.1	0.6	1.9	7.4	1.5
	$-\text{C}\equiv\text{CH}$	7.8	1.3	0.6	1.7	7.6	1.3
	$-\text{phenyl}$	7.8	1.2	0.6	2.0	7.5	1.4
X	$-\text{F}$	8.4	1.1	0.4	2.7	7.5	1.8
	$-\text{Cl}$	8.1	1.1	0.5	2.3	7.5	1.7
	$-\text{Br}$	8.0	1.1	0.5	2.2	7.4	1.8
	$-\text{I}$	7.9	1.1	0.5	1.9	7.5	1.8
O	$-\text{OH}$	8.2	1.1	0.5	2.7	7.4	1.7
	$-\text{OCH}_3$	8.3	1.0	0.4	2.7	7.4	1.8
	$-\text{O-phenyl}$	8.3	1.1	0.5	2.6	7.4	1.7
	$-\text{OCOCH}_3$	8.2	1.1	0.5	2.5	7.5	1.7
N	$-\text{NH}_2$	8.0	1.1	0.5	2.5	7.4	1.6
	$-\text{NHCOCH}_3$	8.2	1.2	0.5	2.4	7.4	1.5
	$-\text{NO}_2$	8.4	1.2	0.5	2.4	7.5	1.5
	$-\text{C}\equiv\text{N}$	7.8	1.3	0.7	1.8	7.7	1.3
S	$-\text{SH}$	7.9	1.2	0.6	2.1	7.5	1.5
	$-\text{S(O)}_2\text{OCH}_3$	8.0	1.2	0.6	2.0	7.6	1.4
O C	$-\text{CHO}$	7.7	1.3	0.6	1.8	7.5	1.3
	$-\text{COCH}_3$	8.0	1.3	0.6	1.8	7.5	1.3
	$-\text{COOH}$	7.9	1.3	0.6	1.9	7.4	1.4
	$-\text{COOCH}_3$	7.9	1.4	0.6	1.8	7.5	1.3
	$-\text{CONH}_2$	7.9	1.2	0.6	2.0	7.5	1.3
	$-\text{COCl}$	8.0	1.2	0.6	2.0	7.5	1.4
M	$-\text{Li}$	6.7	1.5	0.8	0.7	7.4	1.3
	$-\text{MgBr}$	6.9	1.5	0.7	0.7	7.4	1.4
	$-\text{P(phenyl)}_2$	7.6	1.2	0.6	1.7	7.4	1.4
	$-\text{PO(OCH}_3)_2$	7.7	1.4	0.6	1.6	7.6	1.4
	$-\text{Zn-phenyl}$	6.6	2.1	0.7	0.8	7.4	1.5
	$-\text{Hg-phenyl}$	7.5	1.4	0.6	1.1	7.5	1.5



¹H Chemical Shifts and Coupling Constants of Condensed Aromatic-Alicyclic Hydrocarbons (δ in ppm, J in Hz)



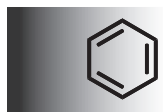
Effect of Substituents in Position 1 on the ¹H Chemical Shifts of Monosubstituted Naphthalenes (in ppm)


for R: H

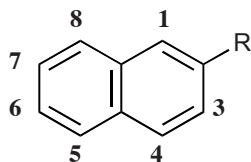
$$\delta_{H_1}, \delta_{H_4}, \delta_{H_5}, \delta_{H_8} = 7.84$$

$$\delta_{H_2}, \delta_{H_3}, \delta_{H_6}, \delta_{H_7} = 7.48$$

	Substituent R	H-2	H-3	H-4	H-5	H-6	H-7	H-8
C	–CH ₃	-0.20	-0.14	-0.17	-0.03	-0.03	0.00	0.12
	–CH ₂ CH ₃	-0.15	-0.08	-0.15	0.00	-0.02	0.01	0.21
	–CH ₂ C≡CH	0.09	-0.23	-0.23	-0.17	-0.13	-0.03	0.52
	–CH ₂ Cl	-0.10	-0.18	-0.11	-0.07	-0.05	0.02	0.22
	–CF ₃	0.51	-0.01	0.01	0.06	0.07	0.13	0.35
	–CH ₂ OH	-0.07	-0.10	-0.09	-0.01	-0.02	0.01	0.18
	–CH ₂ NH ₂	-0.14	-0.13	-0.14	-0.05	-0.07	-0.03	0.10
	–C≡CH	0.22	-0.14	-0.08	-0.08	-0.04	0.05	0.51
	–phenyl	-0.11	-0.04	-0.06	0.05	-0.13	-0.07	-0.02
X	–F	-0.35	-0.10	-0.23	0.00	0.05	0.03	0.27
	–Cl	0.06	-0.14	-0.12	-0.02	0.02	0.09	0.42
	–Br	0.29	-0.25	-0.04	-0.01	0.04	0.11	0.39
	–I	0.20	-0.46	0.13	-0.22	-0.09	-0.02	0.18
O	–OH	-0.75	-0.22	-0.42	-0.05	-0.03	-0.01	0.32
	–OCH ₃	-0.84	-0.25	-0.55	-0.18	-0.12	-0.13	0.33
	–O–phenyl	-0.53	-0.10	-0.22	0.03	0.04	0.00	0.37
	–OCOCH ₃	-0.31	-0.05	-0.27	-0.14	-0.23	-0.09	-0.01
N	–NH ₂	-0.76	-0.22	-0.55	-0.07	-0.05	-0.08	-0.09
	–N(CH ₃) ₂	-0.46	-0.13	-0.36	-0.06	-0.03	-0.06	0.38
	–NHCOCH ₃	0.24	0.01	-0.12	0.08	0.04	0.08	0.27
	–NO ₂	0.71	0.02	0.24	0.08	0.04	0.16	0.69
	–C≡N	0.42	0.04	0.23	0.07	0.14	0.21	0.38
	–NCO	-0.25	-0.13	-0.18	-0.03	0.05	0.01	0.21
	–NCS	-0.13	-0.13	-0.14	-0.04	0.03	0.08	0.21
O C	–CHO	0.46	0.07	0.20	0.05	0.06	0.17	1.39
	–COCH ₃	0.40	-0.02	0.09	0.01	0.02	0.09	0.90
	–COOH	0.72	0.14	0.34	0.20	0.13	0.20	1.09
	–COOCH ₃	0.65	-0.08	0.08	-0.05	-0.02	0.09	1.09
	–COCl	1.03	0.06	0.22	0.02	0.07	0.17	0.88



Effect of Substituents in Position 2 on the ^1H Chemical Shifts of Monosubstituted Naphthalenes (in ppm)



for R: H

$$\delta_{\text{H}_1}, \delta_{\text{H}_4}, \delta_{\text{H}_5}, \delta_{\text{H}_8} = 7.84$$

$$\delta_{\text{H}_2}, \delta_{\text{H}_3}, \delta_{\text{H}_6}, \delta_{\text{H}_7} = 7.48$$

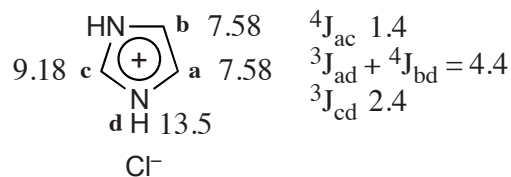
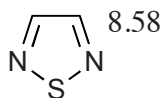
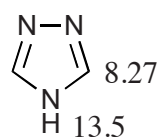
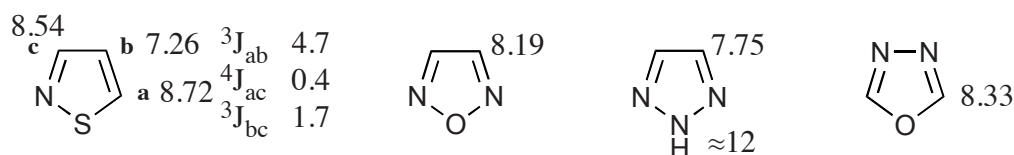
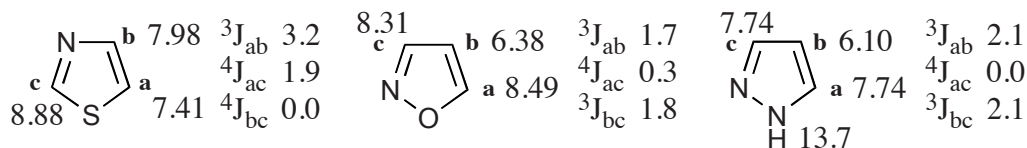
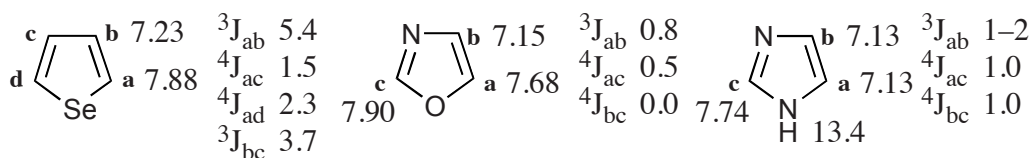
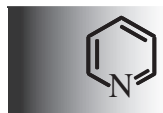
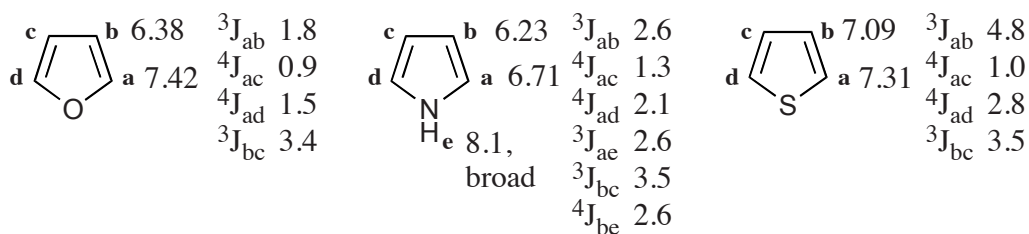
	Substituent R	H-1	H-3	H-4	H-5	H-6	H-7	H-8
C	$-\text{CH}_3$	-0.24	-0.18	-0.11	-0.06	-0.09	-0.05	-0.10
	$-\text{CH}_2\text{CH}_3$	-0.22	-0.14	-0.08	-0.05	-0.08	-0.06	-0.08
	$-\text{CH}(\text{CH}_3)_2$	-0.24	-0.15	-0.12	-0.10	-0.12	-0.10	-0.10
	$-\text{CF}_3$	0.28	0.14	0.06	-0.10	0.09	0.06	-0.10
	$-\text{CH}_2\text{OH}$	-0.13	-0.08	-0.07	-0.05	-0.04	-0.03	-0.08
	$-\text{CH}=\text{CH}_2$	-0.11	0.14	-0.06	-0.06	-0.06	-0.04	-0.06
	$-\text{C}\equiv\text{CH}$	0.19	0.04	-0.05	-0.03	0.02	0.02	-0.03
	-phenyl	0.20	0.25	0.06	0.01	0.02	-0.02	0.05
X	$-\text{Cl}$	-0.04	-0.08	-0.10	-0.05 ^a	-0.03 ^b	-0.01 ^b	-0.12 ^a
	$-\text{Br}$	0.14	0.05	-0.16	-0.12	-0.02	0.00	-0.06
O	$-\text{OH}$	-0.72	-0.39	-0.10	-0.09	-0.16	-0.06	-0.18
	$-\text{OCH}_3$	-0.76	-0.33	-0.14	-0.10	-0.14	-0.06	-0.14
	$-\text{O-phenyl}$	-0.53	-0.22	-0.01	-0.02	-0.08	-0.04	-0.15
	$-\text{OCOCH}_3$	-0.30	-0.27	-0.04	-0.04	-0.04	-0.02	-0.08
N	$-\text{NH}_2$	-0.93	-0.62	-0.23	-0.19	-0.27	-0.15	-0.27
	$-\text{N}(\text{CH}_3)_2$	-1.07	-0.49	-0.30	-0.29	-0.39	-0.24	-0.33
	$-\text{NHCOCH}_3$	0.33	-0.02	-0.10	-0.11	-0.09	-0.06	-0.09
	$-\text{NO}_2$	0.90	0.70	0.05	0.05	0.19	0.15	0.14
	$-\text{C}\equiv\text{N}$	0.40	0.13	0.08	0.06	0.19	0.13	0.06
S O C	$-\text{SH}$	-0.14	-0.19	-0.17	-0.11	-0.09	-0.06	-0.19
	$-\text{CHO}$	0.44	0.45	0.05	0.03	0.14	0.08	0.12
	$-\text{COCH}_3$	0.58	0.51	0.01	0.01	0.08	0.03	0.01
	$-\text{CO-phenyl}$	0.42	0.46	0.09	0.06	0.13	0.06	0.06
	$-\text{COOH}$	0.83	0.57	0.20	0.19	0.20	0.16	0.31
	$-\text{COOCH}_3$	0.66	0.50	-0.08	-0.07	-0.01	-0.05	0.00
	$-\text{COCl}$	0.85	0.58	0.22	0.32	0.17	0.21	0.20

^a interchangeable; ^b interchangeable



5.6 Heteroaromatic Compounds

5.6.1 Non-Condensed Heteroaromatic Rings

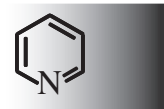
 ^1H Chemical Shifts and Coupling Constants (δ in ppm, $|J|$ in Hz)

	Solvent:		$^3J_{ab}$
	CDCl ₃	DMSO	6.0
	a 8.59	8.58	$^4J_{ac}$ 1.9
	b 7.25	7.38	$^5J_{ad}$ 0.9
	c 7.62	7.75	$^4J_{ae}$ 0.4
			$^3J_{bc}$ 7.6
			$^4J_{bd}$ 1.6

	Solvent:		$^3J_{ab}$
	CDCl ₃ *	DMSO**	6.0
	a 9.00	8.98	$^4J_{ac}$ 1.6
	b 7.97	8.14	$^5J_{ad}$ 0.8
	c 8.43	8.67	$^4J_{ae}$ 1.0
			$^3J_{bc}$ 7.9
			$^4J_{bd}$ 1.4
	* p-toluylsulfonate		
	** HSO ₃ ⁻		

	7.32	$^3J_{ab}$	6.5
		$^4J_{ac}$	1.1
		$^5J_{ad}$	0.6
		$^4J_{ae}$	1.9
		$^3J_{bc}$	7.7
		$^4J_{bd}$	2.1
(in acetone)			

	9.48	$^3J_{ab}$	2.7
		$^4J_{ac}$	0.0
		$^5J_{bc}$	2.2
	9.88		
	8.84		
			9.23

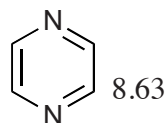


	7.56	$^3J_{ab}$	4.9
		$^4J_{ac}$	2.0
		$^5J_{ad}$	3.5
		$^3J_{bc}$	8.4
	9.22		

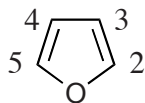
	7.22	$^3J_{ab}$	5.3
		$^4J_{ac}$	1.0
		$^5J_{ad}$	1.0
		$^3J_{bc}$	8.0
		$^4J_{bd}$	2.5
		$^3J_{cd}$	6.5

	7.38	$^3J_{ab}$	5.0
		$^4J_{ac}$	2.5
		$^4J_{ad}$	0
		$^5J_{bd}$	1.5
	8.78		
	9.27		

	8.24	$^3J_{ab}$	6.8
		$^4J_{ac}$	1.6
		$^4J_{ad}$	2.0
		$^3J_{bc}$	4.9
		$^5J_{bd}$	1.0
		$^4J_{cd}$	0



	8.11	$^3J_{ab}$	4.1
		$^5J_{ac}$	0.8
		$^4J_{ad}$	0.6
		$^4J_{bc}$	0.4
	8.44		

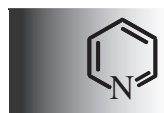
Effect of Substituents on the ¹H Chemical Shifts of Monosubstituted Furans (in ppm)

$$\delta_{\text{H-2}} = 7.42 + Z_{i2}$$

$$\delta_{\text{H-3}} = 6.38 + Z_{i3}$$

$$\delta_{\text{H-4}} = 6.38 + Z_{i4}$$

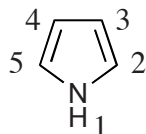
$$\delta_{\text{H-5}} = 7.42 + Z_{i5}$$



Substituent	H ₃	H ₄	H ₅	H ₂	H ₄	H ₅
	in position 2 or 5			in position 3 or 4		
	Z ₂₃ Z ₅₄	Z ₂₄ Z ₅₃	Z ₂₅ Z ₅₂	Z ₃₂ Z ₄₅	Z ₃₄ Z ₄₃	Z ₃₅ Z ₄₂
C						
–CH ₃	-0.45	-0.15	-0.17	-0.25	-0.17	-0.12
–CH ₂ CH ₃	-0.42	-0.12	-0.14			
–CH ₂ OH	-0.12	-0.07	-0.05	-0.07	0.00	-0.06
–CH ₂ SH	-0.22	-0.09	-0.09			
–CH ₂ SCH ₃	-0.21	-0.09	-0.08			
–CH=CHCOCH ₃ (<i>trans</i>)	0.29	0.11	0.08			
X						
–Br	-0.23	-0.17	-0.17			
–I	0.04	-0.21	-0.05	-0.17	-0.04	-0.26
O						
–OCH ₃	-1.26	-0.14	-0.57	-0.50	-0.36	-0.41
N						
–NO ₂	1.13	0.47	0.47			
–C≡N	0.48	-0.02	-0.04	0.41	0.14	-0.06
S						
–SCH ₃	0.05	0.01	0.13	-0.22	-0.13	-0.19
–SCN	0.32	-0.02	0.06	0.15	0.11	-0.01
O						
–CHO	0.92	0.25	0.31	0.92	0.47	0.19
 						
–COCH ₃	0.81	0.16	0.18	0.42	0.28	-0.16
C						
–COCO–2-furyl	1.26	0.27	0.37			
–COOH	0.97	0.19	0.24	0.70	0.40	0.03
–COOCH ₃	0.81	0.14	0.18	0.60	0.37	0.01
–COCl	1.14	0.32	0.46			
M						
–P(–x-furyl) ₂	0.25 ^a	-0.12 ^a	0.03 ^a	-0.16 ^b	-0.10 ^b	-0.09 ^b
–P(O)(–x-furyl) ₂	0.76 ^a	0.15 ^a	0.30 ^a	0.14 ^b	0.19 ^b	0.31 ^b
–P(S)(–x-furyl) ₂	0.77 ^a	0.12 ^a	0.27 ^a	0.10 ^b	0.18 ^b	0.30 ^b
–P ⁺ (CH ₃)(2-furyl) ₂ I [–]	1.53	0.49	0.77			
–HgCl				-0.09	0.02	0.25
–Hg–x-furyl	0.18 ^a	0.24 ^a	0.47 ^a	-0.10 ^b	0.10 ^b	-0.10 ^b

^a x = 2, ^b x = 3

Effect of Substituents on the ^1H Chemical Shifts of Monosubstituted Pyrroles (in ppm)



$\delta_{\text{H-1}} \approx 8$, broad, solvent-dependent

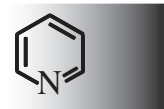
$$\delta_{\text{H-2}} = 6.71 + Z_{i2}$$

$$\delta_{\text{H-3}} = 6.23 + Z_{i3}$$

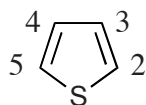
$$\delta_{\text{H-4}} = 6.23 + Z_{i4}$$

$$\delta_{\text{H-5}} = 6.71 + Z_{i5}$$

Substituent in position 1	H₃	H₄
	Z_{12}	Z_{13}
	Z_{15}	Z_{14}
-CH ₃	-0.13	-0.11
-CH ₂ CH ₃	-0.16	-0.12
-CH ₂ CH ₂ CN	-0.05	-0.07
-CH ₂ -phenyl	-0.12	-0.04
-phenyl	0.36	0.11
-N(CH ₃) ₂	0.11	-0.19
-COCH ₃	0.56	0.12
-CO-phenyl	0.57	0.18
-Si(CH(CH ₃) ₂) ₃	0.08	0.08



Substituent	H₃	H₄	H₅	H₂	H₄	H₅
	in position 2 or 5			in position 3 or 4		
	Z_{23} Z_{54}	Z_{24} Z_{53}	Z_{25} Z_{52}	Z_{32} Z_{45}	Z_{34} Z_{43}	Z_{35} Z_{42}
C -CH ₃	5.72	5.89	6.36	-0.33	-0.16	-0.26
N -NO ₂	7.11	6.29	7.05	1.06	0.24	0.43
-C≡N	6.88	6.28	7.13	0.83	0.23	0.51
S -SCH ₃	6.23	6.10	6.72	0.18	0.05	0.10
-SCN	6.53	6.15	6.90	0.48	0.10	0.28
O -CHO	7.01	6.34	7.18	0.78	0.11	0.47
 -COCH ₃	6.93	6.26	7.06	0.70	0.03	0.35
C -COOCH ₃	6.84	6.18	6.91	0.79	0.13	0.29

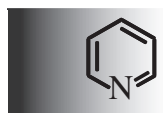
Effect of Substituents on the ¹H Chemical Shifts of Monosubstituted Thiophenes (in ppm)

$$\delta_{\text{H-2}} = 7.31 + Z_{i2}$$

$$\delta_{\text{H-3}} = 7.09 + Z_{i3}$$

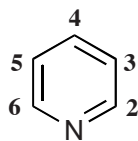
$$\delta_{\text{H-4}} = 7.09 + Z_{i4}$$

$$\delta_{\text{H-5}} = 7.31 + Z_{i5}$$



Substituent	H₃	H₄	H₅	H₂	H₄	H₅
	in position 2 or 5 Z_{23} Z_{54}	Z_{24} Z_{53}	Z_{25} Z_{52}	in position 3 or 4 Z_{32} Z_{45}	Z_{34} Z_{43}	Z_{35} Z_{42}
C -CH ₃	-0.34	-0.20	-0.24	-0.45	-0.22	-0.15
-C≡C	0.02	-0.29	-0.23			
-phenyl				0.11	0.28	0.05
-2-thienyl	0.08	-0.09	-0.11			
-2-pyridyl	0.48	0.01	0.06			
X -F	-0.78	-0.54	-0.86	-0.80	-0.40	-0.31
-Cl	-0.30	-0.35	-0.39	-0.25	-0.17	-0.09
-Br	-0.05	-0.23	-0.10	-0.23	-0.21	-0.21
-I	0.11	-0.34	-0.01	-0.05	-0.13	-0.30
O -OH*	-0.85	0.44	-3.21			
-OCH ₃	-0.93	-0.41	-0.82	-1.10	-0.36	-0.17
N -NH ₂	-1.08	-0.58	-0.96	-1.36	-0.66	-0.36
-NO ₂	0.69	-0.16	0.19	0.84	0.47	-0.08
-C≡N	0.34	-0.13	0.17	0.52	0.07	0.04
S -SH	-0.13	-0.33	-0.18	-0.33	-0.33	-0.21
-SCH ₃	-0.16	-0.31	-0.16	-0.44	-0.23	-0.14
-S(O) ₂ CH ₃	0.90	0.07	0.68	0.85	0.35	0.35
-S(O) ₂ Cl	0.60	-0.07	0.34			
-SCN	0.17	-0.18	0.17	0.14	-0.08	-0.06
O -CHO	0.69	0.13	0.47	0.81	0.44	0.06
 -COCH ₃	0.60	0.03	0.32	0.74	0.45	0.01
C -CO-phenyl	0.55	0.06	0.40			
-COOH	0.67	-0.05	0.29	0.93	0.48	0.03
-COOCH ₃	0.70	0.00	0.22	0.67	0.34	-0.16
-CONHNH ₂	0.63	0.04	0.41	-7.31	-7.09	-7.31
-COCl	0.75	-0.07	0.33	0.94	0.37	-0.08

* Keto form

Effect of Substituents on the ^1H Chemical Shifts of Monosubstituted Pyridines (in ppm)


$$\delta_{\text{H-2}} = 8.59 + Z_{i2}$$

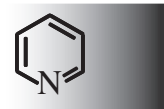
$$\delta_{\text{H-3}} = 7.25 + Z_{i3}$$

$$\delta_{\text{H-4}} = 7.62 + Z_{i4}$$

$$\delta_{\text{H-5}} = 7.25 + Z_{i5}$$

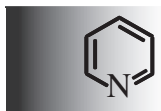
$$\delta_{\text{H-6}} = 8.59 + Z_{i6}$$

		H₃ Z_{23} Z_{65}	H₄ Z_{24} Z_{63}	H₅ Z_{25} Z_{63}	H₆ Z_{26} Z_{62}
C	–CH ₃	-0.11	-0.08	-0.15	-0.11
	–CH ₂ CH ₃	-0.09	0.01	-0.15	0.03
	–CH ₂ –phenyl	0.03	-0.06	0.04	-0.04
	–CH ₂ OH	0.14	0.03	-0.08	-0.14
	–CH=CH ₂	-0.07	-0.14	-0.23	-0.12
	–phenyl	0.42	0.02	-0.09	0.07
	–2-pyridyl	1.27	0.04	-0.11	0.00
X	–F	-0.30	0.16	-0.05	-0.36
	–Cl	0.09	0.02	0.00	-0.10
	–Br	0.26	-0.06	0.03	-0.23
	–I	0.49	-0.29	0.04	-0.23
O	–OH*	-0.63	-0.13	-0.93	-1.17
	–OCH ₃	-0.51	-0.10	-0.41	-0.43
N	–NH ₂	-0.76	-0.24	-0.63	-0.54
	–NHCH ₂ CH ₃	-0.87	-0.22	-0.69	-0.52
	–N(CH ₃) ₂	-0.77	-0.23	-0.73	-0.44
	–NHNH ₂	-0.55	-0.17	-0.58	-0.48
	–NHCOCH ₃	1.00	0.09	-0.19	-0.32
	–NHN=CH–2-pyridyl	0.21	-0.01	-0.42	-0.36
	–NO ₂	0.93	0.44	0.45	0.00
	–C≡N	0.52	0.26	0.35	0.15
S O C	–SH	0.34	-0.20	-0.42	-0.91
	–CHO	0.73	0.26	0.31	0.21
	–COCH ₃	0.80	0.22	0.24	0.10
	–CO–phenyl	0.81	0.27	0.25	0.13
	–COOH	0.87	0.41	0.44	0.17
	–COOCH ₃	0.91	0.24	0.27	0.17
	–CONH ₂	0.98	0.24	0.22	-0.01
	–CH=N–NH–2-pyridyl	0.76	0.05	-0.06	-0.03
	–Si(CH ₃) ₃	0.15	-0.22	-0.24	0.09

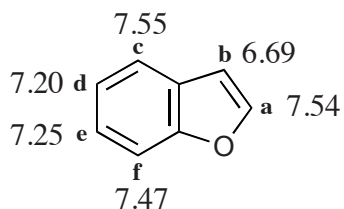


* Keto form (2-pyridone)

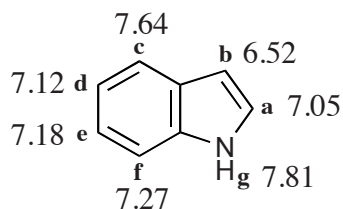
Substituent	H_2	H_4	H_5	H_6	H_2	H_3
	in position 3 or 5				in position 4	
	Z_{32} Z_{56}	Z_{34} Z_{54}	Z_{35} Z_{53}	Z_{36} Z_{52}	Z_{42} Z_{46}	Z_{43} Z_{45}
C $-\text{CH}_3$	-0.15	-0.16	-0.07	-0.17	-0.13	-0.13
$-\text{CH}_2\text{CH}_3$	-0.13	-0.14	-0.06	-0.17	-0.12	-0.14
$-\text{CH}_2\text{-phenyl}$	-0.08	-0.18	-0.04	-0.14	0.00	-0.15
$-\text{phenyl}$	0.25	0.20	0.08	-0.03		
$-\text{CH=CH}_2$					-0.12	-0.08
X $-\text{F}$	-0.05	-0.21	0.04	-0.13	-0.07	-0.03
$-\text{Cl}$	0.09	0.00	0.05	-0.05	0.00	0.05
$-\text{Br}$	0.09	0.18	-0.04	-0.07	0.09	0.35
O $-\text{OH}$	-0.31	-0.29	0.06	-0.50		
$-\text{OCH}_3$	-0.27	-0.37	-0.04	-0.40	-0.16	-0.42
$-\text{OCOCH}_3$	-0.15	-0.15	0.08	-0.13		
N $-\text{NH}_2$	-0.51	-0.65	-0.20	-0.60	-0.15	-0.74
$-\text{N}(\text{CH}_3)_2$					-0.38	-0.77
$-\text{NHCOCH}_3$	0.37	0.50	0.06	-0.16	-0.19	0.16
$-\text{C}\equiv\text{N}$	0.32	0.38	0.25	0.26	0.24	0.32
S $-\text{S-phenyl}$					0.05	-0.16
$-\text{S}(\text{O})_2\text{OH}$	0.70	1.14	0.81	0.70		
O $-\text{CHO}$	0.52	0.58	0.30	0.28	0.31	0.49
$-\text{COCH}_3$	0.58	0.61	0.20	0.20	0.21	0.50
$-\text{CO-phenyl}$					0.23	0.35
C $-\text{COOH}$	0.54	0.57	0.20	0.24	0.20	0.45
$-\text{COOCH}_3$	0.64	0.67	0.16	0.19	0.19	0.61
$-\text{COO-phenyl}$					0.24	0.75
$-\text{CONH}_2$	0.49	0.50	0.15	0.15		
$-\text{CSNH}_2$	0.68	0.67	0.24	0.26	0.35	0.68
$-\text{CH=NOH}$	0.39	0.43	0.19	0.15	0.06	0.32
$-\text{Si}(\text{CH}_3)_3$	0.08	0.00	-0.21	-0.11	-0.08	0.01



5.6.2 Condensed Heteroaromatic Rings

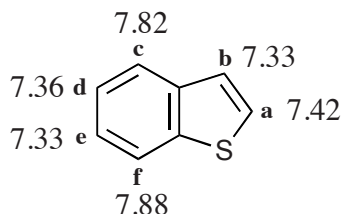
¹H Chemical Shifts and Coupling Constants (δ in ppm, $|J|$ in Hz)

$^3J_{ab}$ 2.5
 $^5J_{bf}$ 0.9
 $^3J_{cd}$ 7.9
 $^4J_{ce}$ 1.2
 $^5J_{cf}$ 0.8
 $^3J_{de}$ 7.3
 $^4J_{df}$ 0.9
 $^3J_{ef}$ 8.4
 (all other coupling constants negligible)

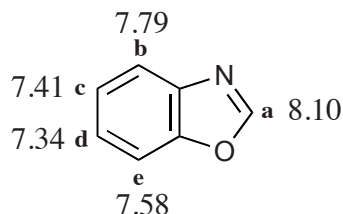


(chemical shifts in $CDCl_3$,
coupling constants in
acetone)

$^3J_{ab}$ 3.1
 $^3J_{ag}$ 2.5
 $^5J_{bf}$ 0.7
 $^4J_{bg}$ 2.0
 $^3J_{cd}$ 7.8
 $^4J_{ce}$ 1.2
 $^5J_{cf}$ 0.9
 $^5J_{cg}$ 0.8
 $^3J_{de}$ 7.1
 $^4J_{df}$ 1.3
 $^3J_{ef}$ 8.1
 (all other coupling constants negligible)

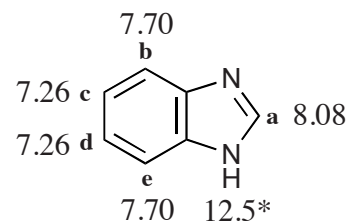


$^3J_{ab}$ 5.5
 $^5J_{bf}$ 0.8
 $^3J_{cd}$ 8.0
 $^4J_{ce}$ 1.1
 $^5J_{cf}$ 0.9
 $^3J_{de}$ 7.2
 $^4J_{df}$ 1.0
 $^3J_{ef}$ 8.0
 (all other coupling constants negligible)



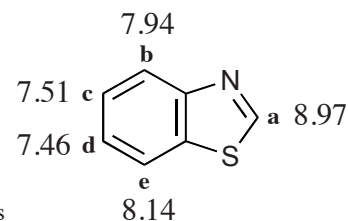
(chemical shifts in $CDCl_3$,
coupling constants in
acetone)

$^5J_{ab}$ 0.2
 $^6J_{ac}$ -0.1
 $^6J_{ad}$ 0.4
 $^5J_{ae}$ 0.0
 $^3J_{bc}$ 8.2
 $^4J_{bd}$ 1.0
 $^5J_{be}$ 0.7
 $^3J_{ce}$ 7.4
 $^4J_{de}$ 1.2
 $^3J_{de}$ 8.3



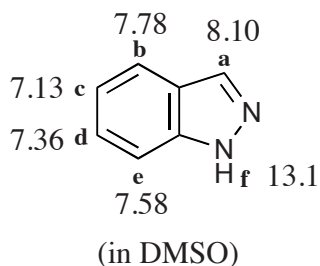
* in DMSO

$^3J_{bc}$ 8.2
 $^4J_{bd}$ 1.4
 $^5J_{be}$ 0.7
 $^3J_{cd}$ 7.1
 (all other coupling constants negligible)



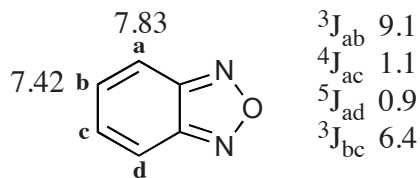
(chemical shifts in $CDCl_3$,
coupling constants in
acetone)

$^5J_{ab}$ 0.1
 $^6J_{ac}$ -0.2
 $^6J_{ad}$ 0.4
 $^5J_{ae}$ 0.1
 $^3J_{bc}$ 8.2
 $^4J_{bd}$ 1.1
 $^5J_{be}$ 0.6
 $^3J_{cd}$ 7.2
 $^4J_{ce}$ 1.1
 $^3J_{de}$ 8.2

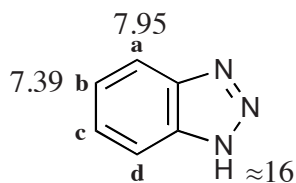


$^5J_{ae}$	0.8
$^3J_{bc}$	7.8
$^4J_{bd}$	1.2
$^5J_{be}$	1.0
$^3J_{cd}$	7.0
$^4J_{ce}$	1.2
$^3J_{de}$	7.9

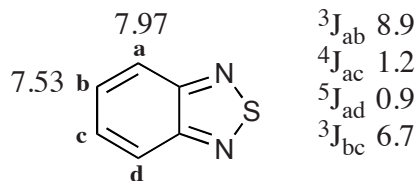
(all other coupling constants negligible)



$^3J_{ab}$	9.1
$^4J_{ac}$	1.1
$^5J_{ad}$	0.9
$^3J_{bc}$	6.4

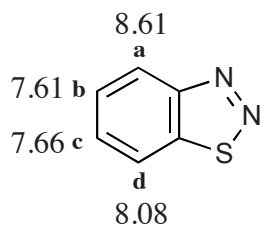
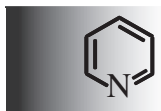


$^3J_{ab}$	8.3
$^4J_{ac}$	1.0
$^5J_{ad}$	0.9
$^3J_{bc}$	7.0

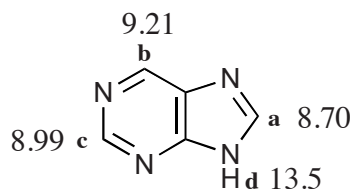


$^3J_{ab}$	8.9
$^4J_{ac}$	1.2
$^5J_{ad}$	0.9
$^3J_{bc}$	6.7

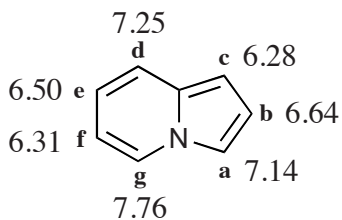
(chemical shifts in CDCl_3 ,
coupling constants in
acetone)



$^3J_{ab}$	8.4
$^4J_{ac}$	1.0
$^5J_{ad}$	0.8
$^3J_{bc}$	7.0
$^4J_{bd}$	1.0
$^3J_{cd}$	7.9

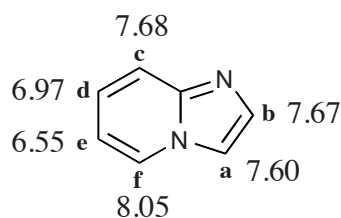


(in DMSO)



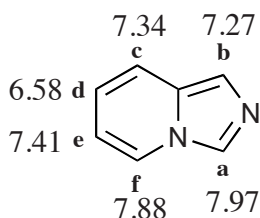
$^3J_{ab}$	2.7
$^4J_{ac}$	1.2
$^5J_{ad}$	0.5
$^3J_{bc}$	3.9
$^6J_{bf}$	0.5
$^5J_{cg}$	1.0
$^3J_{de}$	9.0
$^4J_{df}$	1.0
$^5J_{dg}$	1.2
$^3J_{ef}$	6.4
$^4J_{eg}$	1.0
$^3J_{fg}$	6.8

(all other coupling constants negligible)

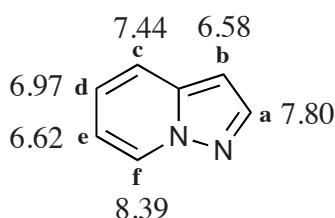


$^3J_{ab}$	1.2
$^4J_{af}$	0.7
$^3J_{cd}$	9.0
$^4J_{ce}$	1.3
$^5J_{cf}$	1.3
$^3J_{de}$	6.6
$^4J_{df}$	1.3
$^3J_{ef}$	6.8

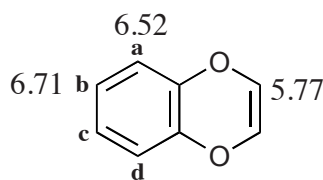
(all other coupling constants negligible)



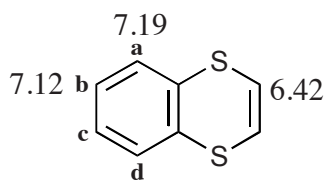
$^5J_{ac}$	1.0
$^5J_{bf}$	0.5
$^3J_{cd}$	9.2
$^4J_{ce}$	1.1
$^5J_{cf}$	1.1
$^3J_{de}$	6.4
$^4J_{df}$	0.9
$^3J_{ef}$	7.1
(all other coupling constants negligible)	



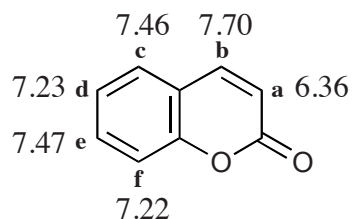
$^3J_{ab}$	2.2
$^6J_{ad}$	0.5
$^5J_{bf}$	0.9
$^3J_{cd}$	8.9
$^4J_{ce}$	1.2
$^5J_{cf}$	1.0
$^3J_{de}$	6.8
$^4J_{df}$	1.0
$^3J_{ef}$	6.9
(all other coupling constants negligible)	



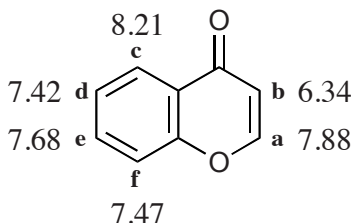
$^3J_{ab}$	7.9
$^4J_{ac}$	1.5
$^5J_{ad}$	0.4
$^3J_{bc}$	7.9



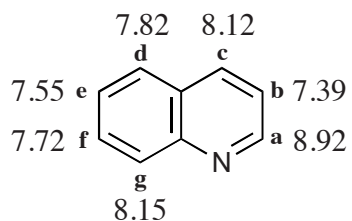
$^3J_{ab}$	7.8
$^4J_{ac}$	1.3
$^5J_{ad}$	1.1
$^3J_{bc}$	7.1



$^3J_{ab}$	9.6
$^3J_{cd}$	7.7
$^4J_{ce}$	1.6
$^3J_{de}$	7.4
$^4J_{df}$	1.1
$^3J_{ef}$	8.4
(all other coupling constants negligible)	

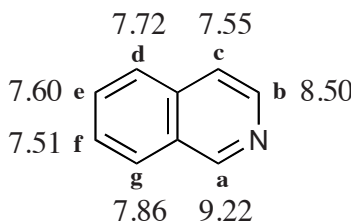


$^3J_{ab}$	6.0
$^3J_{cd}$	8.0
$^4J_{ce}$	1.8
$^5J_{cf}$	0.5
$^3J_{de}$	7.0
$^4J_{df}$	1.1
$^3J_{ef}$	8.4



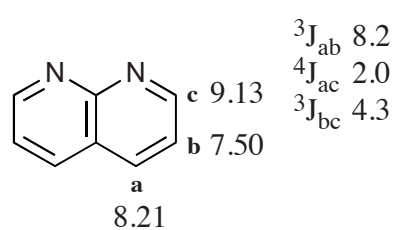
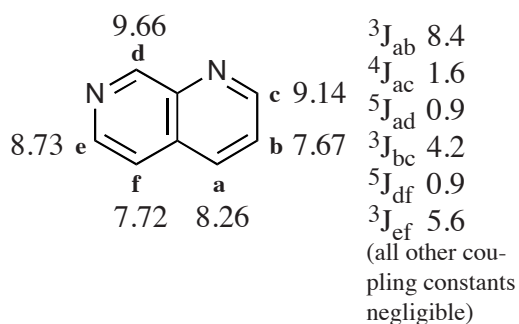
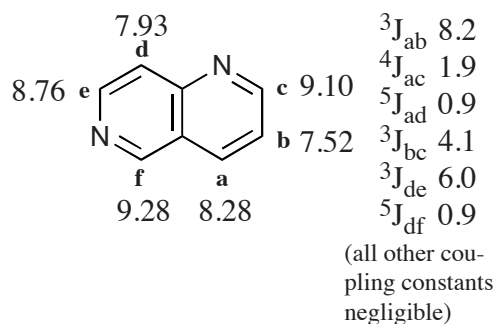
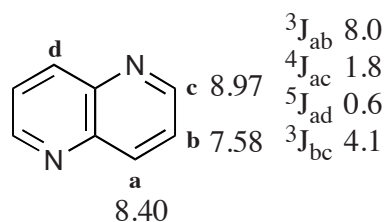
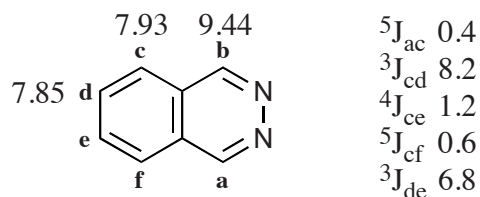
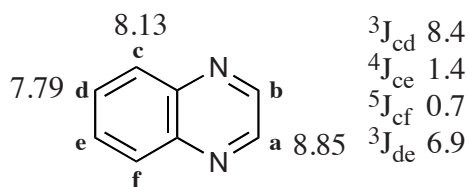
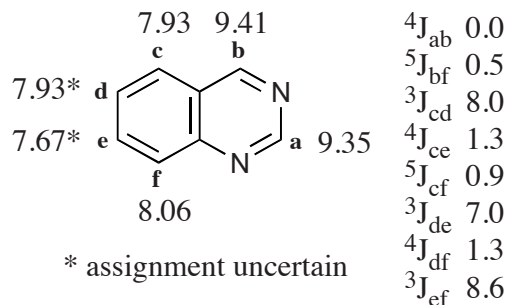
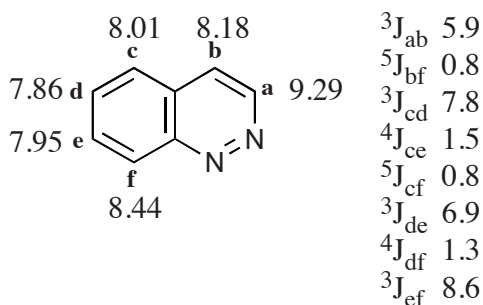
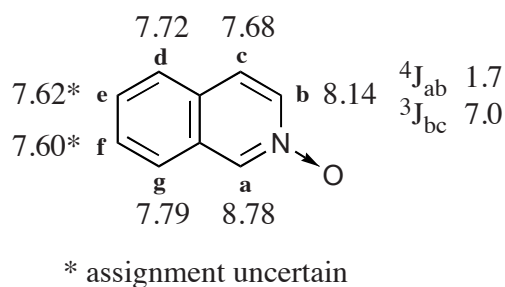
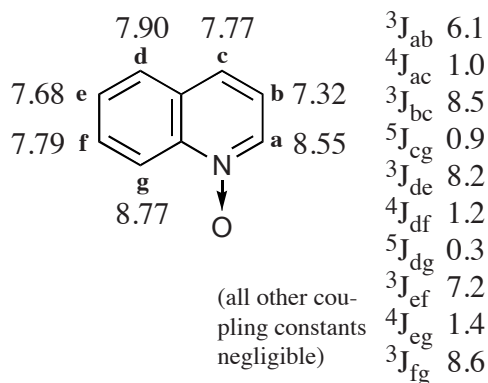
$^3J_{ab}$	4.2
$^4J_{ac}$	1.8
$^3J_{bc}$	8.2
$^5J_{cg}$	0.8
$^3J_{de}$	8.2
$^4J_{df}$	1.4
$^5J_{dg}$	0.7
$^3J_{ef}$	6.9
$^4J_{eg}$	1.2
$^3J_{fg}$	8.5

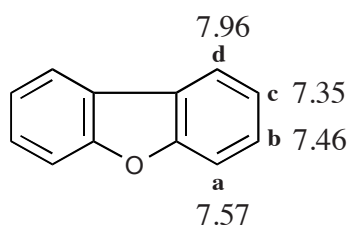
(all other coupling constants negligible)



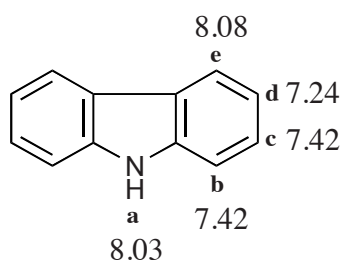
$^5J_{ac}$	1.0
$^5J_{ad}$	0.9
$^3J_{bc}$	5.8
$^5J_{cg}$	0.9
$^3J_{de}$	8.3
$^4J_{df}$	1.2
$^5J_{dg}$	0.8
$^3J_{ef}$	6.9
$^4J_{eg}$	1.2
$^3J_{fg}$	8.3

(all other coupling constants negligible)

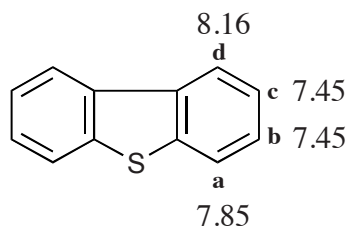




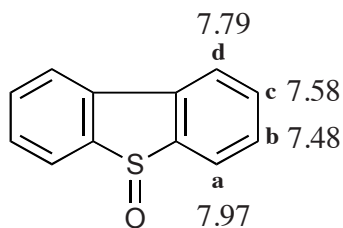
$^3J_{ab}$ 8.5
 $^4J_{ac}$ 0.9
 $^5J_{ad}$ 0.6
 $^3J_{bc}$ 7.3
 $^4J_{bd}$ 1.3
 $^3J_{cd}$ 7.6



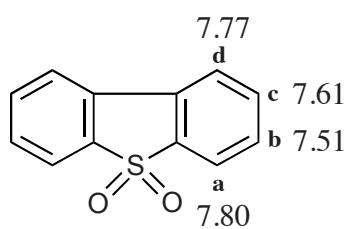
$^5J_{ae}$ 0.7
 $^3J_{bc}$ 8.1
 $^4J_{bd}$ 1.0
 $^5J_{be}$ 0.7
 $^3J_{cd}$ 7.2
 $^4J_{ce}$ 1.2
 $^3J_{de}$ 7.9



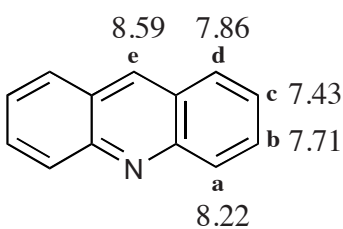
$^3J_{ab}$ 8.0
 $^4J_{ac}$ 1.1
 $^5J_{ad}$ 0.7
 $^3J_{bc}$ 7.2
 $^4J_{bd}$ 1.2
 $^3J_{cd}$ 8.1



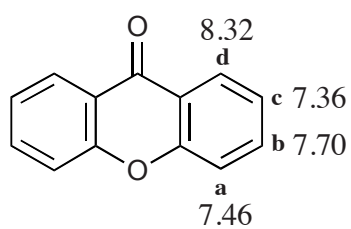
$^3J_{ab}$ 7.7
 $^4J_{ac}$ 1.1
 $^5J_{ad}$ 0.7
 $^3J_{bc}$ 7.5
 $^4J_{bd}$ 1.1
 $^3J_{cd}$ 7.6



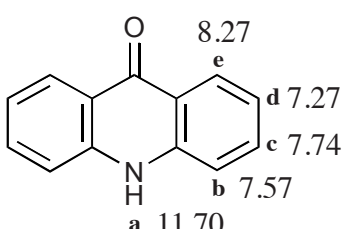
$^3J_{ab}$ 7.7
 $^4J_{ac}$ 1.1
 $^5J_{ad}$ 0.6
 $^3J_{bc}$ 7.5
 $^4J_{bd}$ 1.0
 $^3J_{cd}$ 7.8



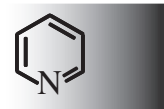
$^3J_{ab}$ 8.9
 $^4J_{ac}$ 1.1
 $^5J_{ad}$ 0.8
 $^5J_{ae}$ 0.9
 $^3J_{bc}$ 6.7
 $^4J_{bd}$ 1.5
 $^3J_{cd}$ 8.4
 $^4J_{de}$ -0.5

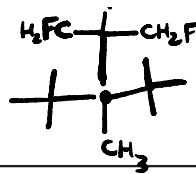


$^3J_{ab}$ 8.4
 $^4J_{ac}$ 1.1
 $^5J_{ad}$ 0.5
 $^3J_{bc}$ 7.1
 $^4J_{bd}$ 1.8
 $^3J_{cd}$ 8.0



$^5J_{ae}$ 0.4
 $^3J_{bc}$ 8.6
 $^4J_{bd}$ 1.0
 $^5J_{be}$ 0.4
 $^3J_{cd}$ 7.0
 $^4J_{ce}$ 1.4
 $^3J_{de}$ 8.2



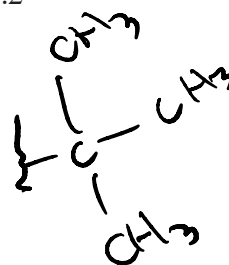
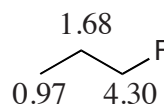
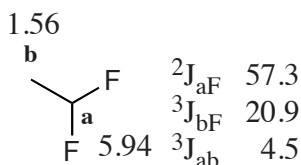
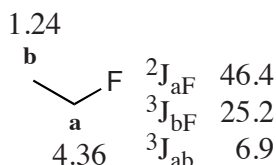
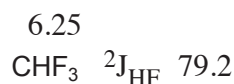
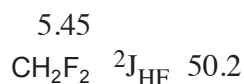


5.7 Halogen Compounds

5.7.1 Fluoro Compounds

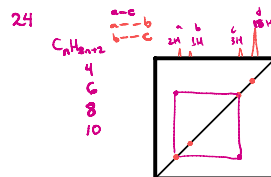
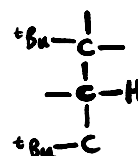
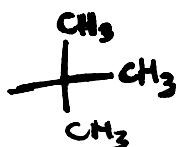
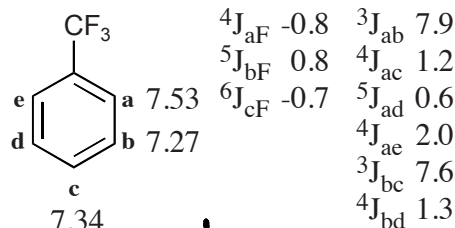
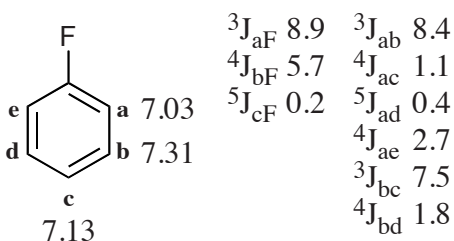
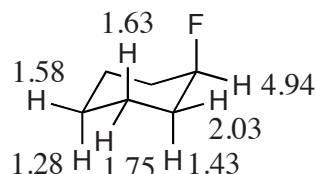
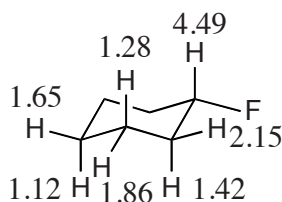
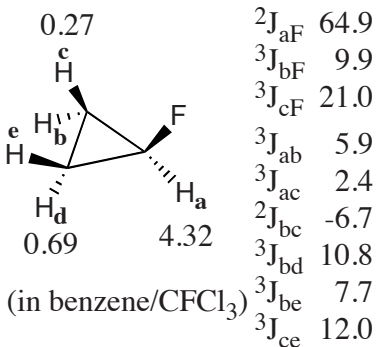
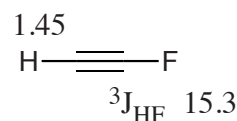
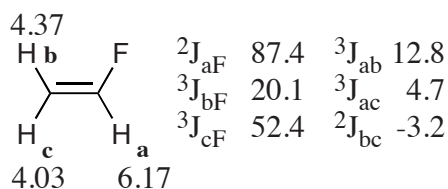
^{19}F (natural abundance 100%) has a spin quantum number I of $1/2$. The signals of ^1H atoms are split by coupling to ^{19}F up to a distance of about four bonds.

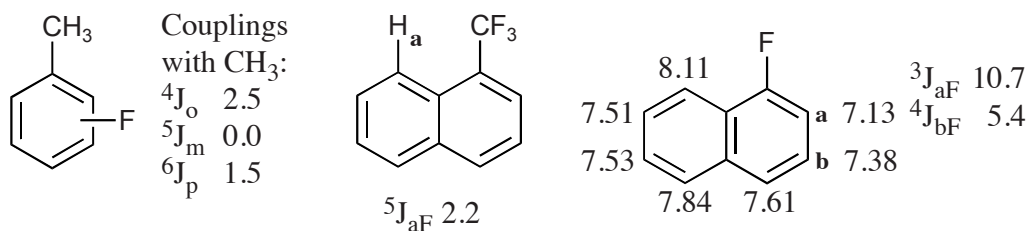
^1H Chemical Shifts and Coupling Constants (δ in ppm, $|J|$ in Hz)



Hal

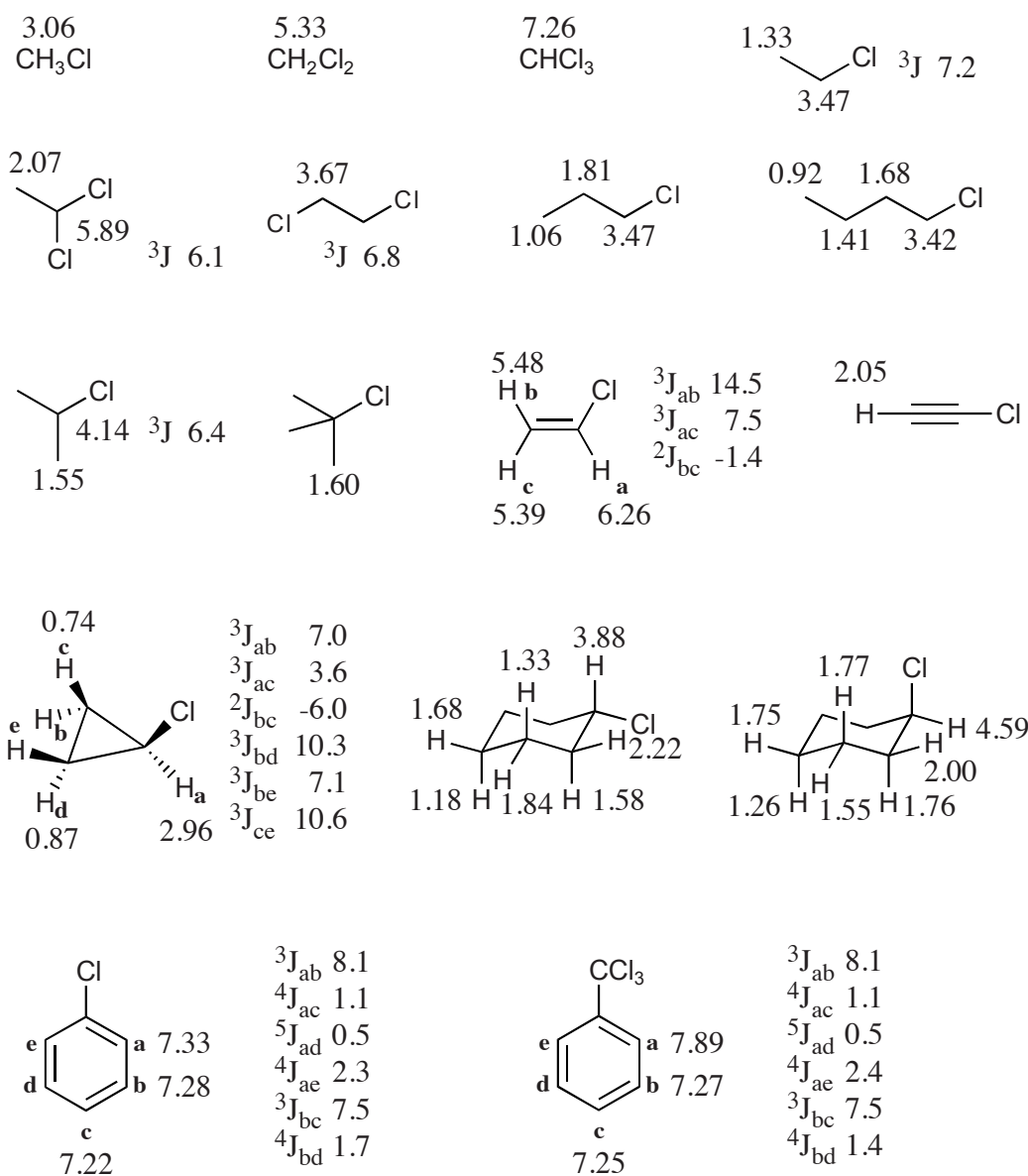
1.34



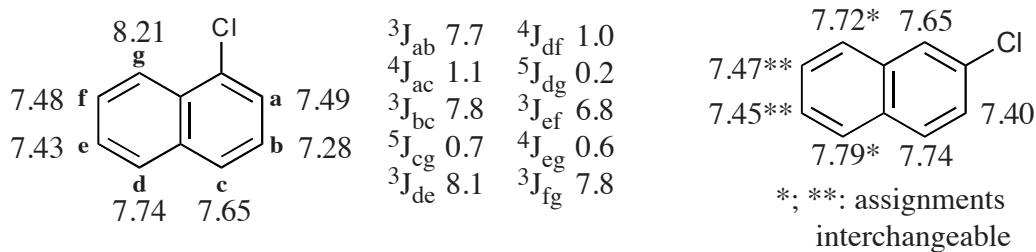


5.7.2 Chloro Compounds

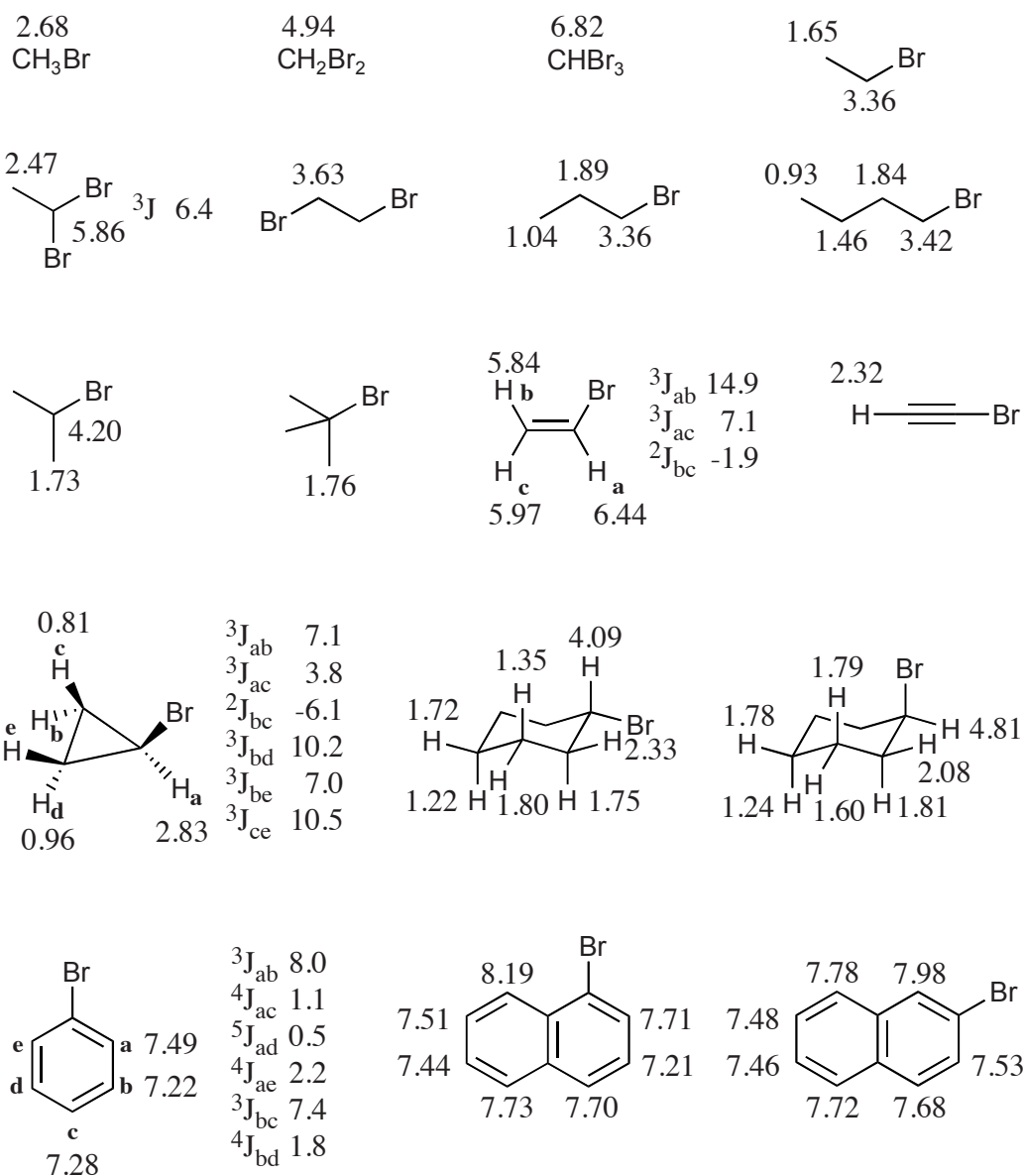
¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)



Hal



5.7.3 Bromo Compounds

 ^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

5.7.4 Iodo Compounds

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)2.16
CI3.90
CI4.91
CI1.86
3.15
CCCI2.96
5.24
 3J 7.0
CC(C)CI3.64
CCCI1.86
1.04 3.16
CCC(C)CI0.93 1.80
1.42 3.20
CCCCI4.24
1.89
CC(C)(C)CI1.95
CC(C)(C)CI6.57
6.23
6.53
 $^3J_{ab}$ 15.9
 $^3J_{ac}$ 7.8
 $^2J_{bc}$ -1.5
C=CI2.06
C#CI0.76
1.04 2.31
 $^3J_{ab}$ 7.5
 $^3J_{ac}$ 4.4
 $^2J_{bc}$ -5.9
 $^3J_{bd}$ 9.9
 $^3J_{be}$ 6.6
 $^3J_{ce}$ 10.0
C1C(CI)C14.18
1.36
1.80
1.30 1.67 1.97 2.45
C1CCC(CI)C11.72
1.73
1.26 1.62 1.53 4.96 2.06
C1CCC(CI)C17.70 7.10 7.32
 $^3J_{ab}$ 8.0
 $^4J_{ac}$ 1.1
 $^5J_{ad}$ 0.5
 $^4J_{ae}$ 2.2
 $^3J_{bc}$ 7.4
 $^4J_{bd}$ 1.8
c1ccc(CI)cc18.02
7.46 7.39 7.68 7.02 7.62 7.97
c1ccc2cc(CI)ccc2c1

Hal

5.8 Alcohols, Ethers, and Related Compounds

5.8.1 Alcohols

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

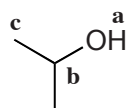
Aliphatic and alicyclic alcohols: $\delta_{\text{OH}} = 0.5\text{--}3.0$ (in DMSO: 4–6) ppm

Phenols: $\delta_{\text{OH}} = 4.0\text{--}8.0$ (in DMSO: 8–12) ppm

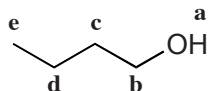
Hydrogen bonds strongly deshield hydroxyl protons. The position of the signal may depend heavily on the experimental conditions including the concentration of the sample. If a compound contains several kinds of hydroxyl protons (–OH, –COOH, H₂O), in general only one signal at an average position is seen because of rapid exchange. In dimethyl sulfoxide (DMSO) as solvent, this exchange in most cases is so slow that isolated signals are obtained. In this case, the chemical shifts of hydroxyl protons are characteristic. However, if the sample contains strong acids or amine bases, the exchange rate increases and, also in DMSO, a single signal at an average position is observed. Frequently, intermediate exchange rates lead to very broad signals extending over several ppm and, therefore, sometimes not discernible in routine spectra.

As a consequence of fast intermolecular exchange of the hydroxyl protons, their coupling with the protons on the adjacent carbon atoms is usually not observed. However, in very pure (acid-free) solutions or in DMSO, the exchange is sufficiently slow so that the H–O–C–H couplings become visible. Their dependence on the conformation is analogous to that shown by the H–C–C–H couplings (Chapter 5.1.2). In case of fast rotation: $^3J_{\text{HOCH}} \approx 5$ Hz. In cyclohexanols, the vicinal coupling constants for axial hydroxyl protons (3.0–4.2 Hz) are lower than those of equatorial ones (4.2–5.7 Hz).

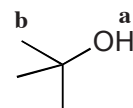
$\begin{array}{cc} \text{b} & \text{a} \\ \text{CH}_3 & \text{OH} \end{array}$			$\begin{array}{cc} & \text{a} \\ \text{c} & \text{OH} \\ & \text{b} \end{array}$			$\begin{array}{cc} & \text{a} \\ & \text{OH} \\ \text{c} & \text{b} \\ \text{d} & \end{array}$		
CDCl ₃	DMSO	D ₂ O	CDCl ₃	DMSO	D ₂ O	CDCl ₃	DMSO	D ₂ O
a 1.13	4.05		a 1.51	4.31		a 1.51	4.31	
b 3.49	3.17	3.34	b 3.71	3.44	3.65	b 3.59	3.34	3.61
	$^3J_{\text{ab}}$ 5.2		c 1.24	1.06	1.17	c 1.59	1.42	1.57
				$^3J_{\text{ab}}$ 4.8		d 0.94	0.84	0.89
				$^3J_{\text{bc}}$ 6.9				



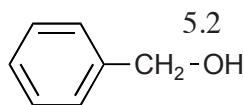
	CDCl ₃	DMSO	D ₂ O
a	1.36	4.30	
b	4.04	3.78	4.02
c	1.22	1.04	1.17
		³ J _{ab}	6.2



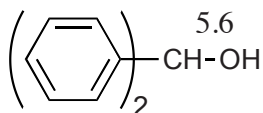
	CDCl ₃	DMSO	D ₂ O
a	1.50	4.30	
b	3.64	3.38	3.61
c	1.56	1.40	1.51
d	1.39	1.30	1.35
e	0.94	0.87	0.91



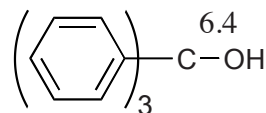
	CDCl ₃	DMSO	D ₂ O
a	1.37	4.19	
b	1.28	1.11	1.24



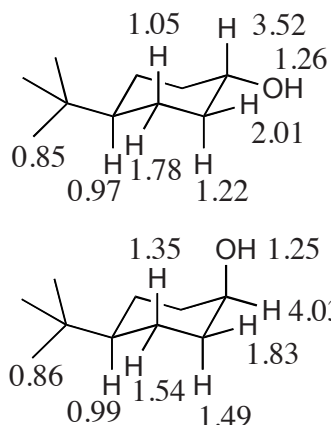
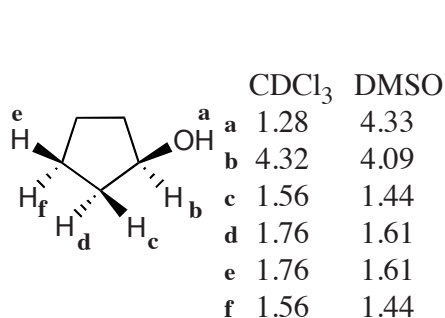
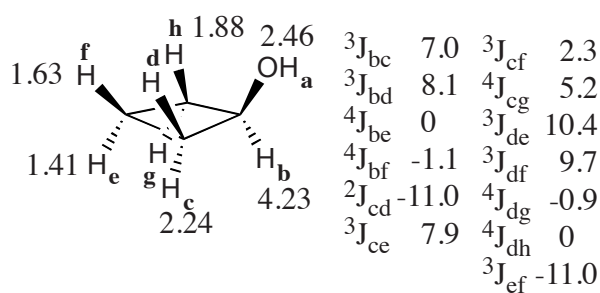
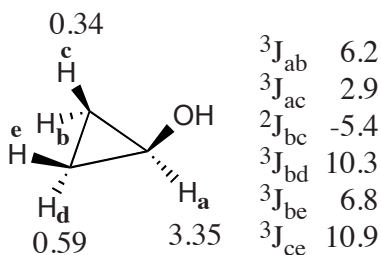
(in DMSO)



(in DMSO)

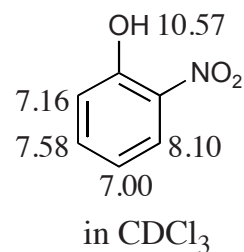
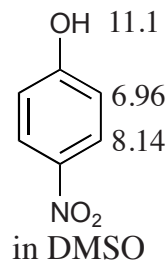
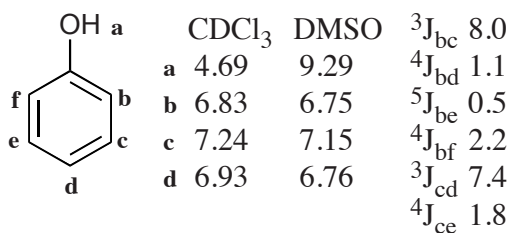


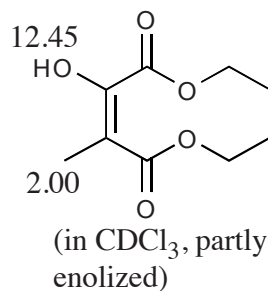
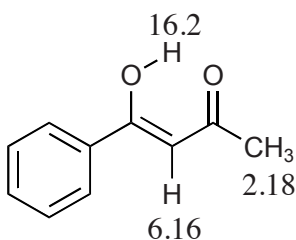
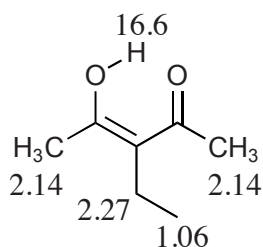
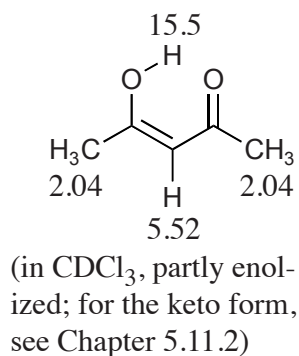
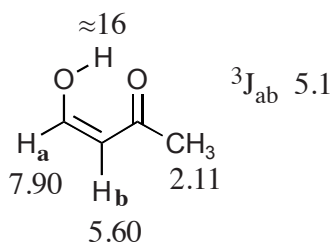
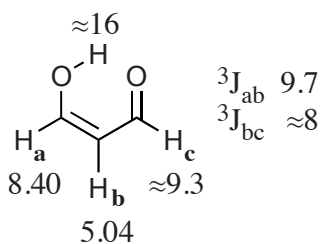
(in DMSO)



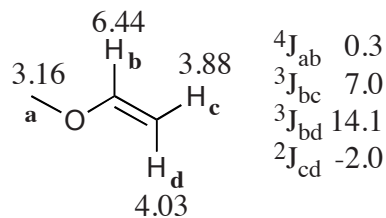
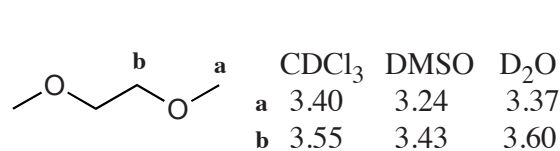
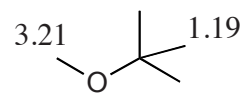
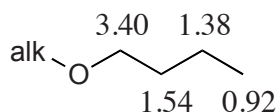
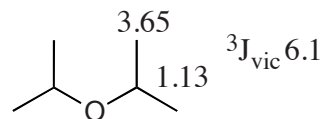
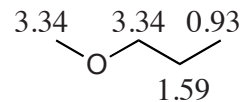
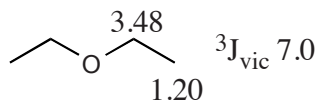
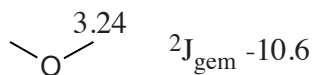
Derivatives
in DMSO:
δ_{OH} 4.0–4.5
J_{CH,OH} 4.2–5.7

Derivatives
in DMSO:
δ_{OH} 3.8–4.2
J_{CH,OH} 3.0–4.2

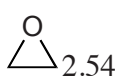
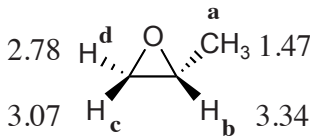
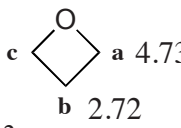


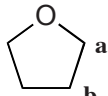
^1H Chemical Shifts and Coupling Constants of Enols (δ in ppm, J in Hz)

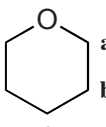
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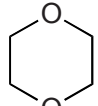
5.8.2 Ethers **^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)**

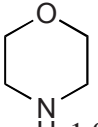
¹H Chemical Shifts and Coupling Constants of Cyclic Ethers (δ in ppm, J in Hz)

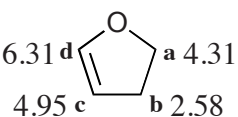
		
In derivatives:		
$^2J_{gem}$ 5–6	$^3J_{ab}$ 5.2	$^2J_{a,gem}$ -5.8
$^3J_{cis}$ 4.5	$^3J_{bc}$ 4.2	$^2J_{b,gem}$ -11.0
$^3J_{trans}$ 3.1	$^3J_{bd}$ 3.3	$^3J_{cis}$ 8.7
Throughout:	$^2J_{cd}$ 4.2	$^3J_{trans}$ 6.6
$J_{cis} > J_{trans}$		$ ^4J_{ac} < 0.3$

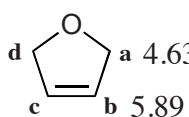
	CDCl ₃	DMSO
 a	a 3.74	3.60
b	b 1.85	1.76

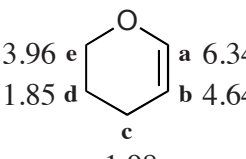
	CDCl ₃	DMSO
 a	a 3.65	3.53
b	b 1.57	1.47
c	c 1.64	1.58

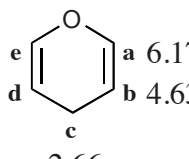
	CDCl ₃	DMSO	D ₂ O
	3.71	3.57	3.75

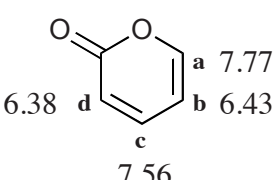
	CDCl ₃	DMSO
 a	3.67	3.88
b	2.87	2.57
c	1.92	

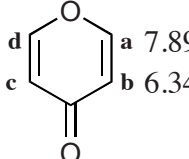
	$^3J_{ab,cis}$ 8.3
6.31 d	$^3J_{ab,trans}$ 10.7
4.95 c	$^3J_{bc}$ 2.5
a 4.31	$^4J_{bd}$ 2.6
b 2.58	$^3J_{cd}$ 2.6

	$^3J_{ab}$ 1.6
a 4.63	$^4J_{ac}$ -2.5
b 5.89	$^4J_{ad,cis}$ 7.1
c	$^4J_{ad,trans}$ 4.6
d	$^3J_{bc}$ 6.3

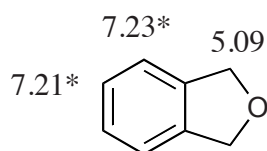
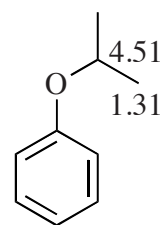
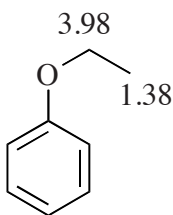
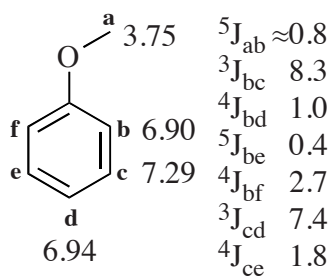
	$^3J_{ab}$ 6.2
3.96 e	$^4J_{ac}$ 2.0
1.85 d	$^3J_{bc}$ 3.8
a 6.34	$^4J_{bd}$ 0.6
b 4.64	
c 1.98	

	$^3J_{ab}$ 7.0
e 6.17	$^4J_{ac}$ 1.7
d 4.63	$^4J_{ae}$ 1.5
a 6.17	$^3J_{bc}$ 3.4
b 4.63	
c 2.66	

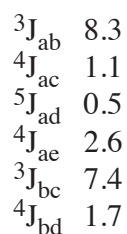
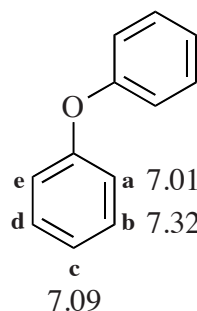
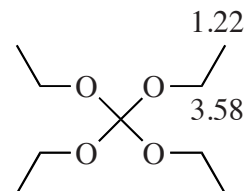
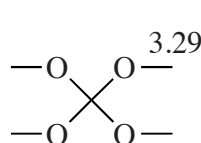
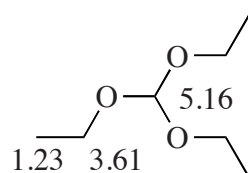
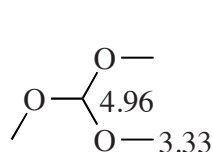
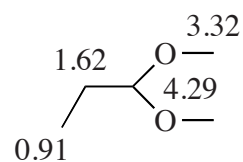
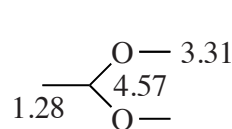
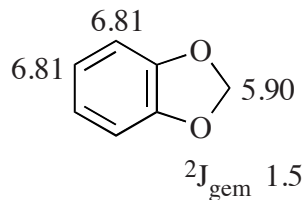
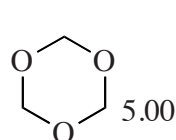
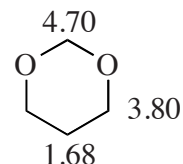
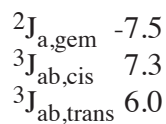
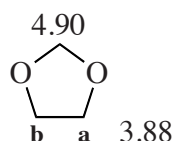
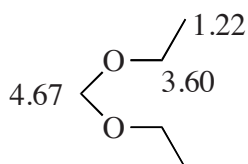
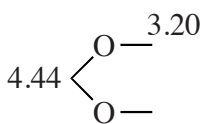
	$^3J_{ab}$ 5.0
6.38 d	$^4J_{ac}$ 2.4
a 7.77	$^5J_{ad}$ 1.2
b 6.43	$^3J_{bc}$ 6.3
c 7.56	$^4J_{bd}$ 1.5
	$^3J_{cd}$ 9.4

	$^3J_{ab}$ 6.0
d 7.89	$^5J_{ac}$ 0.3
a 7.89	$^4J_{ad}$ 2.7
b 6.34	$^4J_{bc}$ 1.1
c 6.34	

O

^1H Chemical Shifts and Coupling Constants of Aromatic Ethers (δ in ppm, J in Hz)

* assignment uncertain

 **^1H Chemical Shifts and Coupling Constants of Acetals, Ketals, and Ortho Esters (δ in ppm, J in Hz)**

5.9 Nitrogen Compounds

5.9.1 Amines

Amine and Ammonium Protons (δ in ppm, $|J|$ in Hz)

Chemical shifts of amine protons lie around 0.5–6 ppm depending on solvent, concentration, and hydrogen bonding. Those of ammonium protons are found between ca. 7 and 12 ppm. Neighboring H bond acceptors lead to deshielding in all cases.

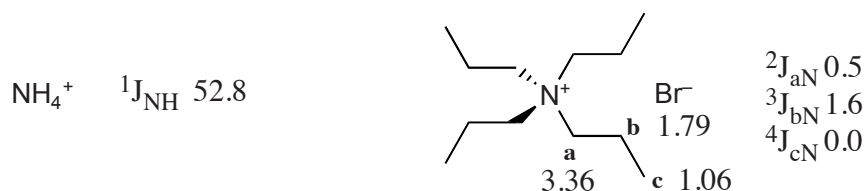
			in CDCl ₃	in DMSO
Amines:	$\delta_{\text{NH}_2}, \delta_{\text{NH}}$	aliphatic	<1–2	2–4
		aromatic	3–4	4–7
Ammonium:	$\delta_{\text{NH}_3^+}, \delta_{\text{NH}_2^+}, \delta_{\text{NH}^+}$	aliphatic	7–11	7–11
		aromatic	8–12	8–12

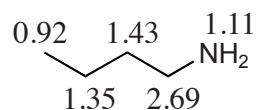
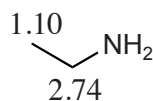
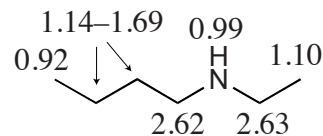
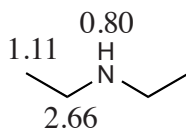
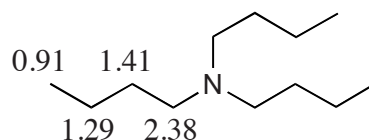
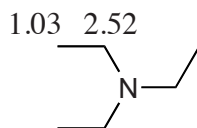
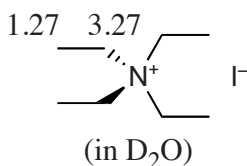
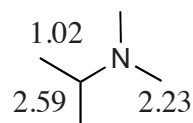
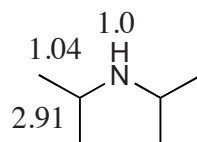
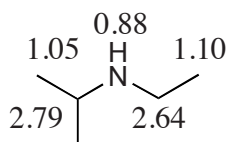
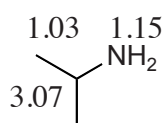
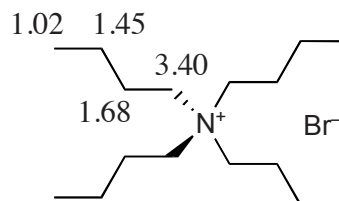
Coupling of amine protons with vicinal H atoms is usually not seen in aliphatic amines because of their rapid intermolecular exchange. However, for =C–NH–CH moieties (enamines, aromatic amines, amides, etc.), the exchange rate is slower and splitting (or line broadening at intermediate rates) is often observed. The H–C–N–H coupling depends on the conformation in a similar way as the H–C–C–H coupling (see Chapter 5.1.2). For N–CH₃ and N–CH₂ groups: $^3J_{\text{HCNH}} \approx 5\text{--}6\text{ Hz}$.

In acidic media (e.g., in trifluoroacetic acid as solvent), the exchange of the ammonium protons is slowed down to such an extent that the vicinal coupling H–N⁺–C–H generally becomes observable. In other media, signals are usually broad owing to intermediate exchange rates.

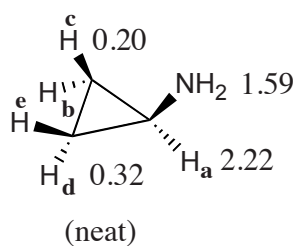
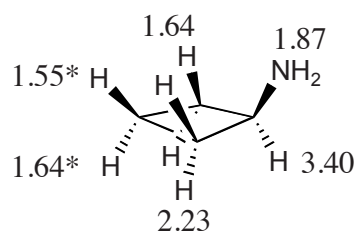
The signals of amine and especially of ammonium protons are often broadened additionally because the ^{14}N – ^1H coupling is only partly eliminated by the quadrupole relaxation of ^{14}N (spin quantum number, $I = 1$; natural abundance, 99.6%; $^1J_{\text{NH}} \approx 60\text{ Hz}$). This line broadening has no effect on the vicinal H–C–N–H coupling so that sharp multiplets can be observed for neighboring H atoms even when the NH proton exhibits a broad signal. In ammonium compounds of high symmetry, the quadrupole relaxation is slow and the coupling with ^{14}N leads to triplets of equal intensity for all three lines.

N

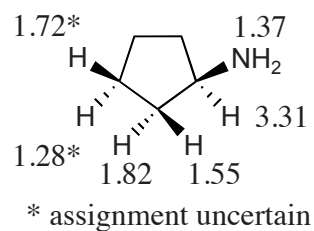


¹H Chemical Shifts and Coupling Constants of Amines and Ammonium Salts (δ in ppm, J in Hz)2.47
CH₃NH₂ ≈ 2.3
(CH₃)₂NH2.22
(CH₃)₃N
²J_{gem} -11.73.21 (in D₂O)
(CH₃)₄N⁺ I⁻3.46 (in CDCl₃)
(CH₃)₄N⁺ AcO⁻

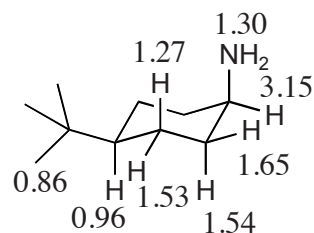
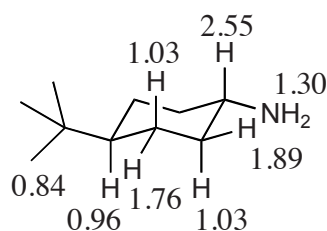
N

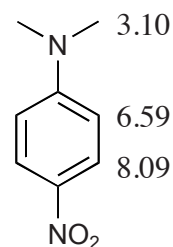
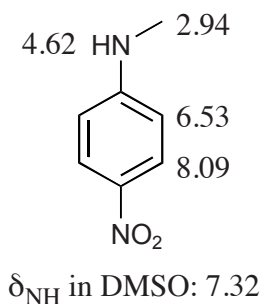
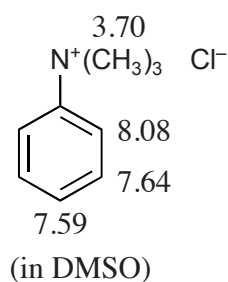
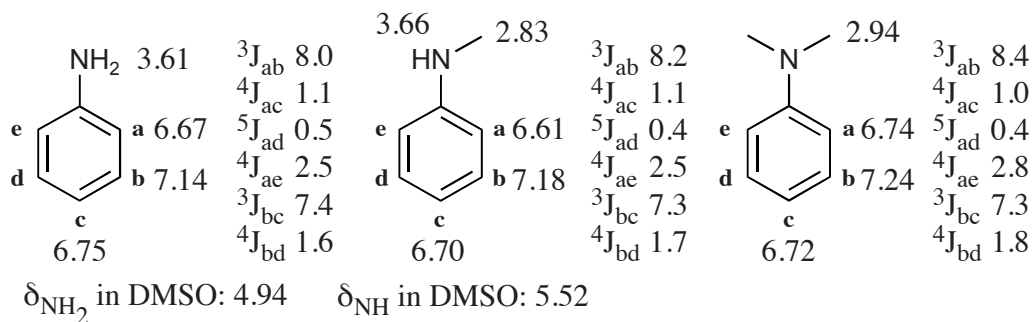

³J_{ab} 6.6
³J_{ac} 3.6
²J_{bc} -4.3
³J_{bd} 9.7
³J_{be} 6.2
³J_{ce} 9.9


* assignment uncertain

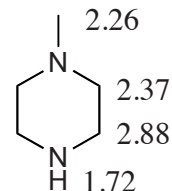
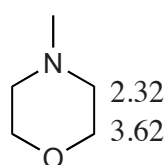
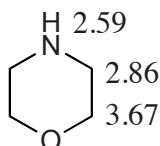
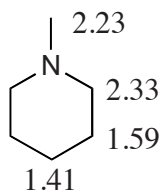
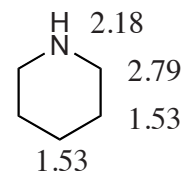
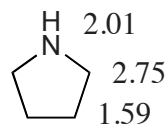
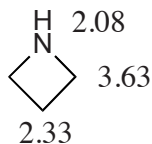
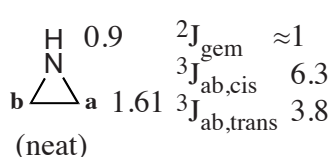


* assignment uncertain





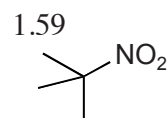
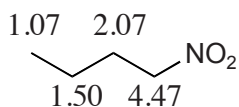
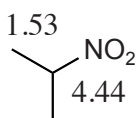
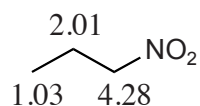
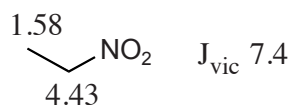
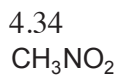
¹H Chemical Shifts and Coupling Constants of Cyclic Amines (δ in ppm, J in Hz)

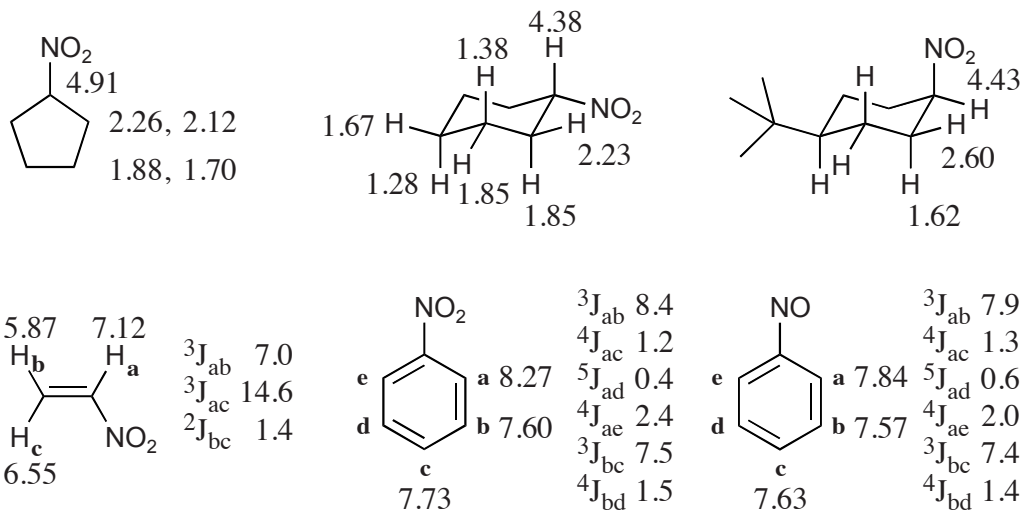


N

5.9.2 Nitro and Nitroso Compounds

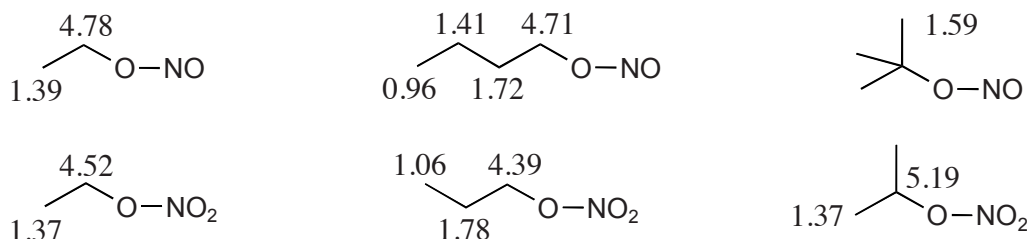
¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)





5.9.3 Nitrites and Nitrates

¹H Chemical Shifts (δ in ppm)

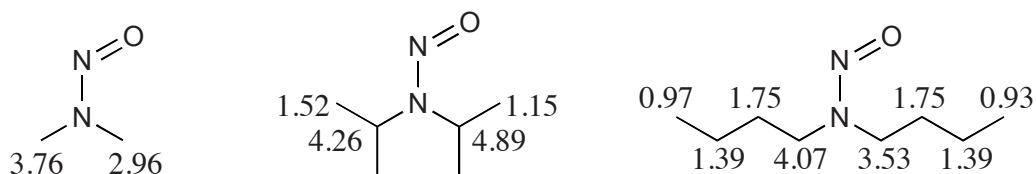


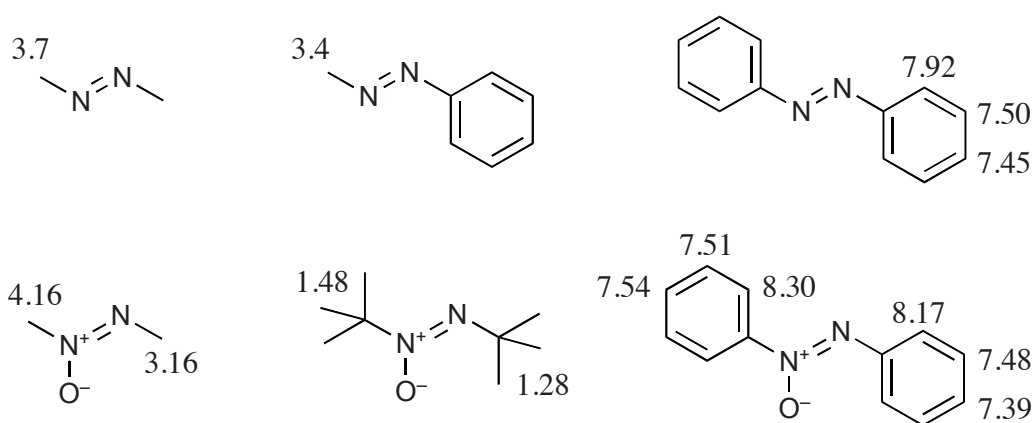
5.9.4 Nitrosamines, Azo and Azoxy Compounds

¹H Chemical Shifts (δ in ppm)

Owing to hindered rotation around the N–NO bond, corresponding protons in *cis* and *trans* positions have different chemical shifts in the neighborhood of the N=O group.

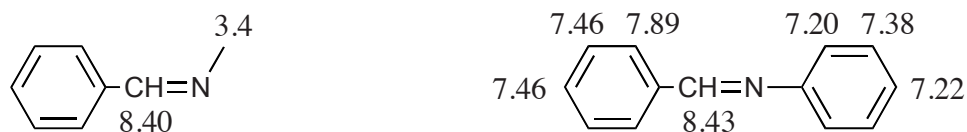
In general: $\delta_{\text{cis}} < \delta_{\text{trans}}$ for $\alpha\text{-CH}_3$, $\alpha\text{-CH}_2$, and $\beta\text{-CH}_3$
 $\delta_{\text{cis}} > \delta_{\text{trans}}$ for $\alpha\text{-CH}$



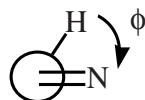


5.9.5 Imines, Oximes, Hydrazones, and Azines

^1H Chemical Shifts (δ in ppm)

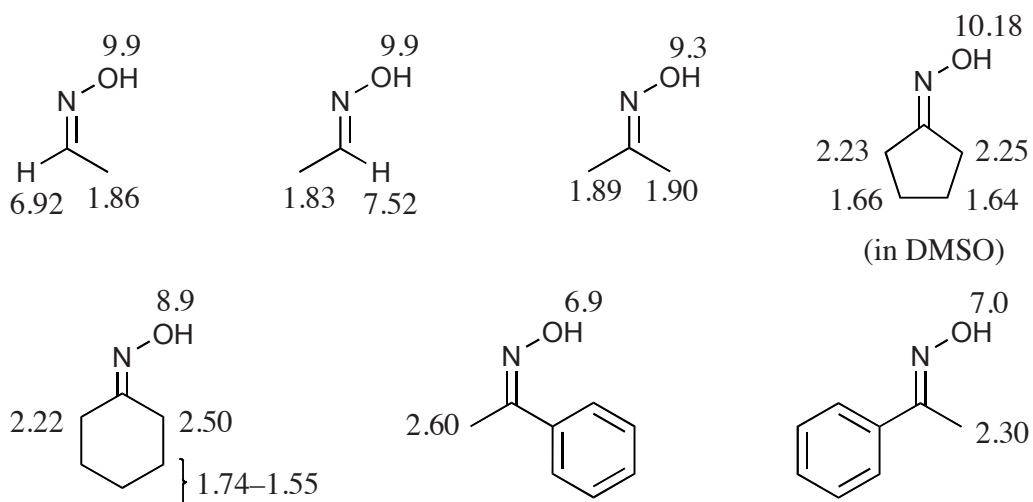


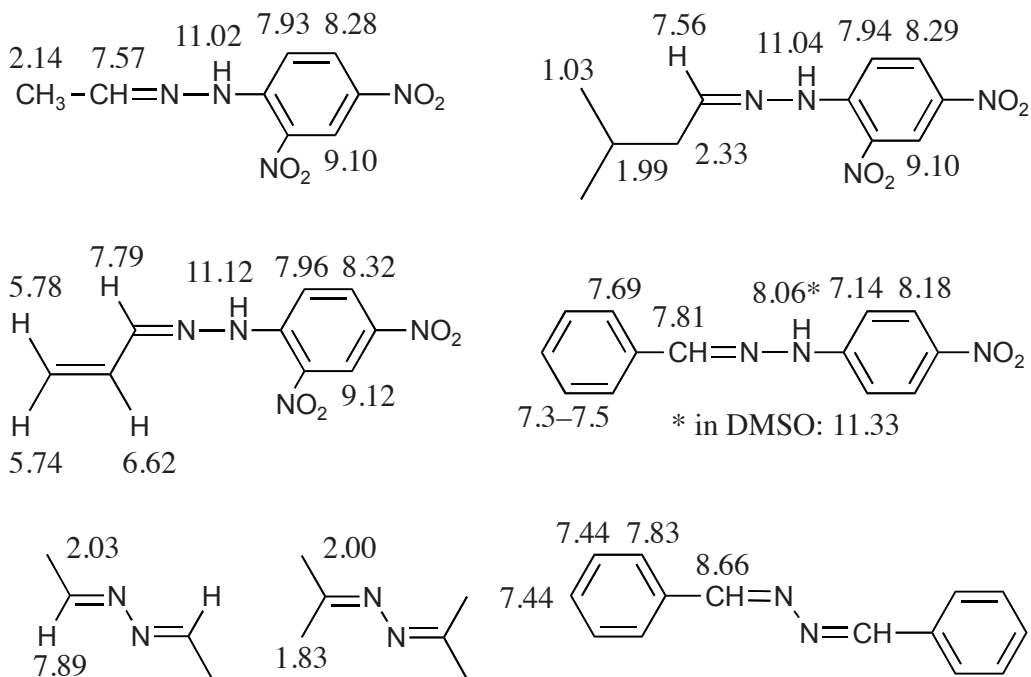
In aldoximes and ketoximes, the chemical shift difference between *syn* and *anti* protons at the α -CH group, $\Delta\delta = \delta_{\text{syn}} - \delta_{\text{anti}}$, depends on the dihedral angle, $\phi_{\text{H-C-C=N}}$:



ϕ	$\Delta\delta = \delta_{\text{syn}} - \delta_{\text{anti}}$
0°	1
60°	0
115°	-0.3

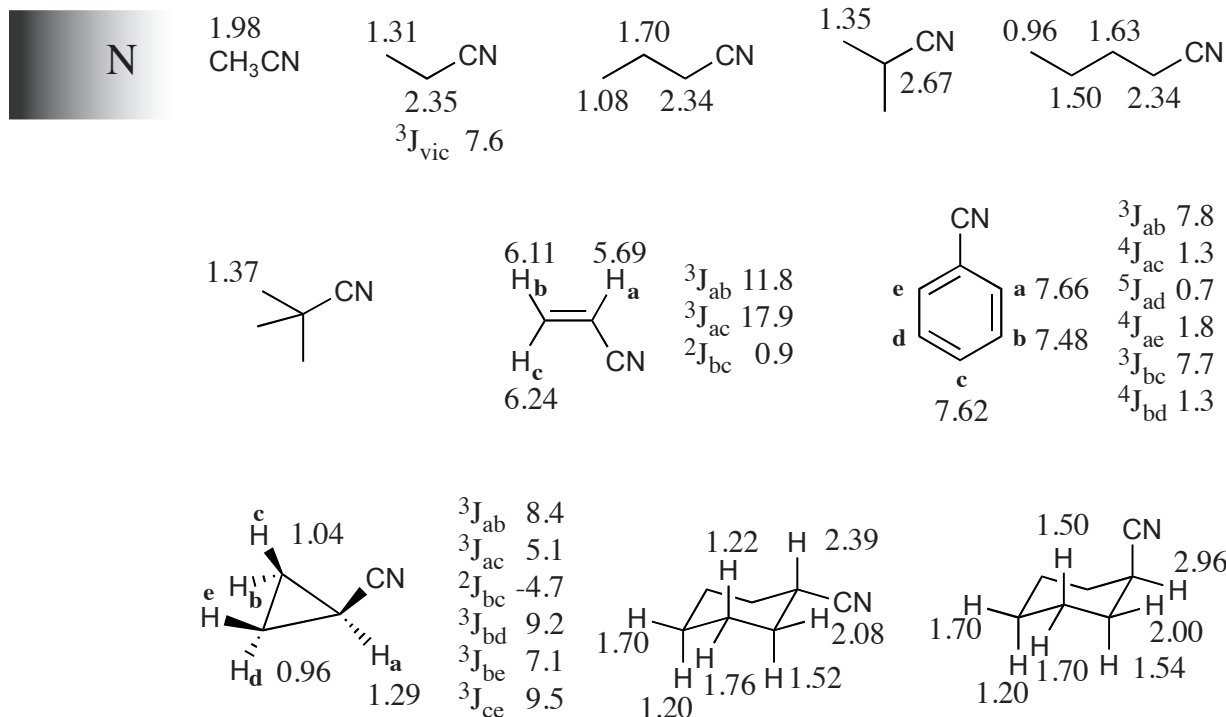
N





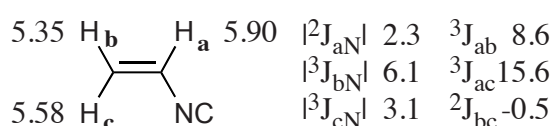
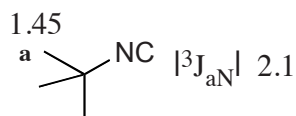
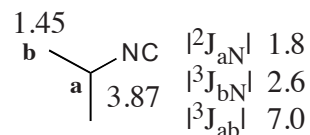
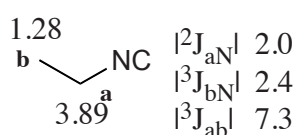
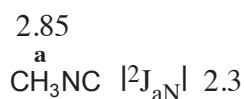
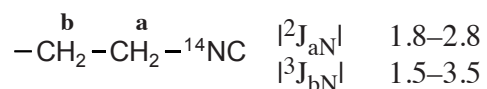
5.9.6 Nitriles and Isonitriles

¹H Chemical Shifts and Coupling Constants of Nitriles (δ in ppm, J in Hz)



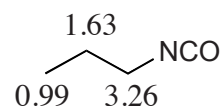
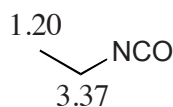
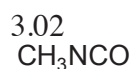
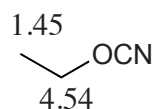
¹H Chemical Shifts and Coupling Constants of Isonitriles (δ in ppm, |J| in Hz)

Because of the symmetrical electron distribution around the N atom, the quadrupole relaxation of the nitrogen nucleus is so slow that the ¹⁴N-¹H coupling becomes observable and leads to triplets with relative intensities of 1:1:1 (spin quantum number of ¹⁴N: I = 1; natural abundance, 99.6%):

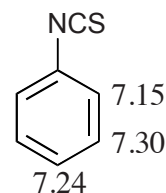
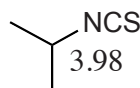
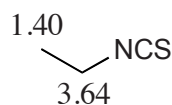
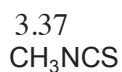
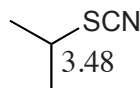
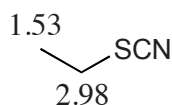
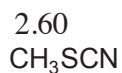
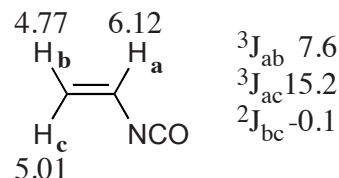
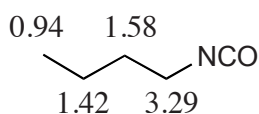
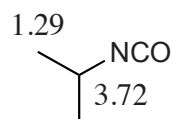


5.9.7 Cyanates, Isocyanates, Thiocyanates, and Isothiocyanates

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)



N

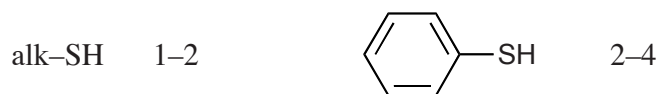


5.10 Sulfur Compounds

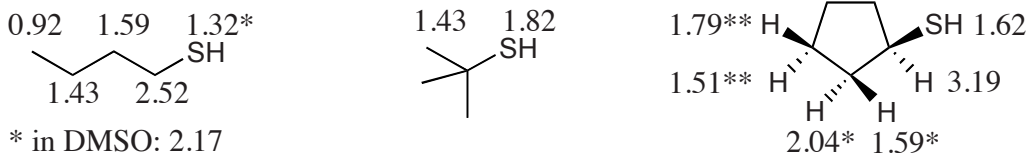
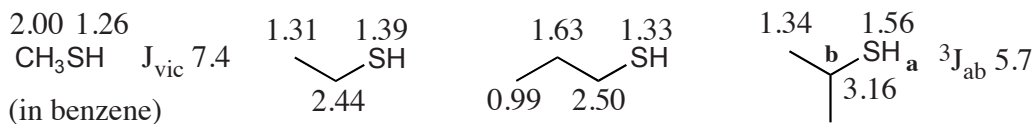
5.10.1 Thiols

 ^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

Typical ranges of SH chemical shifts:

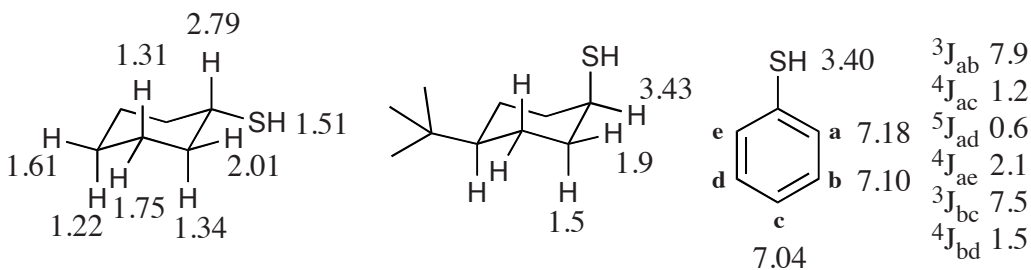


The exchange with other SH, OH, NH, or COOH protons is generally so slow that the chemical shift is characteristic and the vicinal coupling with SH protons becomes visible (5–9 Hz in aliphatic systems with fast rotation).

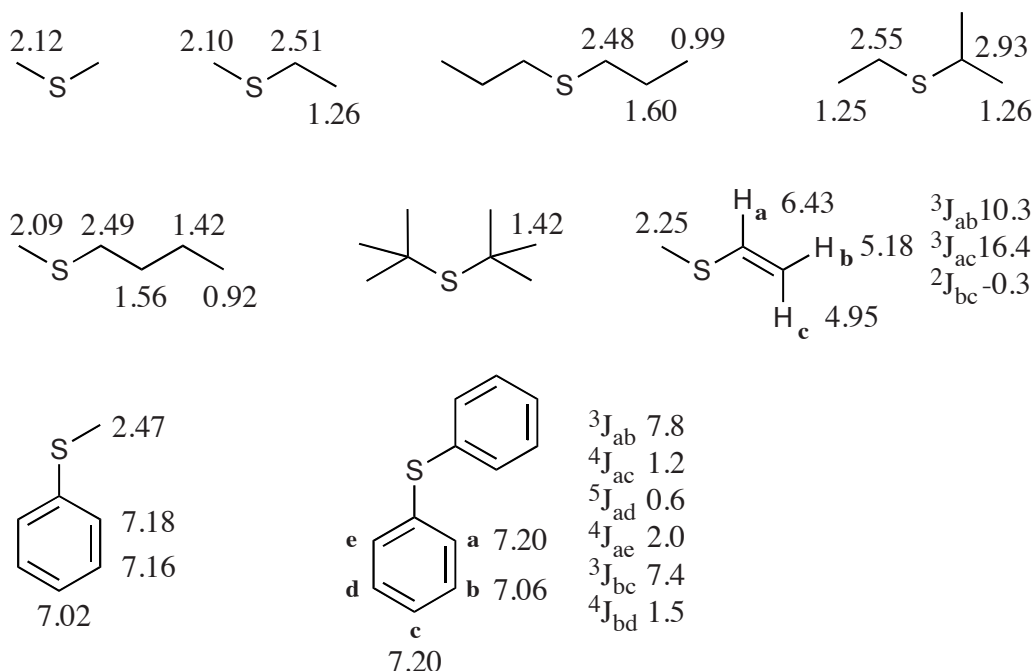
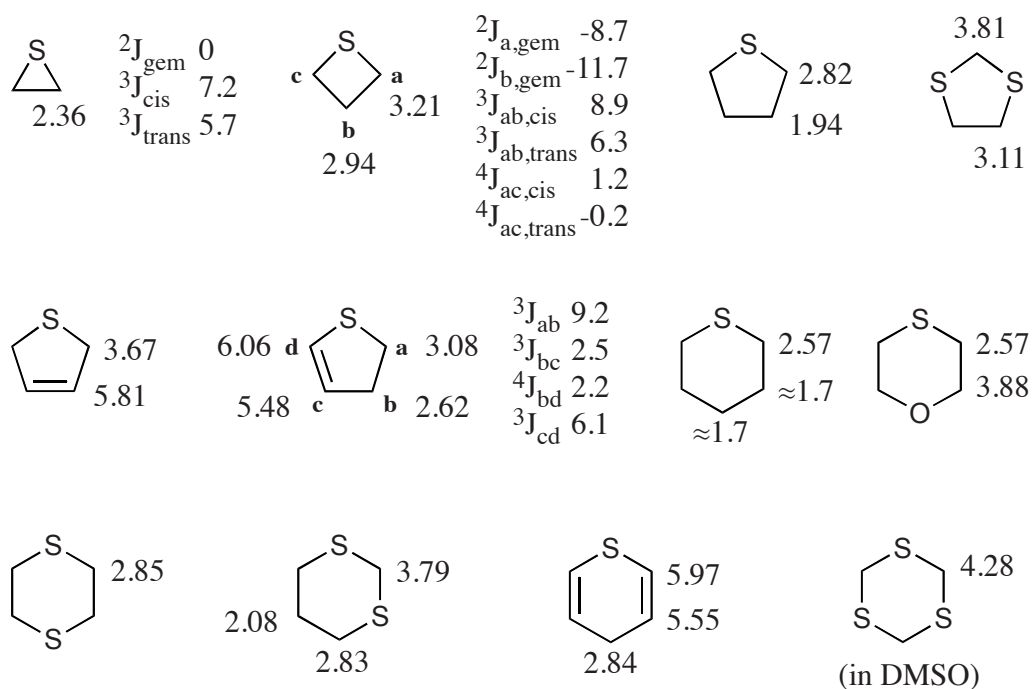


*, **: assignments uncertain

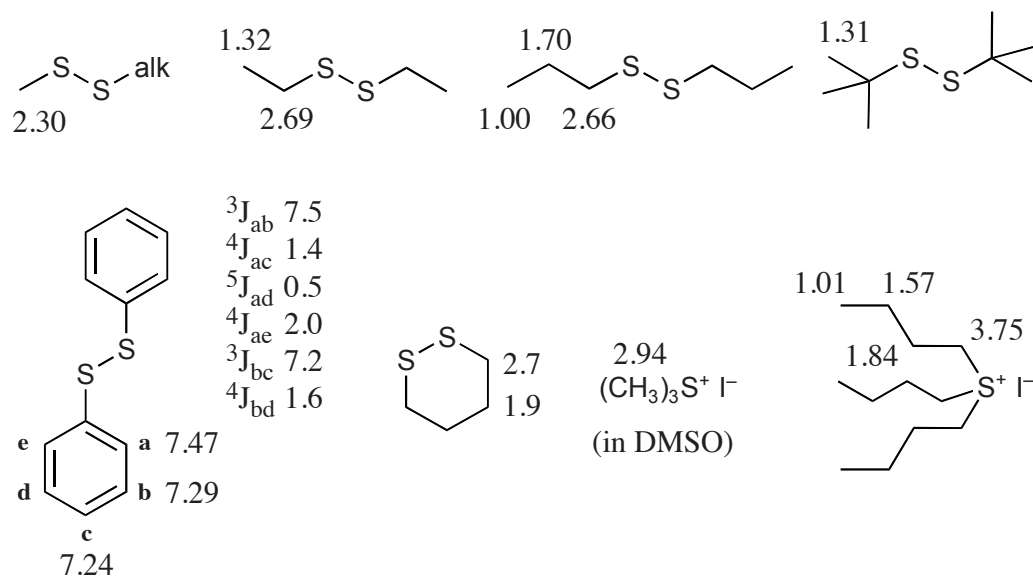
S



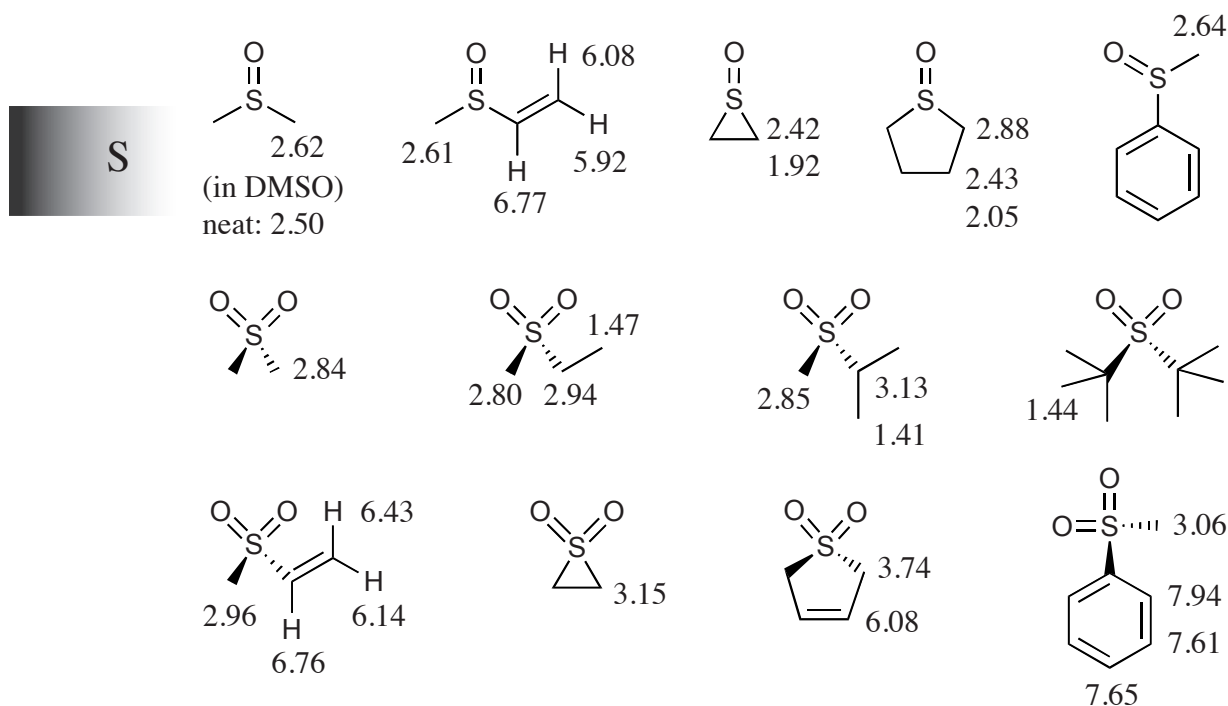
5.10.2 Sulfides

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)***¹H Chemical Shifts and Coupling Constants of Cyclic Sulfides (δ in ppm, J in Hz)***

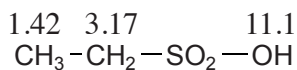
5.10.3 Disulfides and Sulfonium Salts

 ^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

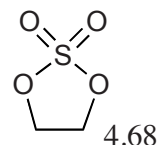
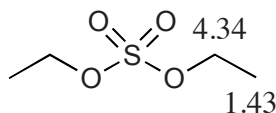
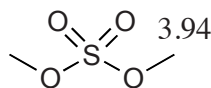
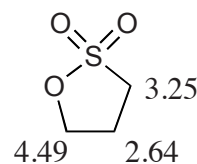
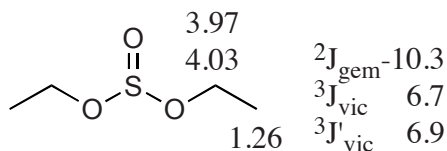
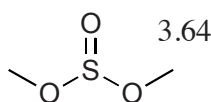
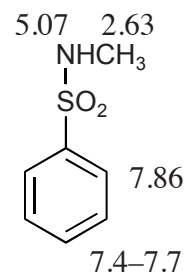
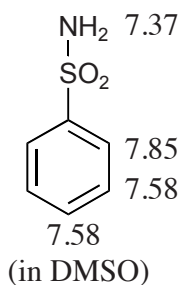
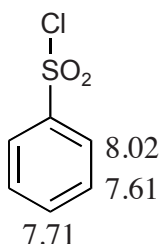
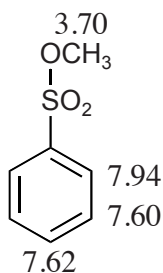
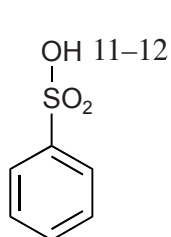
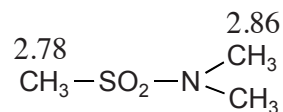
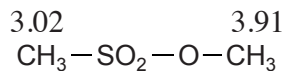
5.10.4 Sulfoxides and Sulfones

 ^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

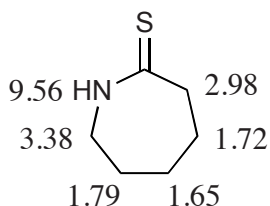
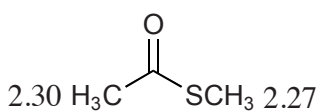
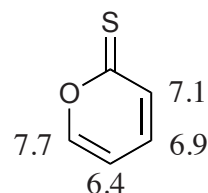
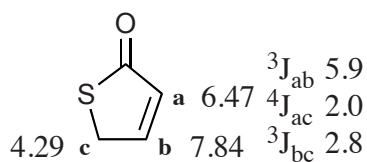
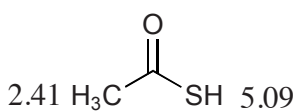
5.10.5 Sulfonic, Sulfurous, and Sulfuric Acids and Derivatives

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

Apparently lower δ_{OH}
values in DMSO due to
fast exchange with H_2O

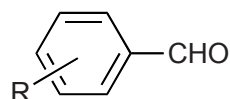


5.10.6 Thiocarboxylate Derivatives

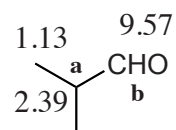
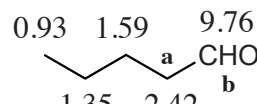
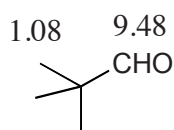
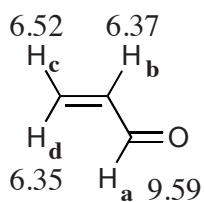
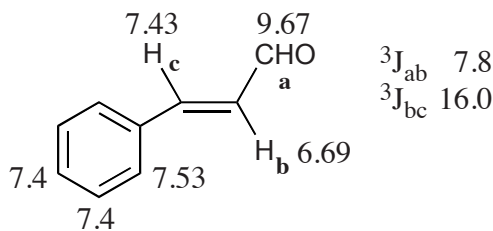
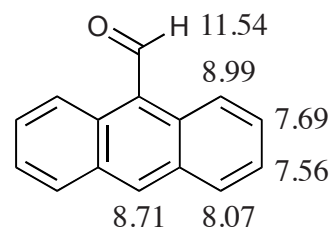
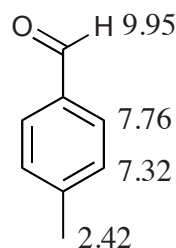
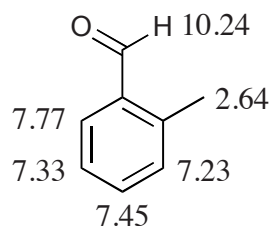
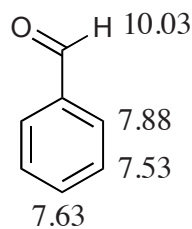
¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

5.11 Carbonyl Compounds

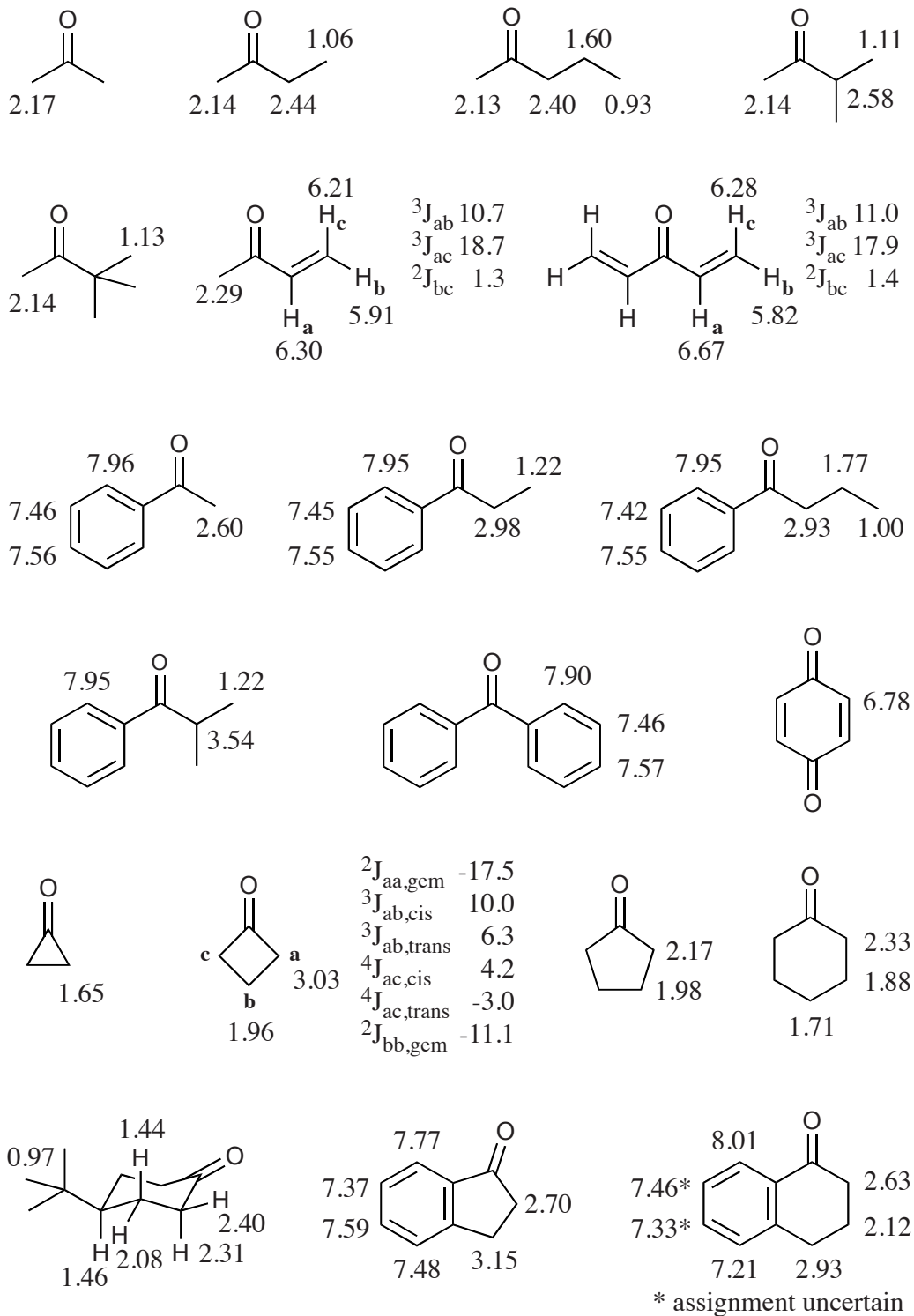
5.11.1 Aldehydes

 ^1H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)alk-CHO 9–10 $^3J_{\text{vic}}$ 0–3alken-CHO 9–10 $^3J_{\text{vic}}$ ≈ 8 *ortho*-substituted: 10–10.5*meta*-, *para*-substituted: 9.5–10.29.60
CH₂=O

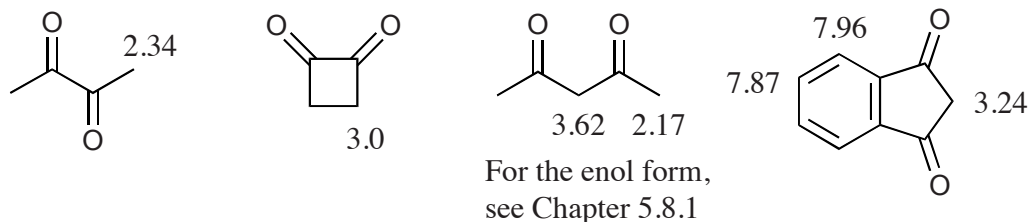
(in TMS)

 $|^2J_{\text{gem}}|$ 42.42.20 9.79
H₃C-CHO**a** **b** $^3J_{\text{ab}}$ 3.01.13 9.79
CHO**a** **b** $^3J_{\text{ab}}$ 1.41.67 9.74
CHO0.97 2.42 **a** **b** $^3J_{\text{ab}}$ 2.0 $^3J_{\text{ab}}$ 1.1 $^3J_{\text{ab}}$ 1.9
 $^3J_{\text{ab}}$ 4.7 $^3J_{\text{bc}}$ 10.0
 $^4J_{\text{ac}}$ < 1 $^4J_{\text{bd}}$ 17.4
 $^4J_{\text{ad}}$ < 1 $^2J_{\text{cd}}$ 1.0

 $^3J_{\text{ab}}$ 7.8
 $^3J_{\text{bc}}$ 16.0


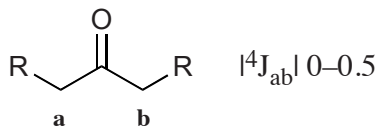
5.11.2 Ketones

¹H Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

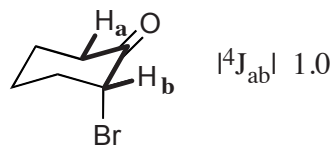
C = X

^1H Chemical Shifts of Diketones (δ in ppm)**Long-Range Coupling in Ketones ($|J|$ in Hz)**

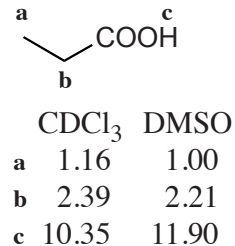
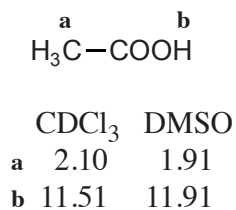
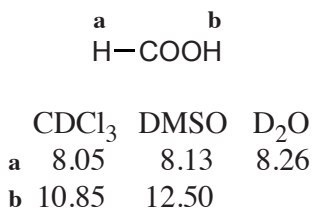
For fixed conformations, the coupling over the $\text{C}=\text{O}$ group is often detectable for W-arrangement of the coupling path.



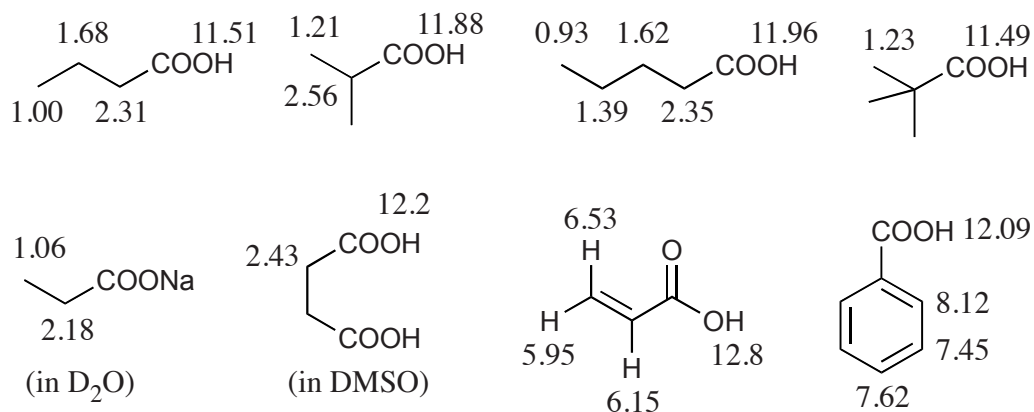
No W-arrangement



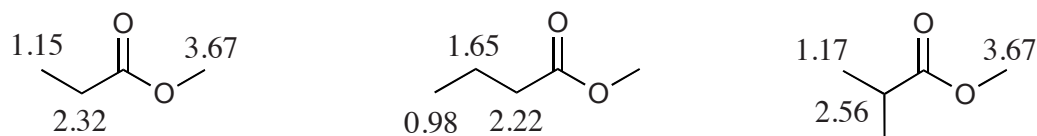
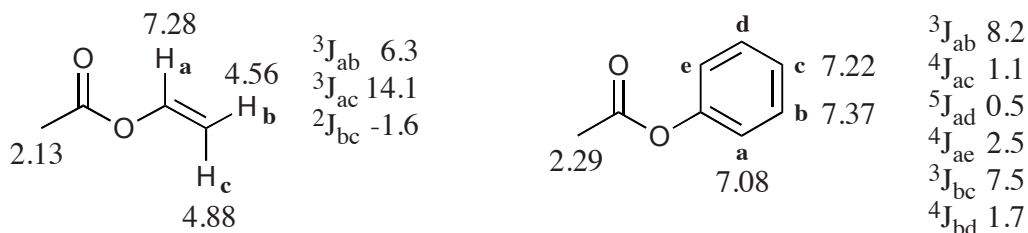
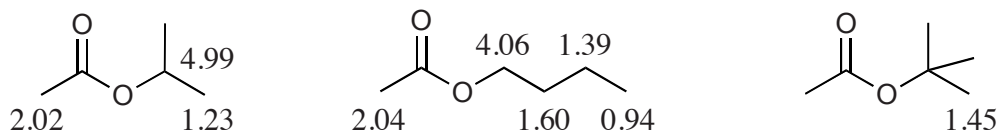
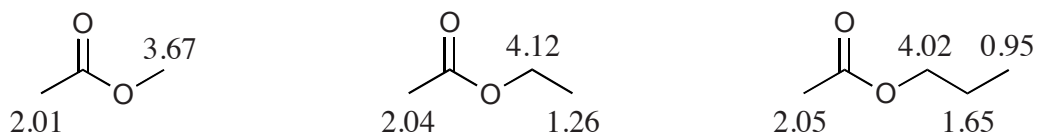
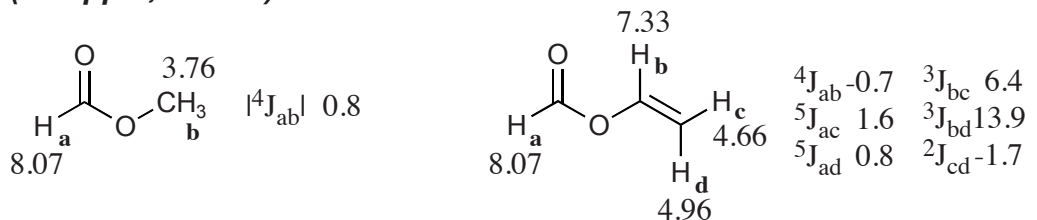
W-arrangement

5.11.3 Carboxylic Acids and Carboxylates **^1H Chemical Shifts (δ in ppm)**

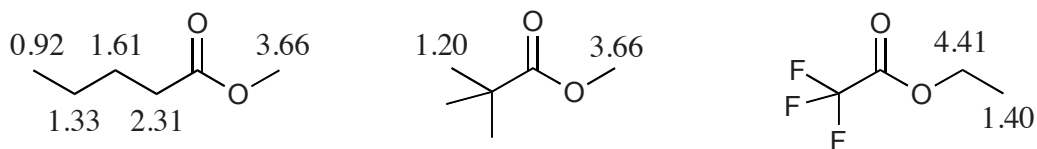
$\text{C}=\text{X}$

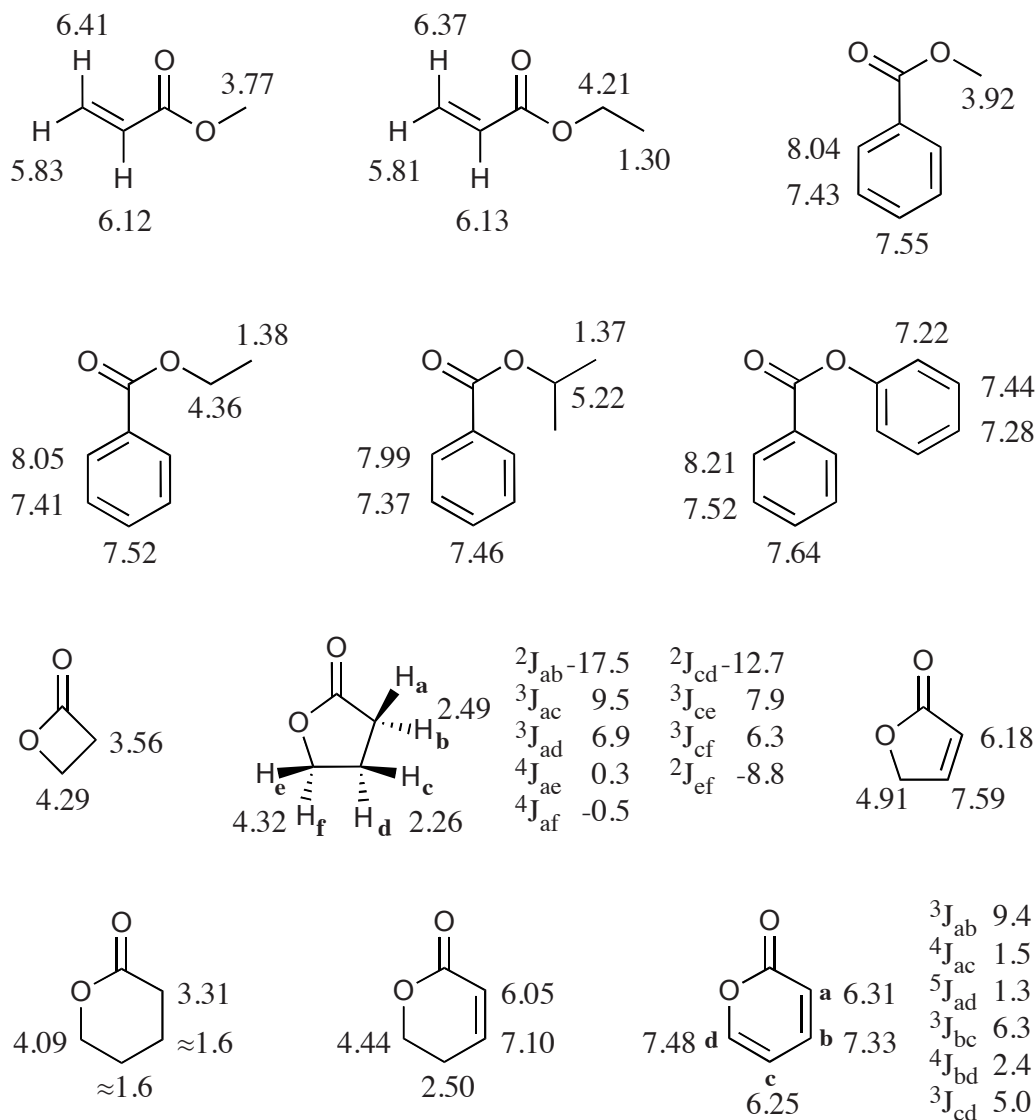


5.11.4 Esters and Lactones

¹H Chemical Shifts and Coupling Constants of Carboxylic Acid Esters
(δ in ppm, J in Hz)

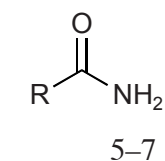
C = X



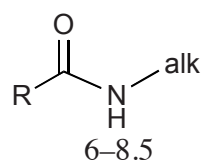


C=X

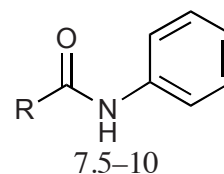
5.11.5 Amides and Lactams

Amide Protons (δ in ppm, J in Hz)

R: alk or ar



R: alk or ar



R: alk or ar

Higher values in DMSO or with H bond acceptors in the neighborhood.

The signals of the NH protons are often broad because the ^{14}N – ^1H coupling is only partly eliminated by the quadrupole relaxation of ^{14}N (spin quantum number, $I = 1$; $^1J_{\text{NH}} \approx 60$). In primary amides, the hindered rotation around the CO–N bond is another reason for line broadening. At slow rotation, the chemical shifts of the two primary amide protons differ by about 0.4–1 ppm. Therefore, at intermediate rotation rates, line widths of up to 1 ppm may be observed.

Due to the slow intermolecular exchange of amide protons, their coupling to neighboring hydrogen atoms is usually detectable. The splitting of the C–H signal is clearly observed even in those cases where the signal of the NH proton is broad and featureless. The H–N–C–H coupling depends on the conformation in a similar way as the H–C–C–H coupling (see Chapter 5.1.2). For N–CH₃ and N–CH₂ groups: $^3J_{\text{HNCH}} \approx 7$ Hz.

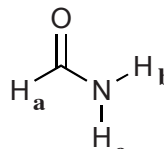
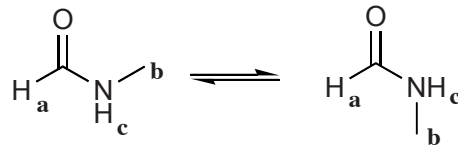
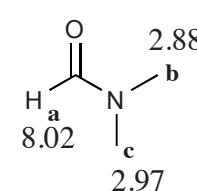
Tertiary Alkylamides

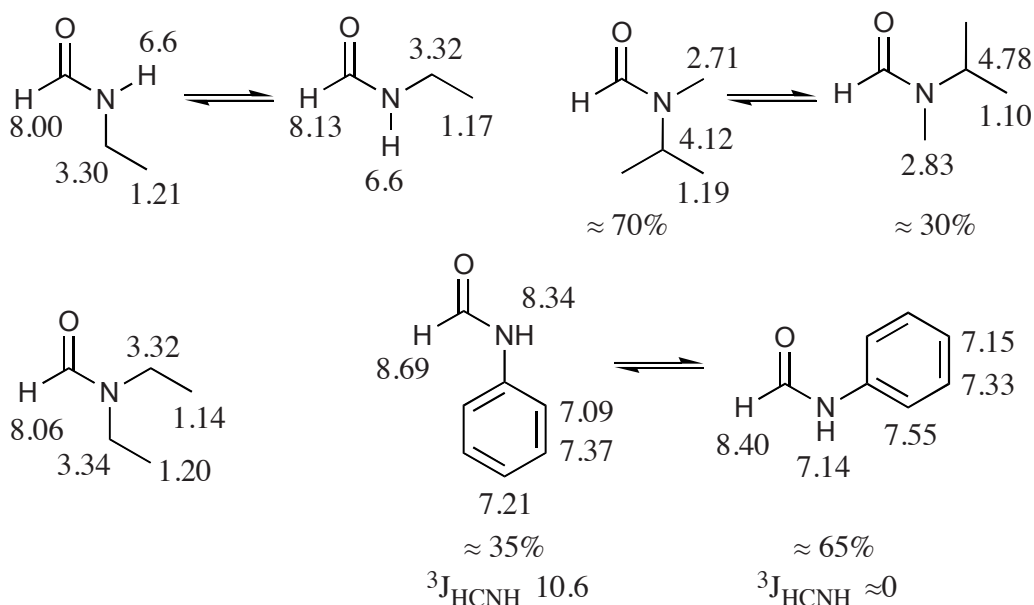
The rotation around the CO–N bond is usually so slow that, for identical substituents, two separate signals are observed for *cis* and *trans* positions. With different N-substituents, two separate pairs of signals are observed for the two conformers. In general, the following relationships hold:

$$\begin{array}{ll} \text{for NCH}_3, \text{NCH}_2\text{CH}_3, \text{and NCH(CH}_3)_2 & \delta_{\text{cis to O}} \leq \delta_{\text{trans to O}} \\ \text{for NCH(CH}_3)_2 \text{ and NC(CH}_3)_3 & \delta_{\text{trans to O}} \leq \delta_{\text{cis to O}} \\ \text{for NCH}_2 & \delta_{\text{cis to O}} \approx \delta_{\text{trans to O}} \end{array}$$

Formamides (δ in ppm, J in Hz)

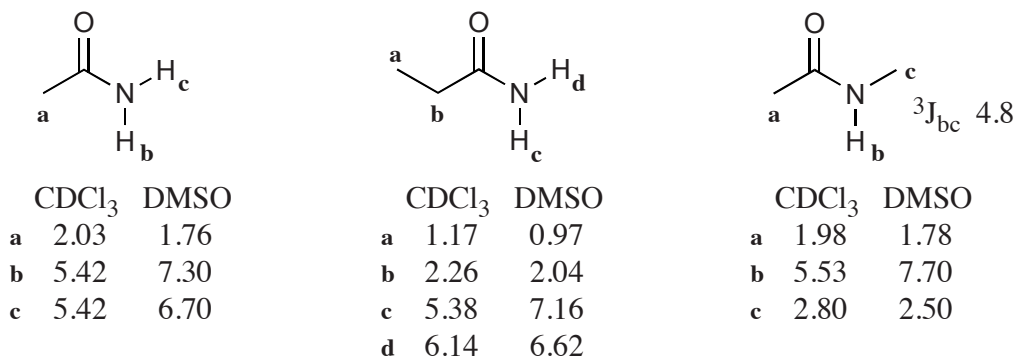
In the more stable conformer of monosubstituted formamides, the substituent occupies the *cis* position relative to the carbonyl oxygen. In the more stable conformer of asymmetrically disubstituted formamides, the larger substituent occupies the *trans* position relative to the carbonyl oxygen.

						C = X			
		$\approx 90\%$ in CDCl ₃	$\approx 10\%$ in CDCl ₃						
		CDCl ₃	DMSO	CDCl ₃	DMSO				
a	8.23	7.98	a	8.19	8.01	a	8.06	7.81	$^4J_{ab} \approx 0.3$
b	5.80	7.14	b	2.86	2.59	b	2.94	2.72	$^4J_{ac} \approx 0.7$
c	5.48	7.41	c	5.55	7.90	c	5.86	7.90	

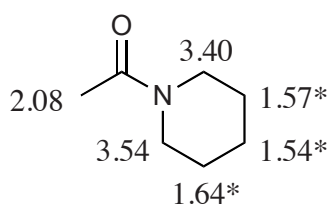
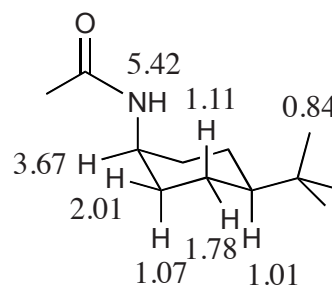
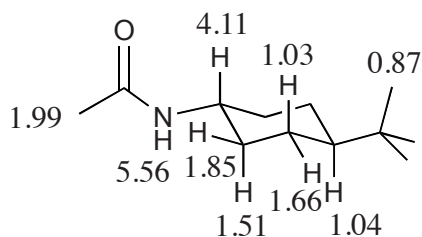
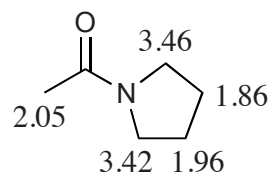
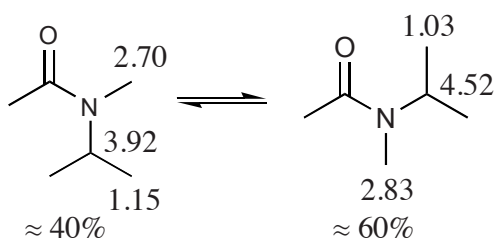


Amides of Aliphatic Carboxylic Acids (δ in ppm, J in Hz)

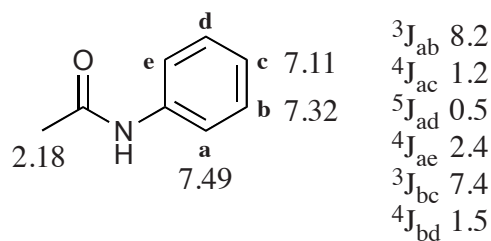
In *monosubstituted* acetamides, the substituent of the only observable conformer is *cis* to the carbonyl oxygen. In *disubstituted* acetamides, the more stable conformer has the larger substituent *cis* to the carbonyl oxygen.



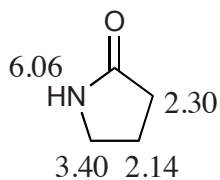
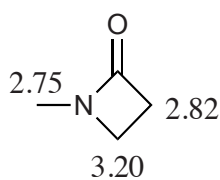
C=X



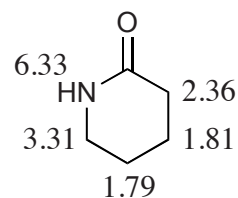
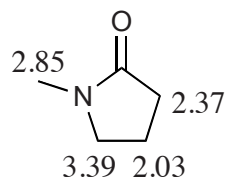
* assignment uncertain



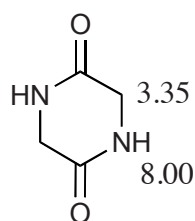
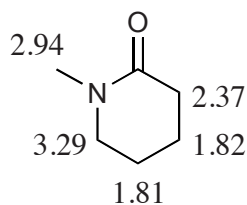
Lactams (δ in ppm, J in Hz)



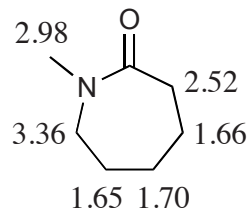
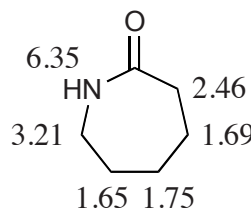
$^3J_{\text{HNCH}_2}$ 0.8



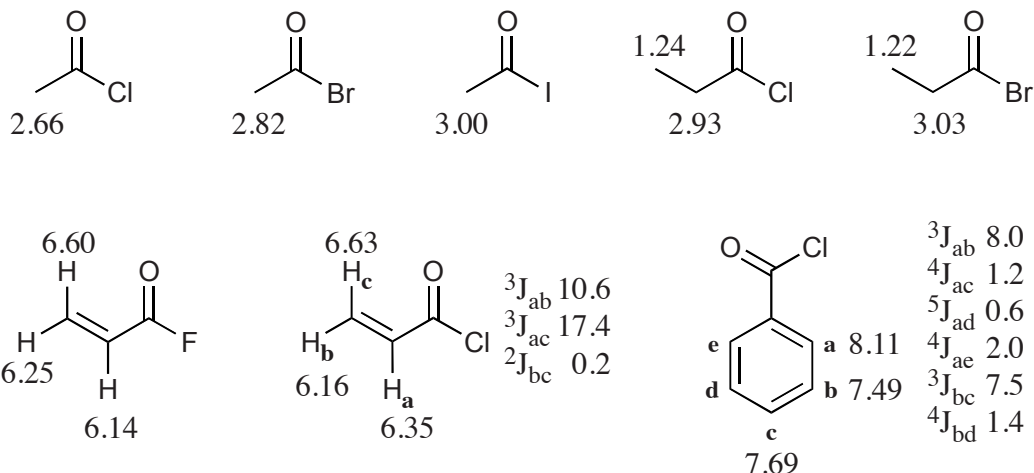
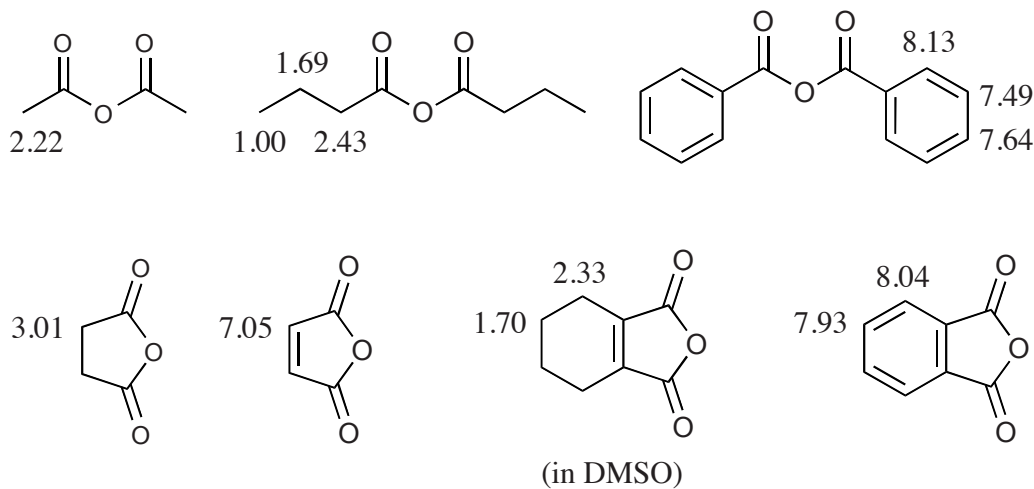
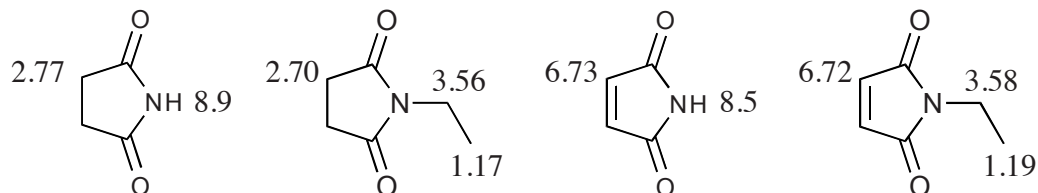
C = X

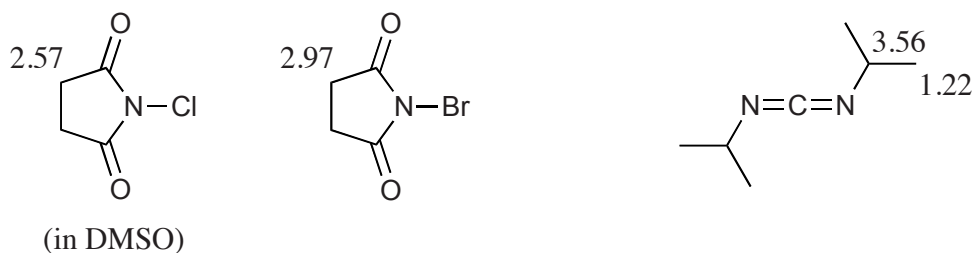
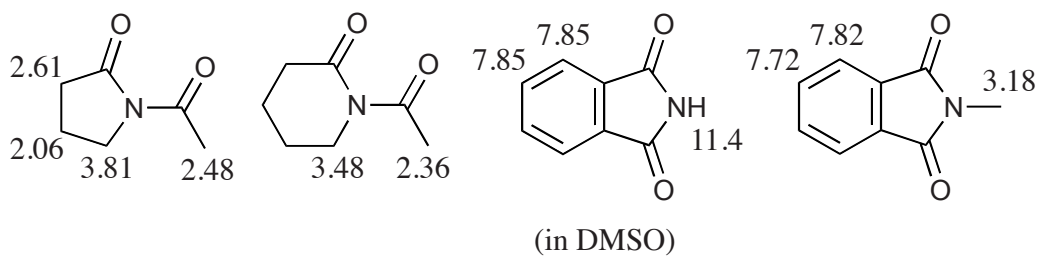


$^3J_{\text{HNCH}_2}$ 2.2

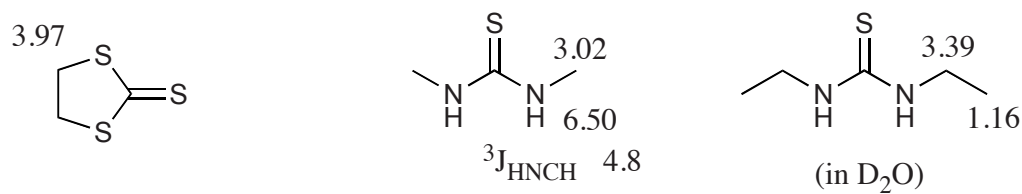
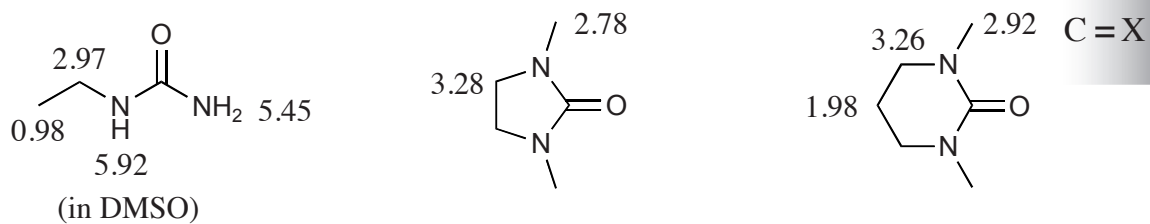
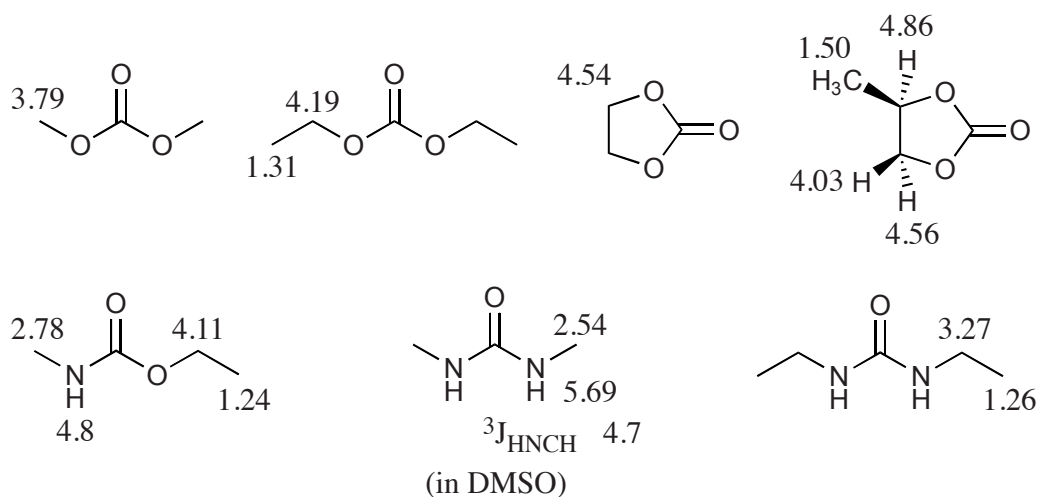


5.11.6 Miscellaneous Carbonyl Derivatives

Carboxylic Acid Halides (δ in ppm, J in Hz)*Carboxylic Acid Anhydrides (δ in ppm)**Carboxylic Acid Imides (δ in ppm, J in Hz)*



Carbonyl Acid Derivatives (δ in ppm, J in Hz)



C = X

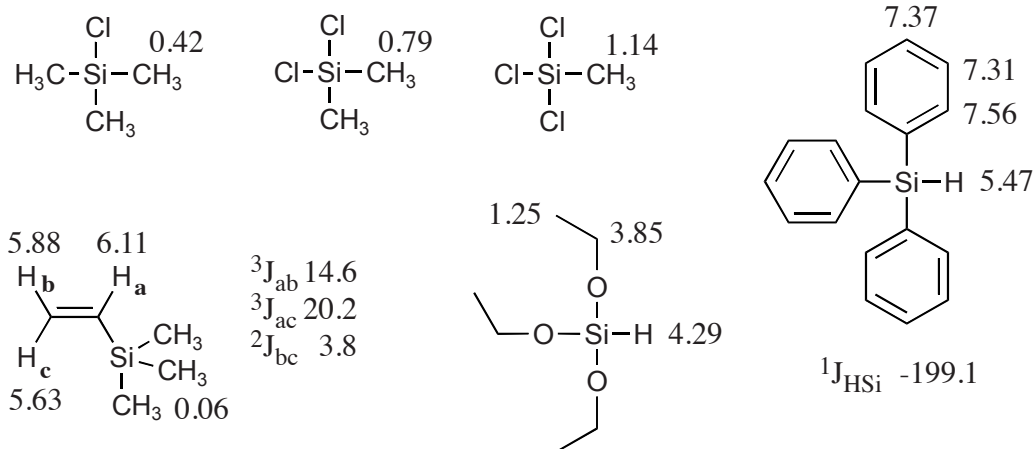
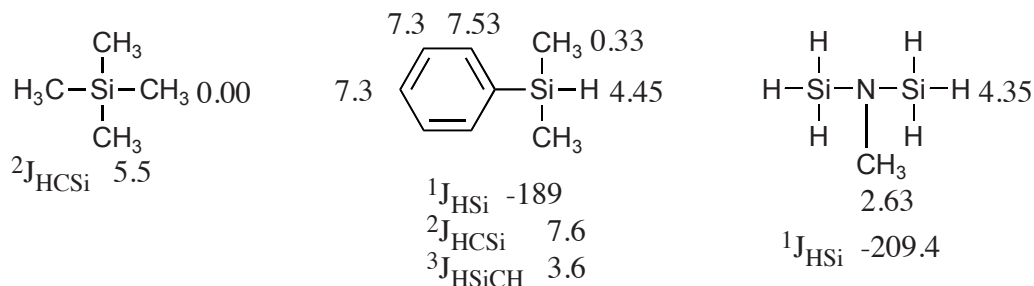
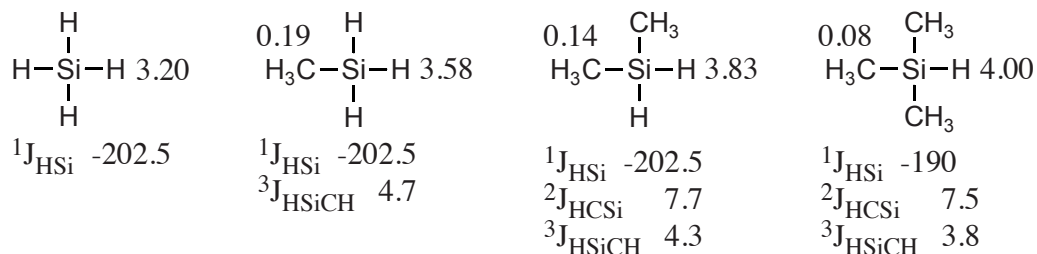
5.12 Miscellaneous Compounds

5.12.1 Compounds with Group IV Elements

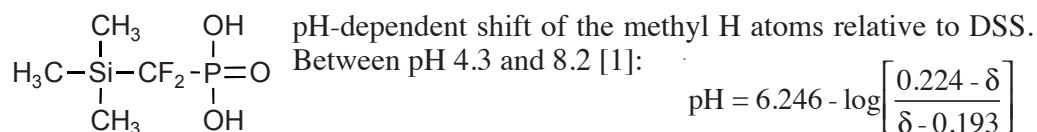
Silicon Compounds (δ in ppm, J in Hz)

Coupling with silicon: The isotope ^{29}Si (natural abundance, 4.7 %) has a spin quantum number I of 1/2. Doublets with the corresponding intensity ("Si satellites") are usually observed. Typical coupling constants: $^1J_{\text{HSi}}$ -150 to -380 Hz

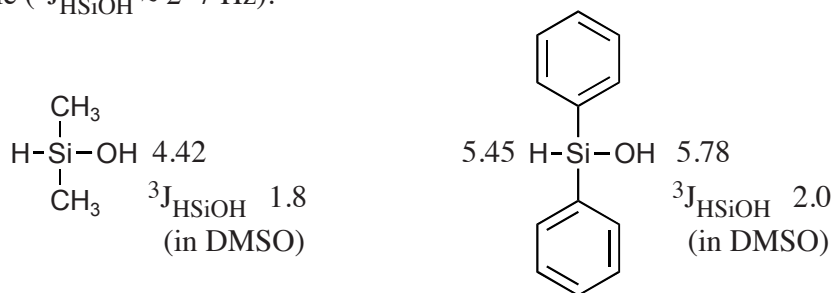
$^2J_{\text{HCSi}}$ 5 to 10 Hz



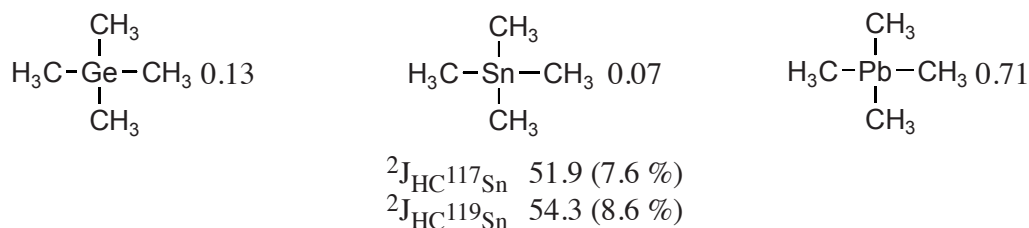
P Si



The silanol hydrogen is exchangeable with D₂O. Slow intermolecular exchange is observed in DMSO as solvent so that the vicinal coupling in H–Si–O–H is detectable ($^3J_{\text{HSiOH}} \approx 2\text{--}7\text{ Hz}$).



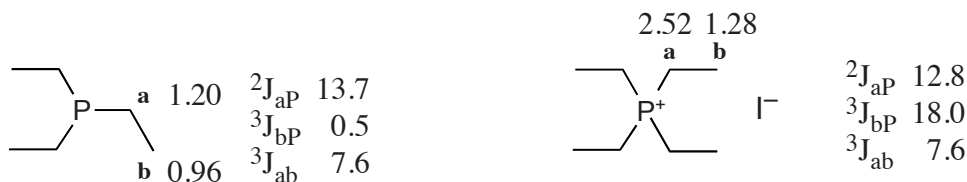
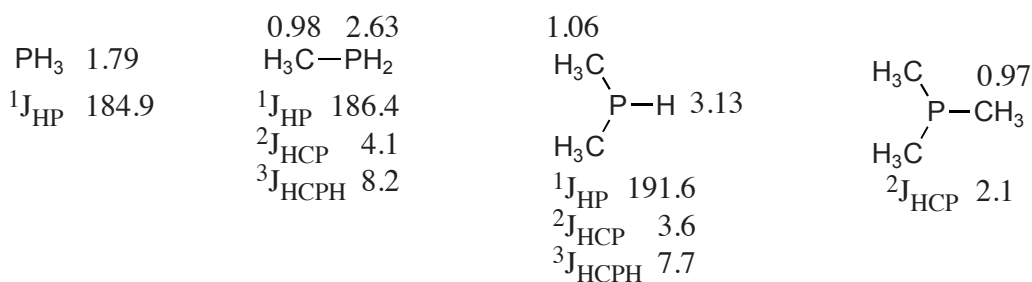
Germanium, Tin, and Lead Compounds (δ in ppm, J in Hz)



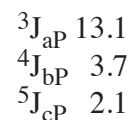
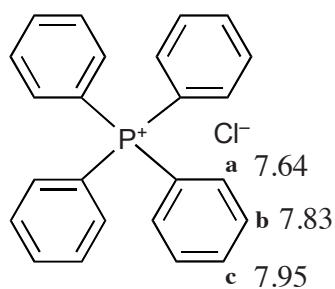
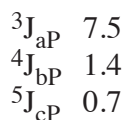
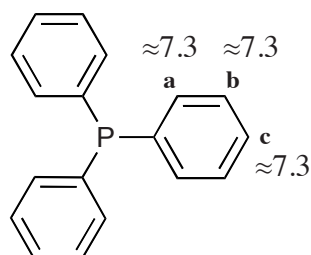
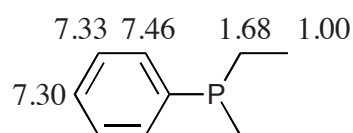
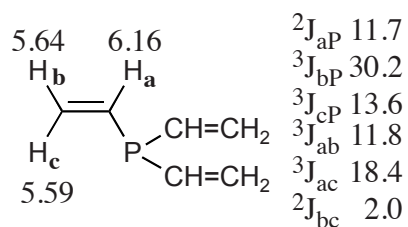
5.12.2 Phosphorus Compounds

^{31}P (natural abundance, 100%) has a spin quantum number I of $1/2$. Couplings to protons through up to 5 bonds are usually observed.

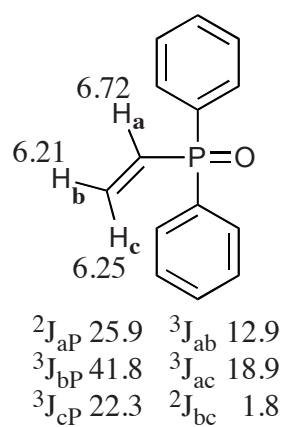
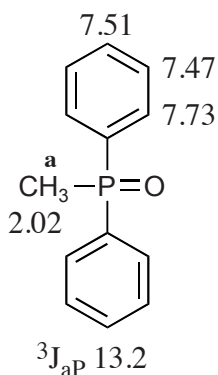
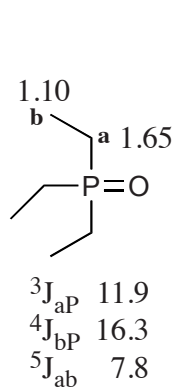
Phosphines and Phosphonium Compounds (δ in ppm, J in Hz)



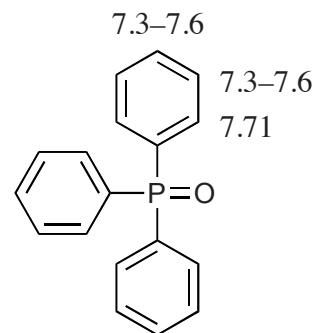
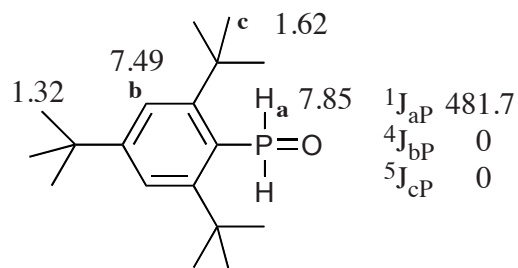
P Si

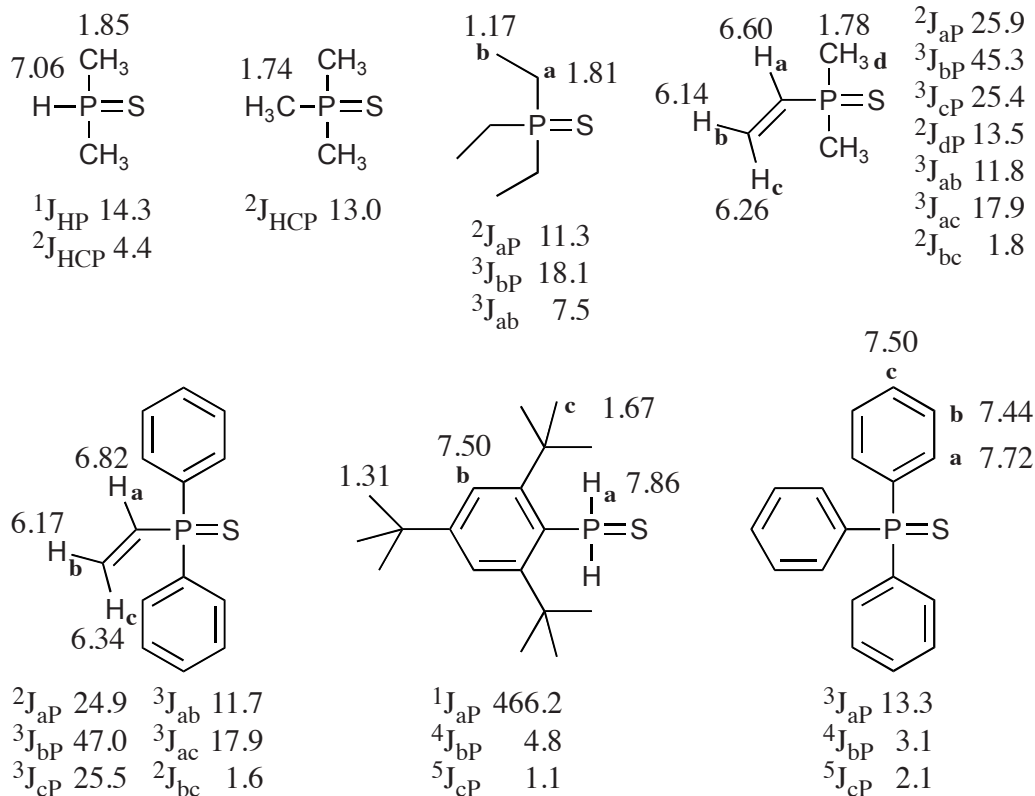


Phosphine Oxides and Sulfides (δ in ppm, J in Hz)

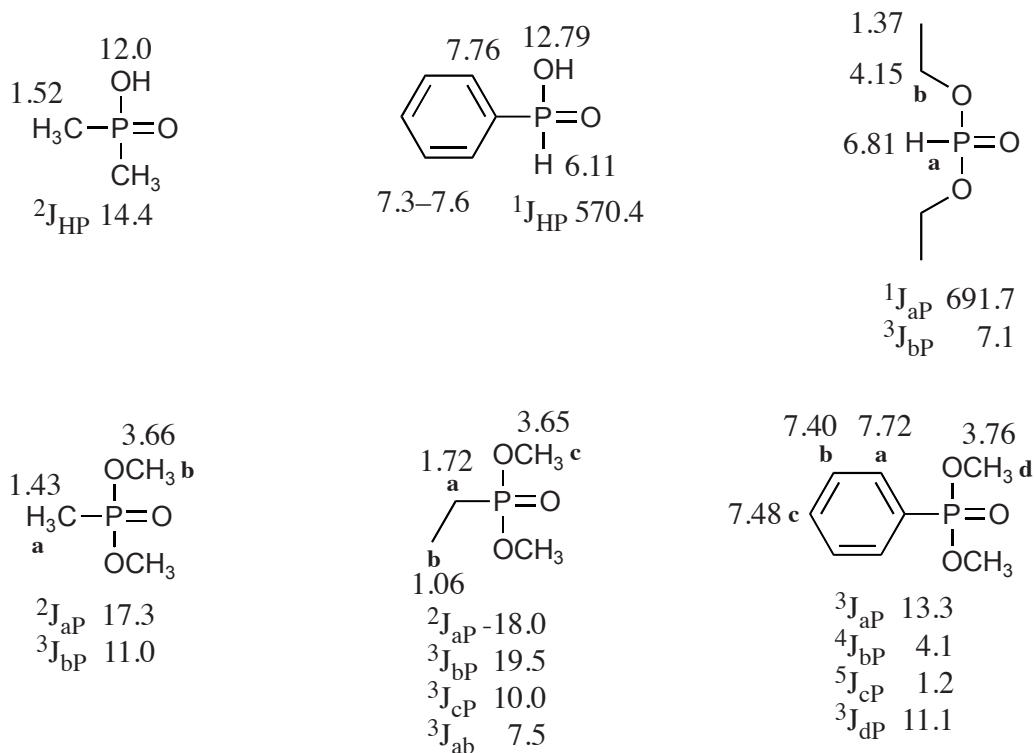


P Si

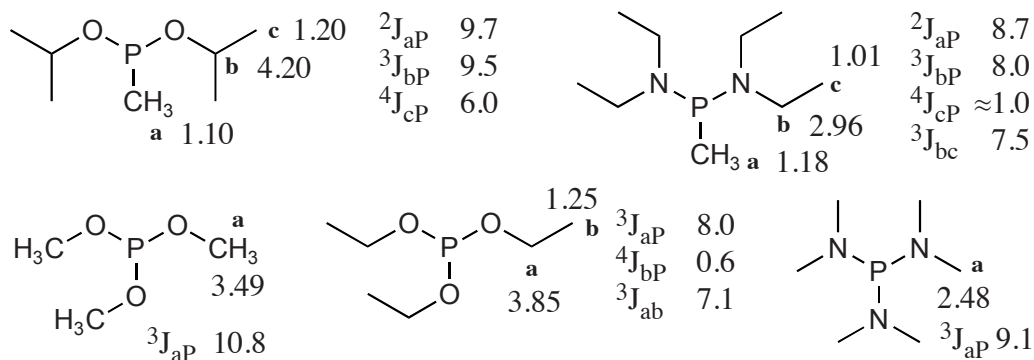
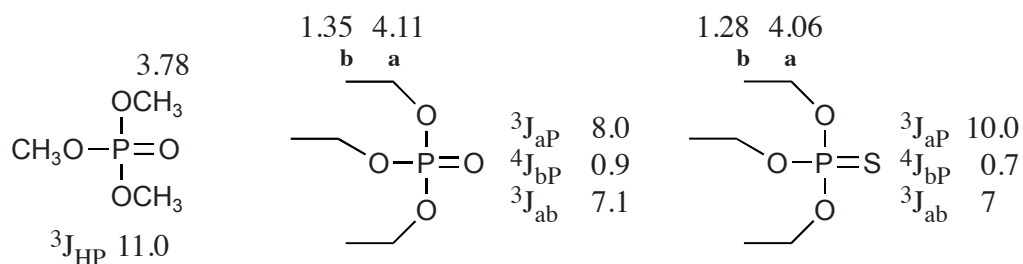
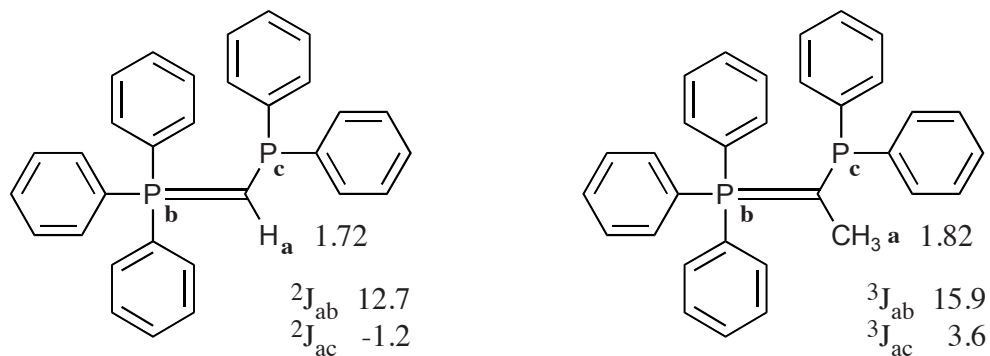




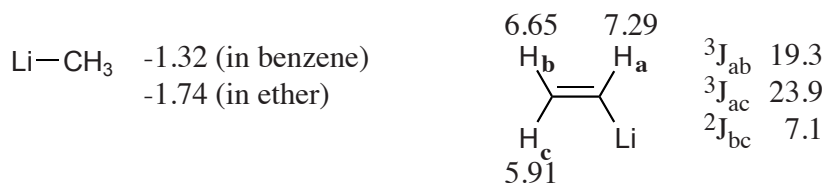
Phosphinic and Phosphonic Acid Derivatives (δ in ppm, J in Hz)

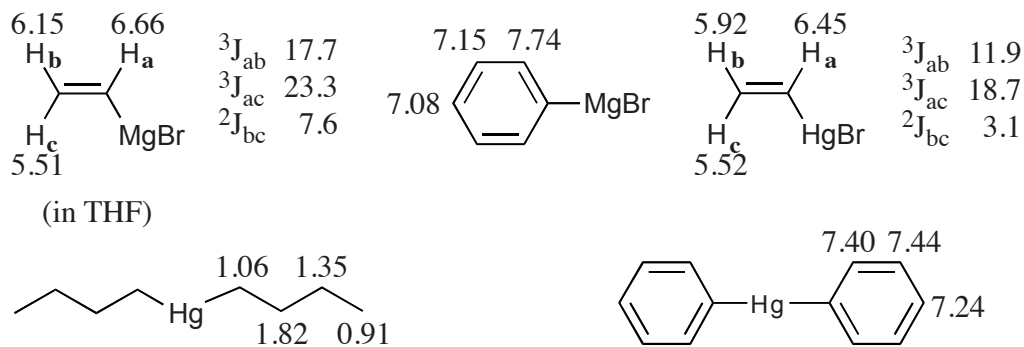


P Si

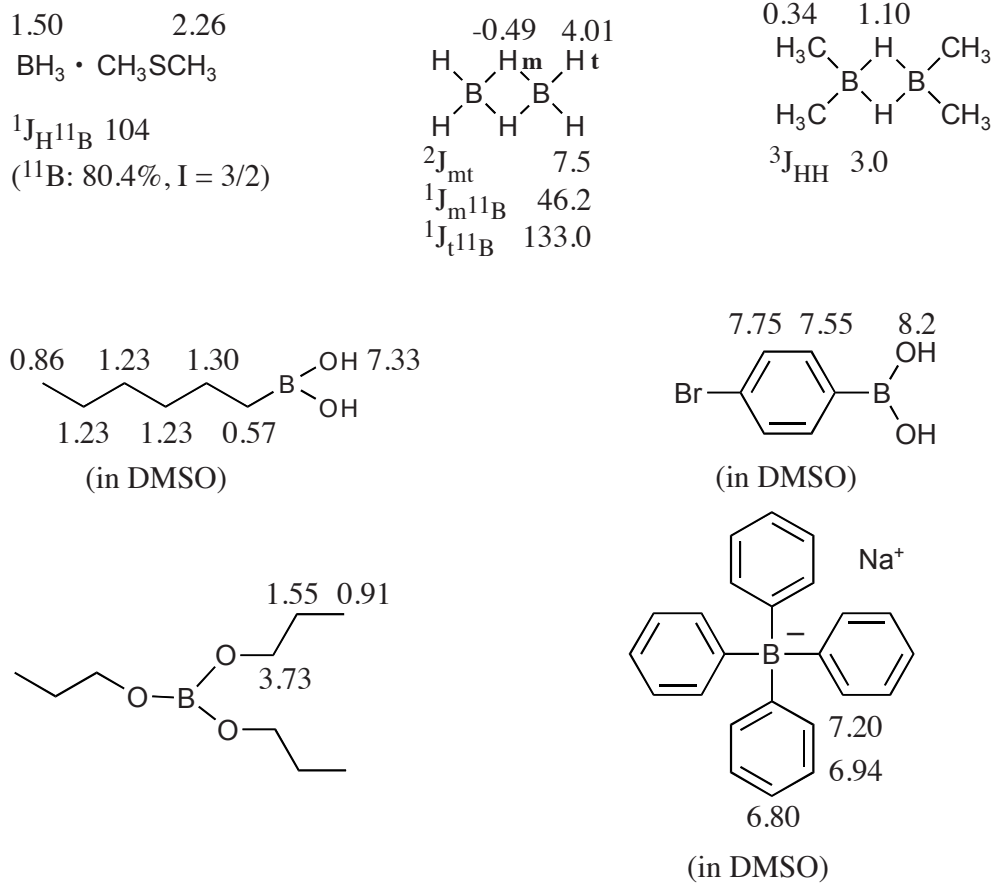
Phosphonous and Phosphorous Acid Derivatives (δ in ppm, J in Hz)**Phosphoric Acid Derivatives (δ in ppm, J in Hz)****Phosphorus Ylids (δ in ppm, J in Hz)**

P Si

5.12.3 Miscellaneous Compounds**Organometallic Compounds (δ in ppm, J in Hz)**



Boron Compounds (δ in ppm, J in Hz)



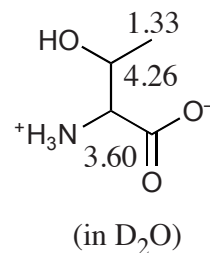
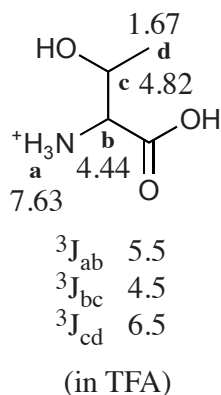
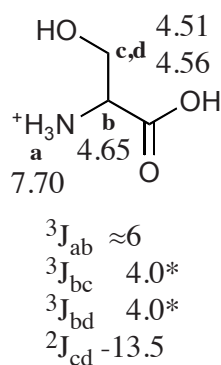
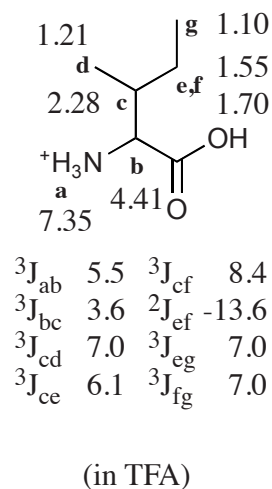
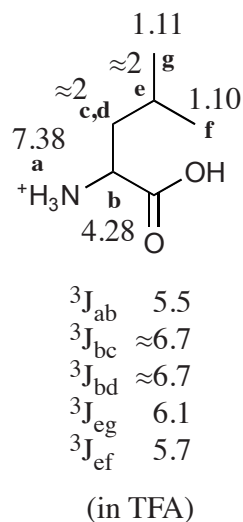
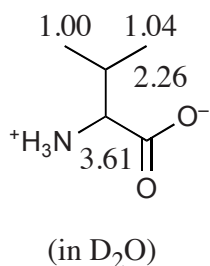
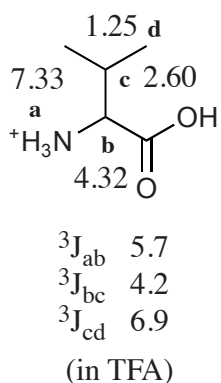
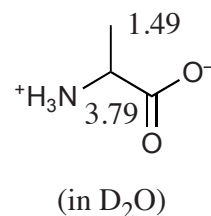
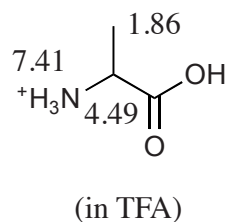
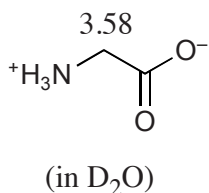
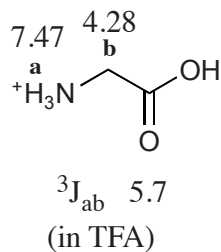
P Si

5.12.4 References

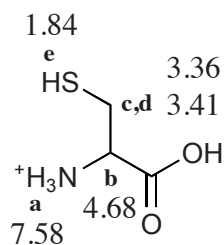
[1] M.D. Reily, L.C. Robosky, M.L. Manning, A. Butler, J.D. Baker, R.T. Winters, DFTMP, an NMR reagent for assessing the near-neutral pH of biological samples, *J. Am. Chem. Soc.* **2006**, 128, 12360.

5.13 Natural Products

5.13.1 Amino Acids

Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)Natural
Products

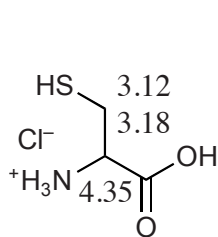
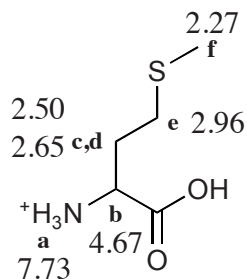
* average value



³J_{ab} 5.3 ²J_{cd}-15.5
³J_{bc} 5.0* ³J_{ce} 9.1*
³J_{bd} 5.0* ³J_{de} 9.1*

(in TFA)

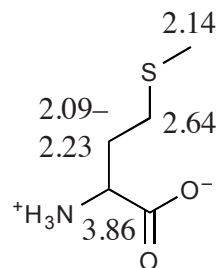
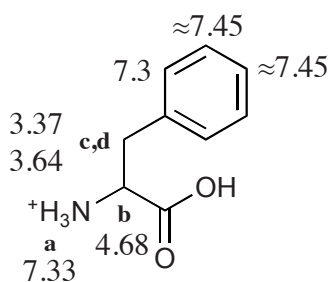
* average value

(in D₂O)

³J_{ab} 5.5 ²J_{cd}-15.7
³J_{bc} 7.7 ³J_{ce} 6.5*
³J_{bd} 4.4 ³J_{de} 6.5*

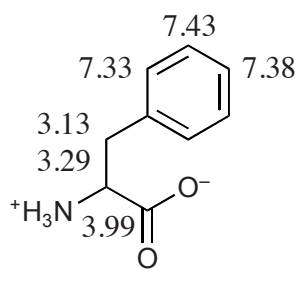
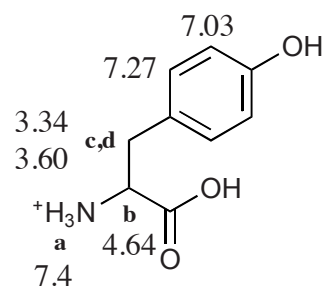
(in TFA)

* average value

(in D₂O)

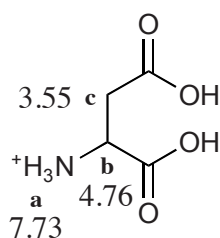
³J_{bc} 8.5
³J_{bd} 4.5
²J_{cd}-14.5

(in TFA)

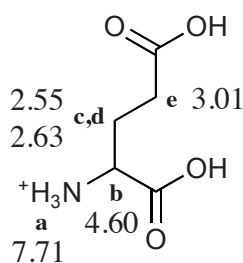
(in D₂O)

³J_{bc} 8.5
³J_{bd} 4.5
²J_{cd}-15.0

(in TFA)



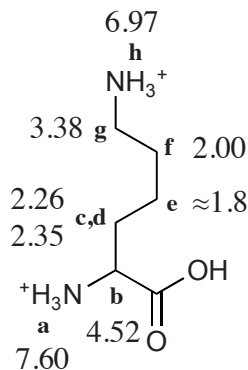
³J_{ab} 5.1
³J_{bc} 4.7
 (in TFA)



³J_{ab} 5.5 ²J_{cd}-15.5
³J_{bc} 5.6* ³J_{ce} 6.2*
³J_{bd} 5.6* ³J_{de} 6.2*

(in TFA)

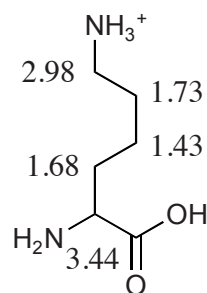
* average value

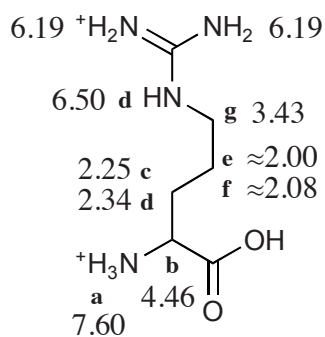


³J_{ab} 5.8 ²J_{cd}-15.0
³J_{bc} 5.6* ³J_{ce} 6.0*
³J_{bd} 5.6* ³J_{de} 6.0*

(in TFA)

* average value

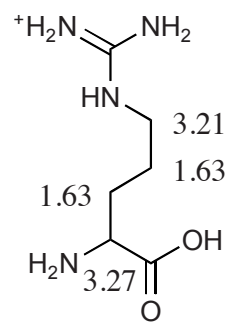
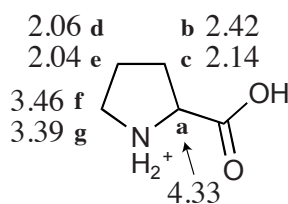
(in D₂O)Natural
Products



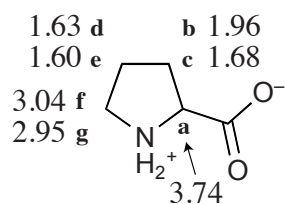
(in TFA)

* average value

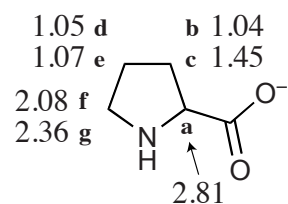
$^3J_{ab}$	5.5
$^3J_{bc}$	5.3*
$^3J_{bd}$	5.3*
$^2J_{cd}$	≈ -15.0
$^3J_{eg}$	6.5*
$^3J_{fg}$	6.5*
$^3J_{gh}$	5.3

(in D₂O)

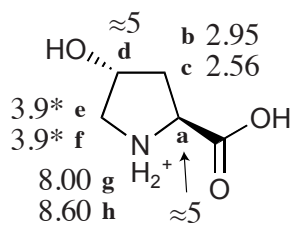
$^3J_{ab}$	8.5	$^3J_{cd}$	5.5
$^3J_{ac}$	6.5	$^3J_{ce}$	7.5
$^2J_{bc}$	-13.5	$^2J_{de}$	-13.0
$^3J_{bd}$	7.5	$^3J_{df}$	5.5
$^3J_{be}$	5.5	$^3J_{dg}$	7.5
$^4J_{bf}$	-0.4	$^3J_{ef}$	7.5
$^4J_{bg}$	0.0	$^3J_{eg}$	5.5
		$^2J_{fg}$	-11.0

(in D₂O, pH 2.0)

$^3J_{ab}$	8.4	$^3J_{cd}$	5.6
$^3J_{ac}$	6.2	$^3J_{ce}$	7.8
$^2J_{bc}$	-13.5	$^2J_{de}$	-13.0
$^3J_{bd}$	7.6	$^3J_{df}$	5.7
$^3J_{be}$	5.4	$^3J_{dg}$	7.9
$^4J_{bf}$	-0.4	$^3J_{ef}$	7.9
$^4J_{bg}$	0.0	$^3J_{eg}$	5.7
		$^2J_{fg}$	-11.0

(in D₂O, pH 7.0)

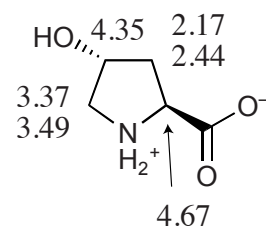
$^3J_{ab}$	8.6	$^3J_{cd}$	6.7
$^3J_{ac}$	6.6	$^3J_{ce}$	8.5
$^2J_{bc}$	-12.0	$^2J_{de}$	-11.0
$^3J_{bd}$	8.1	$^3J_{df}$	5.5
$^3J_{be}$	5.9	$^3J_{dg}$	8.1
$^4J_{bf}$	-0.6	$^3J_{ef}$	7.7
$^4J_{bg}$	0.0	$^3J_{eg}$	5.7
		$^2J_{fg}$	-10.5

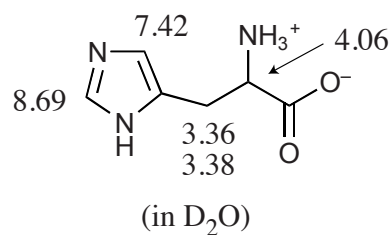
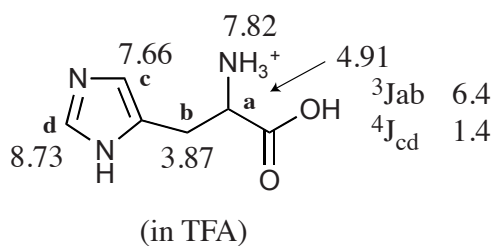
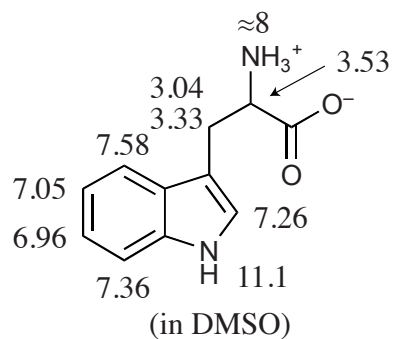
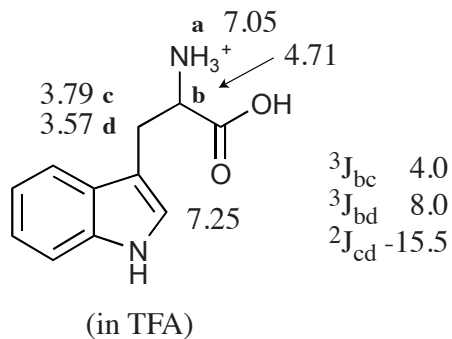
(in D₂O, pH 13.0)

(in TFA)

* average value

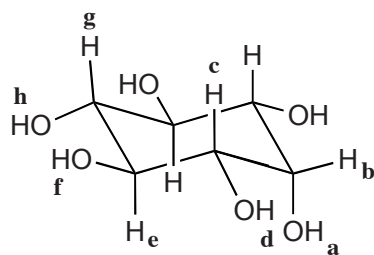
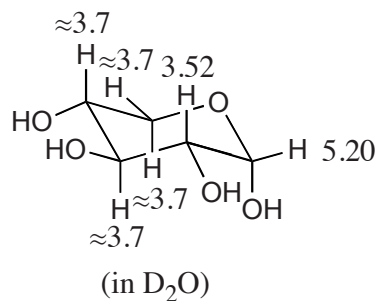
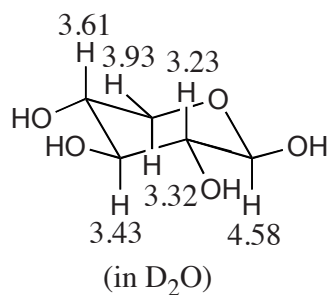
$^3J_{ab}$	8.2
$^3J_{ac}$	10.4
$^2J_{bc}$	-15.0
$^3J_{bd}$	<2
$^3J_{cd}$	4.2

(in D₂O)Natural
Products

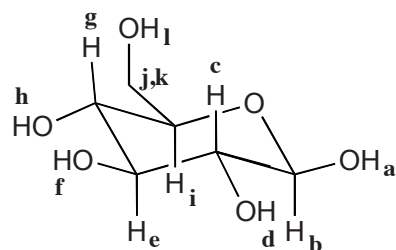


5.13.2 Carbohydrates

Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

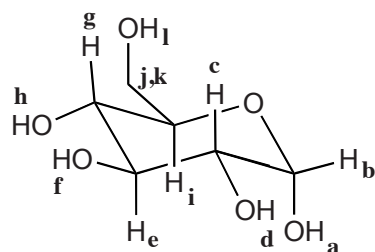


	D ₂ O	DMSO
a		4.55
b	4.07	3.72
c	3.54	3.14
d		4.51
e	3.64	3.37
f		4.46
g	3.27	2.93
h		4.31



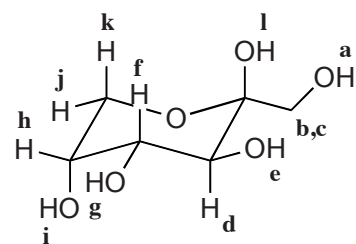
(in DMSO ca. 80%)

	D ₂ O*	DMSO	D ₂ O	DMSO
a		6.58	$^3J_{bc}$ 7.8	$^3J_{ab}$ 6.5
b	4.51	4.27	$^3J_{ce}$ 9.5	$^3J_{cd}$ 4.5–6
c	3.13	2.89	$^3J_{eg}$ 9.5	$^3J_{ef}$ 4.5–6
d		4.84	$^3J_{gi}$ 9.5	$^3J_{gh}$ 4.5–6
e	3.37	3.10	$^3J_{ij}$ 2.8	$^3J_{jl}$ 5.5
f		4.84	$^3J_{ik}$ 5.7	$^3J_{kl}$ 6.0
g	3.30	3.10	$^2J_{jk}$ -12.8	
h		4.84		
i	3.35	3.04		
j	3.60	3.42		
k	3.75	3.66		
l		4.48		

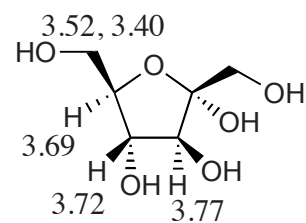
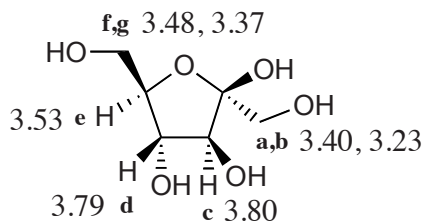
(* relative to internal
acetone at $\delta = 2.12$)

(in DMSO ca. 20%)

	D ₂ O*	DMSO	D ₂ O	DMSO
a		6.20	$^3J_{bc}$ 3.6	$^3J_{ab}$ 4.5
b	5.09	4.91	$^3J_{ce}$ 9.5	$^3J_{cd}$ 6.8
c	3.41	3.10	$^3J_{eg}$ 9.5	$^3J_{ef}$ 4.8
d		4.84	$^3J_{gi}$ 9.5	$^3J_{gh}$ 5.5
e	3.61	3.42	$^3J_{ij}$ 2.8	$^3J_{jl}$ 5.7
f		4.64	$^3J_{ik}$ 5.7	$^3J_{kl}$ 6.2
g	3.29	3.04	$^2J_{jk}$ -12.8	
h		4.77		
i	3.72	3.57		
j	3.72	3.57		
k	3.63	3.42		
l		4.37		

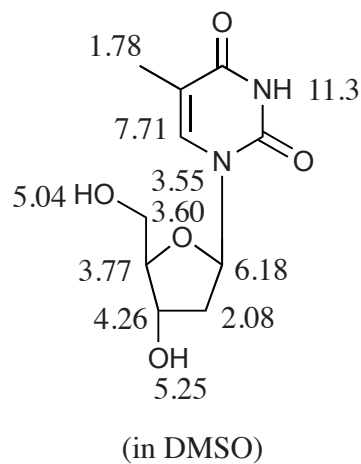
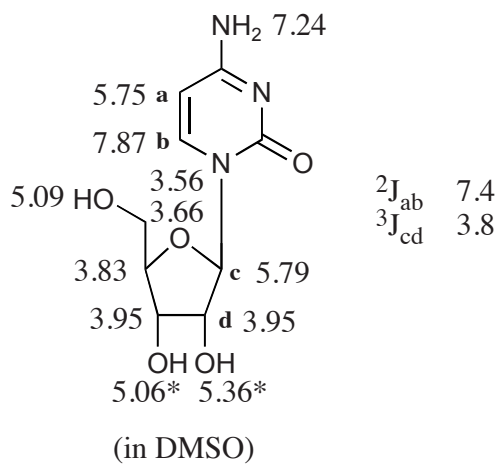
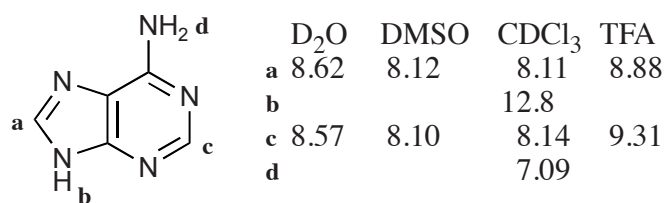
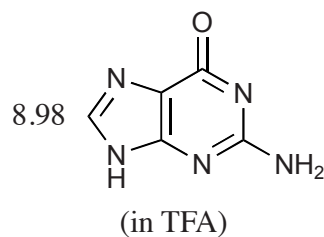
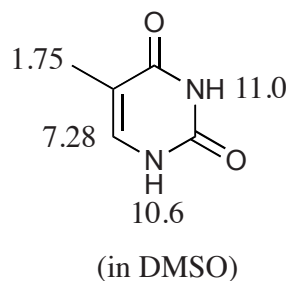
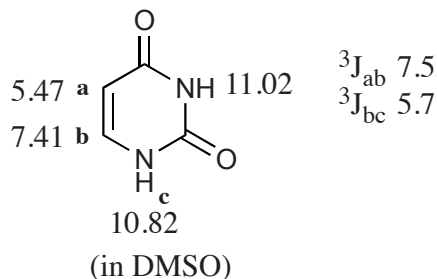
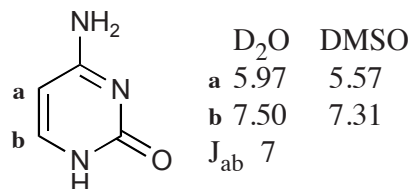
(* relative to internal
acetone at $\delta = 2.12$)a 25% β -D
b 75% β -D

	D ₂ O ^a	DMSO ^b	DMSO	
a		4.48	$^3J_{ab}$ 7.4 ^c	
b	3.68	3.39	$^3J_{ac}$ 5.4 ^c	
c	3.53	3.25	$^2J_{bc}$ -11.3 ^d	
d	3.76	3.55	$^3J_{de}$ 6.8 ^c	
e		4.23	$^3J_{df}$ 10.1 ^d	
f	3.86	3.58	$^3J_{fg}$ 5.8 ^c	
g		4.38	$^3J_{fh}$ 4.0 ^d	
h	3.96	3.62	$^3J_{hi}$ 3.8 ^c	
i		4.32	$^3J_{hj}$ 1.9 ^d	
j	4.00	3.77	$^3J_{hk}$ 1.6 ^d	^c at 25 °C
k	3.68	3.41	$^2J_{jk}$ -12.1 ^d	^d at 70 °C
l		5.14		

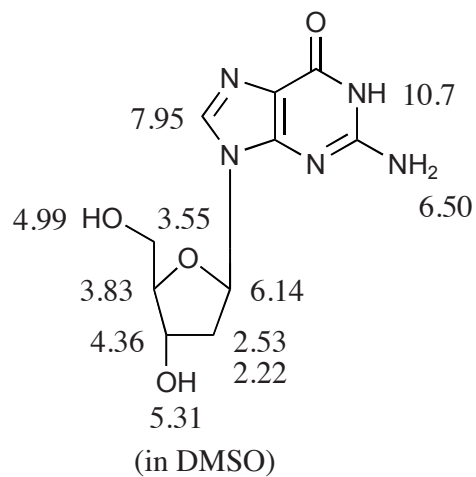
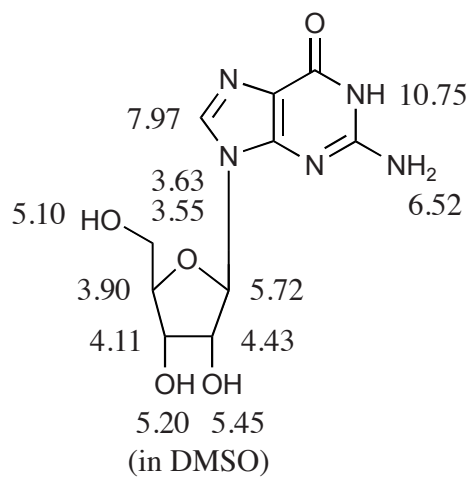
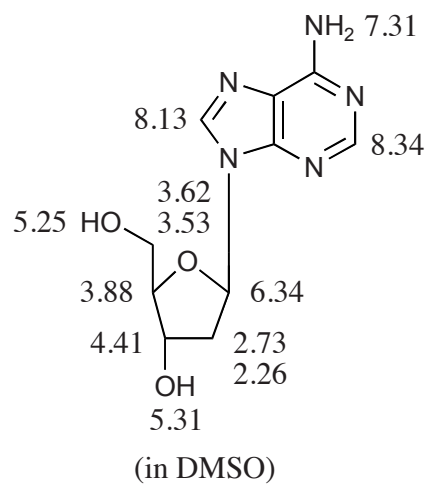
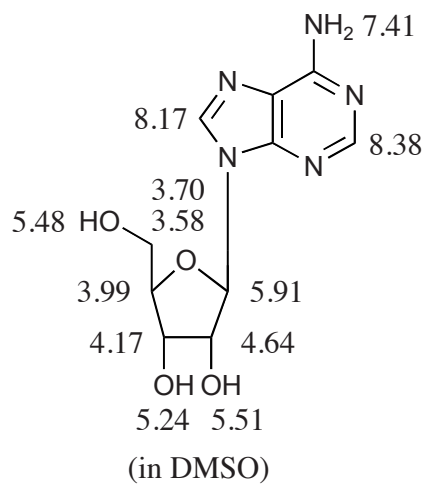
Natural
Products(in DMSO, 20% α -D)(in DMSO, 55% β -D)

$^2J_{ab}$	-11.0
$^3J_{cd}$	7.1
$^3J_{de}$	5.9
$^3J_{ef}$	2.3
$^3J_{eg}$	3.6
$^2J_{fg}$	-11.3

5.13.3 Nucleotides and Nucleosides

Chemical Shifts and Coupling Constants (δ in ppm, J in Hz)

* interchangeable

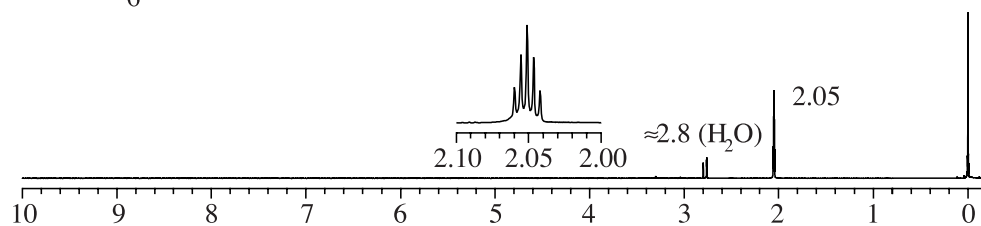


5.14 Spectra of Solvents and Reference Compounds

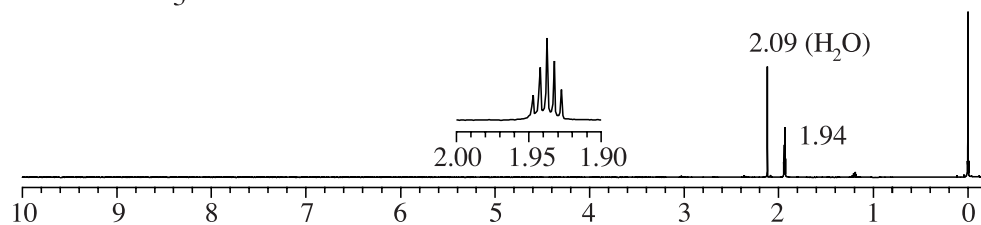
5.14.1 ^1H NMR Spectra of Common Deuterated Solvents

500 MHz; $\approx 1\,000$ data points per 1 ppm; δ in ppm relative to TMS

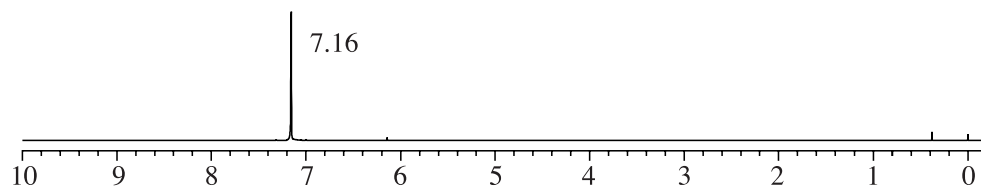
Acetone- d_6



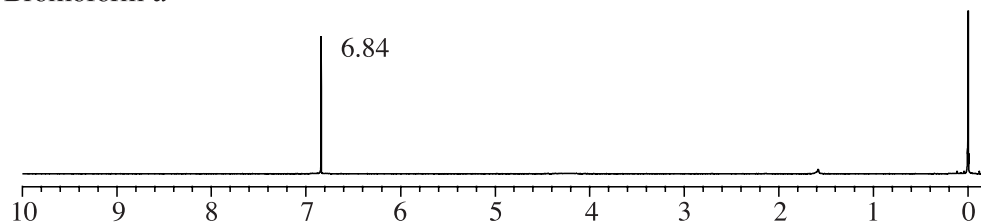
Acetonitrile- d_3



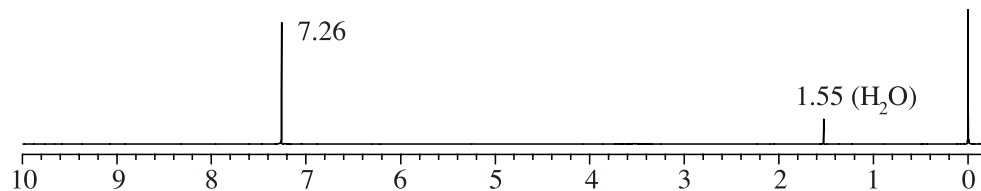
Benzene- d_6



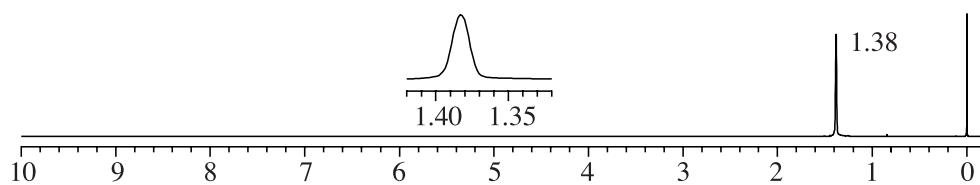
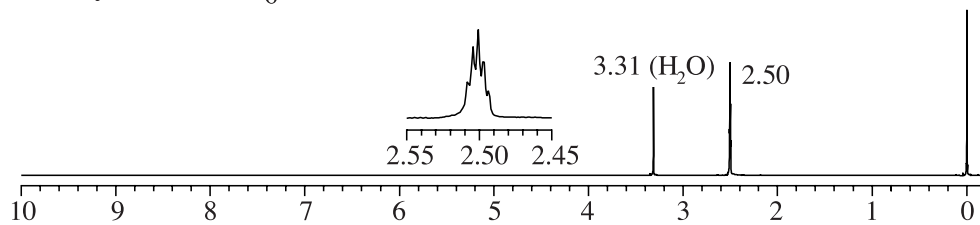
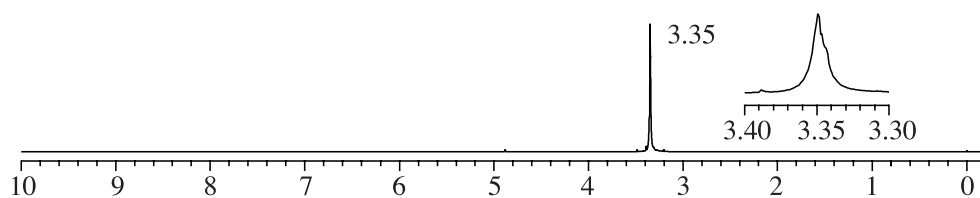
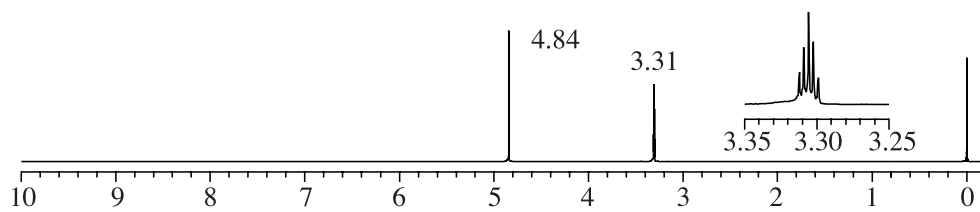
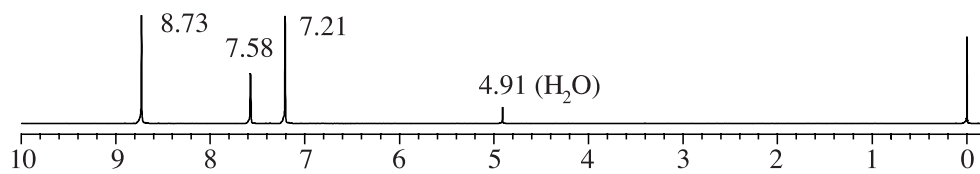
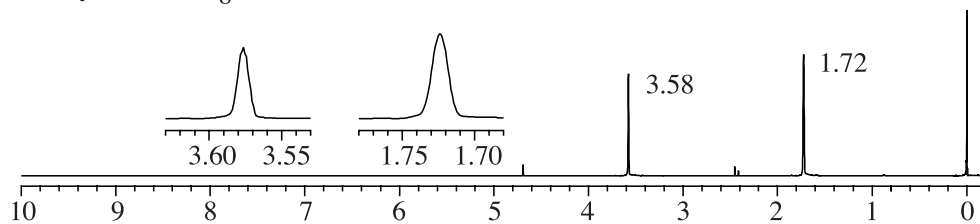
Bromoform- d

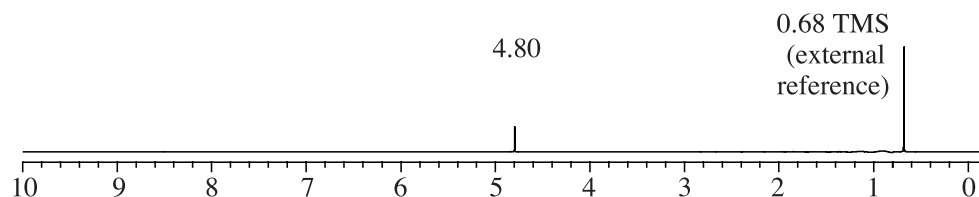


Chloroform- d



Solvents

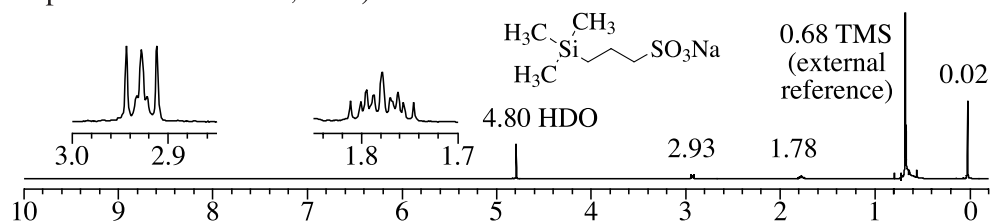
Cyclohexane- d_{12} Dimethyl sulfoxide- d_6 Methanol- d_1 Methanol- d_4 Pyridine- d_5 Tetrahydrofuran- d_8 

Water- d_2 

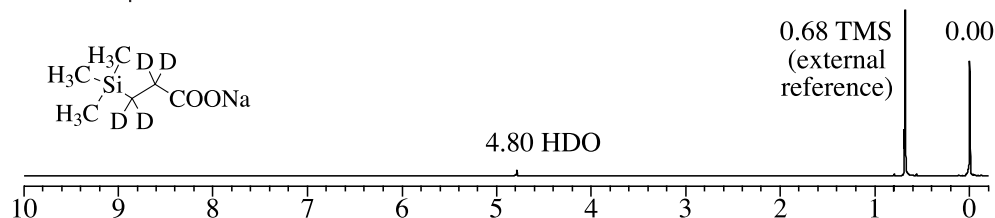
5.14.2 ^1H NMR Spectra of Secondary Reference Compounds

Chemical shifts in ^1H NMR spectra are usually reported relative to the peak position of tetramethylsilane (TMS) added to the sample as an internal reference. If TMS is not sufficiently soluble, a capillary with TMS may be used as external reference. In this case, owing to the different volume susceptibilities, the local magnetic fields in the sample and reference differ, and the peak position of the reference must be corrected. For a D_2O solution in a cylindrical sample and neat TMS in a capillary, the correction amounts to +0.68 and -0.34 ppm for superconducting and electromagnets, respectively. These values must be subtracted from the chemical shifts relative to the external TMS signal if its position is set to 0.00 ppm. Alternatively, secondary references with $(\text{CH}_3)_3\text{SiCH}_2$ groups may be used. The following spectra of two such secondary reference compounds in D_2O were measured at 500 MHz with TMS as external reference. Chemical shifts are reported in ppm relative to TMS upon correction for the difference in the volume susceptibilities of D_2O and TMS. As a result, the peak for the external TMS appears at 0.68 ppm.

3-(Trimethylsilyl)-1-propanesulfonic acid sodium salt (sodium 4,4-dimethyl-4-silapentane-1-sulfonate; DSS)



2,2,3,3- D_4 -3-(Trimethylsilyl)propionic acid sodium salt



Solvents

5.14.3 ^1H NMR Spectrum of a Mixture of Common Nondeuterated Solvents

The following ^1H NMR spectrum (500 MHz, δ in ppm relative to TMS) of CDCl_3 containing 18 common solvents (0.05–0.4 vol%) is shown as a guide for the identification of possible impurities. Where the signals of several solvents overlap, insets show signals for the individual compounds from separate spectra. Peaks in these insets are labeled with the corresponding chemical shifts from their main spectrum but their values may differ by up to 0.03 ppm. Signals that are particularly prone to vary in their position are marked with *. THF: tetrahydrofuran; EGDME: ethylene glycol dimethyl ether.

